B.Tech.

IN

MATHEMATICS AND COMPUTING

CURRICULUM
AND
SYLLABI

(Applicable from 2026 Admission onwards)

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

Pala – 686635, KERALA, INDIA

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Programme Educational Objectives (PEOs) of B.Tech. in MATHEMATICS AND COMPUTING

PEO1	Develop strong foundations in mathematics, computing, and analytical reasoning to formulate, model, and solve complex scientific, engineering, and data-driven problems.
PEO2	Apply advanced concepts from optimization, machine learning, artificial intelligence, cryptography, numerical computing, and data analytics to develop innovative computational solutions for real-world challenges.
PEO3	Pursue successful careers, higher studies, research, entrepreneurship, and interdisciplinary professional practice through ethical conduct, effective communication, teamwork, leadership, and lifelong learning.

Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) of B.Tech. in MATHEMATICS AND COMPUTING

PO1	Apply knowledge of mathematics, statistics, computing, and engineering fundamentals to identify, formulate, and solve complex computational and real-world problems.
PO2	Analyze and model complex systems using mathematical reasoning, abstraction, probability, optimization, and algorithmic thinking to derive meaningful insights and effective solutions.
PO3	Design and develop efficient algorithms, mathematical models, software systems, and intelligent computational frameworks that satisfy specified requirements and constraints.
PO4	Conduct investigations using theoretical analysis, experimentation, simulation, numerical methods, machine learning techniques, and data analytics to generate valid conclusions and support decision making.
PO5	Select, adapt, and apply modern computational tools, programming environments, mathematical software, data science platforms, and emerging technologies for solving multidisciplinary problems.
PO6	Function effectively as an individual and as a member or leader of multidisciplinary teams, communicate technical ideas clearly, practice professional ethics, and engage in lifelong learning and innovation.
PSO1	Apply advanced mathematical concepts including algebra, analysis, probability, graph theory, optimization, topology, numerical methods, and measure theory to model and solve problems arising in computing, engineering, and data science.
PSO2	Develop computational solutions using algorithms, programming, machine learning, artificial intelligence, cryptography, database systems, and scientific computing techniques for intelligent information processing and decision making.
PSO3	Integrate mathematical modelling, data analytics, optimization, and computational methods to address interdisciplinary challenges in areas such as artificial intelligence, cybersecurity, network science, computational finance, scientific computing, and emerging technologies.

CURRICULUM

Total credits for completing B.Tech. in MATHEMATICS AND COMPUTING is 160. (In addition, 10 mandatory Activity Credits must be earned; these are not counted towards the CGPA.)

COURSE CATEGORIES AND CREDIT REQUIREMENTS

The structure of the B.Tech. programme shall have the following course categories:

Sl. No.	Course Category	Number of Courses	Minimum Credits
1	Institute Core (IC)	14 Theory + 9 Lab	57
2	Programme Core (PC)	19	65
3	Programme Electives (PE)	6	18
4	Open Electives (OE)	3	8
5	Projects (P)	2	12
		Total credits (counted for CGPA)	**160**
6	Activity Credits (AC) – *mandatory, not counted for CGPA*	–	10

COURSE REQUIREMENTS. The effort to be put in by the student is indicated in the tables as follows:
L: Lecture (one unit is of 50 minutes) **T**: Tutorial (one unit is of 50 minutes) **P**: Practical (one unit is of one hour)

1. Institute Core (IC)

(a) Engineering Foundation Courses

Sl.	Code	Course Title	L	T	P	Credits
1	CSC111	Computer Programming	3	1	0	4
2	ECC111	Electronic Devices and Digital Logic Design	3	1	0	4
3	CSC112	Foundations of Computing Systems	3	0	0	3
4	CSL111	Computer Programming Lab	0	0	2	1
5	ECL111	Electronic Devices and Digital Logic Design Lab	0	0	2	1
6	CSC121	Data Structures	3	1	0	4
7	CSC122	Computer Organization and Architecture	3	1	0	4
8	CSL121	Data Structures lab (C, C++)	0	0	2	1
9	CSC211	Computer Networks	2	1	0	3
10	CSC212	Design and Analysis of Algorithms	2	1	0	3
11	CSC213	Theory of Computation	3	1	0	4
12	CSL211	Computer Networks Lab	0	0	2	1
13	CSL212	Algorithm Design Lab	0	0	2	1
14	CAC221	Machine Learning and Data Analysis	2	1	0	3
15	CSC223	Database Management Systems	2	1	0	3
16	CSC224	Advanced Data Structures	2	1	0	3
17	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1
18	CSL223	Database Management Systems Lab	0	0	2	1
19	CAC312	Artificial Intelligence	3	0	0	3
20	CAC313	Advanced Machine Learning	3	0	0	3
21	CSC313	Cryptography and Network Security	3	1	0	4
22	CAL313	Advanced Machine Learning Lab	0	0	2	1
22	CSL313	Cryptography and Network Security Lab	0	0	2	1
		Total	**38**	**10**	**18**	**57**

2. Programme Core (PC)

(a) Computational Mathematics

Sl.	Code	Course Title	L	T	P	Credits
1	MAC111	Calculus and Linear Algebra	3	1	0	4
2	MAC122	Modern Algebra	3	0	0	3
3	MAC123	Real and Complex Analysis	3	1	0	4
4	MAC124	Vector Calculus and Applications	3	1	0	4
5	MAC211	Probability, Statistics and Random Processes	3	1	0	4
6	MAC212	Graphs and Networks in Computing Systems	3	1	0	4
7	MAC221	Differential Equations and Transforms	3	1	0	4
8	MAC222	Numerical Linear Algebra	3	1	0	4
9	MAC311	Soft Computing	3	0	0	3
10	MAC312	Optimization Techniques for Data Science	3	0	0	3
11	MAC321	Linear Programming and Combinatorial Optimization	3	0	0	3
12	MAC322	Numerical Methods and Scientific Computing	3	1	0	4
13	MAC323	Topological Data Analysis	3	1	0	4
14	MAC324	Applied Functional Analysis	3	1	0	4
15	MAC411	Measure and Integration	3	0	0	3
		Total	**45**	**10**	**0**	**55**

(b) Economics, Management & Entrepreneurship

Sl.	Code	Course Title	L	T	P	Credits
1	HMC211	Engineering Management and Entrepreneurship	2	0	0	2
2	HMC221	Engineering Economics, Finance and Business Analytics	2	0	0	2
		Total	**4**	**0**	**0**	**4**

(c) Communication Skills and Personality Development

Sl.	Code	Course Title	L	T	P	Credits
1	HMC111	Communication Skills for the Digital Age	2	0	2	3
2	HMC121	Personality Development & Soft Skills	3	0	0	3
		Total	**5**	**0**	**2**	**6**

3. List of Programme Electives (PE)

Sl.	Code	Course Title	L	T	P	Credits
1	MAEXXX	Applied Functional Analysis	3	1	0	4
2	MAEXXX	Applied Complex Analysis for Engineering	2	1	0	3
3	MAEXXX	Markov Decision Process and Reinforcement Learning	2	0	0	2
4	MAEXXX	Computational Algebra	3	1	0	4
5	MAEXXX	Approximation Algorithms	3	1	0	4
6	MAEXXX	Advanced Convex Optimization	3	1	0	4
7	MAEXXX	Algebraic Graph Theory	3	1	0	4
8	MAEXXX	Algorithmic Graph Theory	3	1	0	4
9	MAEXXX	Fractional Calculus	3	0	0	3
10	MAEXXX	Mathematical Physics	3	1	0	4
11	MAEXXX	Stochastic Processes and Applications	3	0	0	3
12	MAEXXX	Wavelet Theory and Applications	3	1	0	4
13	MAEXXX	Fuzzy Mathematics and Intelligent Systems	3	0	0	3
14	MAEXXX	Operations Research	3	0	0	3
15	MAEXXX	Introduction to Dynamical Systems Theory	3	0	0	3
16	MAEXXX	Mathematical and Computational Finance	3	1	0	4
17	MAEXXX	Computational Fluid Dynamics	3	1	0	4
18	MAEXXX	Statistical Learning	3	1	0	4
19	MAEXXX	Number Theory and Mathematical Theory of Coding	3	0	0	3
20	MAEXXX	Advanced Cryptography for Cyber Security	3	1	0	4
21	MAEXXX	Computational Neuroscience	3	0	0	3
22	CAE410	Scientific Machine Learning	3	0	0	3

4. List of Open Electives (OE)

Courses offered by other departments or approved online platforms (e.g. NPTEL), subject to the maximum permitted under the B.Tech. Ordinances and Regulations. PE courses offered by the parent department are also available to its own students under this category.

Sl.	Code	Course Title	L	T	P	Credits
1	CSO003	Quantum Computing for Engineers	2	1	0	3
2	CAO002	Deep Learning	3	0	0	3
3	CAO003	Design Thinking and Innovation	3	0	0	3
4	CAO004	Large Language Models	3	0	0	3
5	IHS314	Project Management and Quality Assurance	3	0	0	3
6	IHS315	Introduction to Indian Knowledge Systems	3	0	0	3
7	OE	Foreign Language	2	0	0	2
8	IHS316	Fundamentals of Sanskrit	3	0	0	3
9	CSO006	Internet of Things: Concepts and Applications	3	0	0	3

5. Projects (P)

The major project represents the culmination of study towards the B.Tech. degree. It is offered in two parts – Project-1 and Project-2 – and gives students the opportunity to apply and extend the knowledge gained across the programme. The work may be analytical, simulation-based, hardware design, or a combination, in an emerging area of MATHEMATICS AND COMPUTING, carried out under the supervision of a department faculty member. Project work may be carried out individually or in groups (size decided by the department) and may build on an industrial problem identified during an internship.

6. Activity Credits (AC)

A minimum of 40 Activity Points must be earned to obtain the 10 credits required in the curriculum; these credits are not counted towards CGPA. The scheme of activity points, as approved by the Senate, is announced at the start of the first semester. A minimum of 20 points must be earned by the end of the 4th semester and 40 points by the end of the 7th semester.

PROGRAMME STRUCTURE

Semester-wise Course Structure (I–VIII)

Semester I						**Semester II**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC111	Calculus and Linear Algebra	3	1	0	4	MAC122	Modern Algebra	3	0	0	3
CSC111	Computer Programming	3	1	0	4	MAC 123	Real and Complex Analysis	3	1	0	4
ECC111	Electronic Devices and Digital Logic Design	3	1	0	4	MAC 124	Vector Calculus and Applications	3	1	0	4
HMC 111	Communication Skills for the Digital Age	2	0	2	3	CSC121	Data Structures	3	1	0	4
CSC112	Foundations of Computing Systems	3	0	0	3	HMC 121	Personality Development & Soft Skills	3	0	0	3
CSL111	Computer Programming Lab	0	0	2	1	CSC122	Computer Organization and Architecture	3	1	0	4
ECL111	Electronic Devices and Digital Logic Design Lab	0	0	2	1	CSL121	Data Structures lab (C, C++)	0	0	2	1
Total		14	3	6	20	**Total**		18	4	2	23
Cumulative Credits at the End of First Year: 43											
Semester III						**Semester IV**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC211	Probability, Statistics and Random Processes	3	1	0	4	MAC221	Differential Equations and Transforms	3	1	0	4
MAC212	Graphs and Networks in Computing Systems	3	1	0	4	MAC 222	Numerical Linear Algebra	3	1	0	4
HMC 211	Engineering Management and Entrepreneurship	2	0	0	2	HMC 221	Engineering Economics, Finance and Business Analytics	2	0	0	2
CSC211	Computer Networks	2	1	0	3	CAC221	Machine Learning and Data Analysis	2	1	0	3
CSC212	Design and Analysis of Algorithms	2	1	0	3	CSC223	Database Management Systems	2	1	0	3
CSC213	Theory of Computation	3	1	0	4	CSC224	Advanced Data Structures	2	1	0	3
CSL211	Computer Networks Lab	0	0	2	1	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1
CSL212	Algorithm Design Lab	0	0	2	1	CSL223	Database Management Systems Lab	0	0	2	1
Total		15	5	4	22	**Total**		14	5	4	21
Cumulative Credits at the End of Second Year: 86											
Semester V						**Semester VI**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC311	Soft Computing	3	0	0	3	MAC321	Linear Programming and Combinatorial Optimization	3	0	0	3
MAC312	Optimization Techniques for Data Science	3	0	0	3	MAC322	Numerical Methods and Scientific Computing	3	1	0	4
OE	OE-1	2	1	0	3	MAC323	Topological Data Analysis	3	1	0	4
CAC312	Artificial Intelligence	3	0	0	3	OE	Foreign Language	2	0	0	2
CAC313	Advanced Machine Learning	3	0	0	3	CAO003	Design Thinking and Innovation	3	0	0	3
CSC313	Cryptography and Network Security	3	1	0	4	MAC324	Applied Functional Analysis	3	1	0	4
CAL313	Advanced Machine Learning Lab	0	0	2	1	MAE321	Elective I	3	0	0	3
CSL313	Cryptography and Network Security Lab	0	0	2	1	Honours	Advanced Course I	3	0	0	3
Total		17	3	4	21	**Total**		20	3	0	23
Cumulative Credits at the End of Third Year: 130											
Semester VII						**Semester VIII**					
Code	Course Title	L	T	P	C	Code	Course Title	L	T	P	C
MAC411	Measure and Integration	3	0	0	3	MAP421	BTP II	0	0	18	9
MAE411	Elective II	3	0	0	3	IHP421	Honours Project II	0	0	22	11
MAE412	Elective III	3	0	0	3						
MAE413	Elective IV	3	0	0	3						
MAE414	Elective V	3	0	0	3						
MAE415	Elective VI	3	0	0	3						
MAP411	BTP I	0	0	6	3						
Honours	Advanced Course II	3	0	0	3						
Honours	Honours Project Phase I	0	0	6	3						
Total		18	0	6	21	**Total**		0	0	18	9
Total Credits at the End of Fourth Year: **160** (excluding Activity Credits)											

Detailed Semester-wise Structure

Semester I

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC111	Calculus and Linear Algebra	3	1	0	4	PC
2	CSC111	Computer Programming	3	1	0	4	IC
3	ECC111	Electronic Devices and Digital Logic Design	3	1	0	4	IC
4	HMC111	Communication Skills for the Digital Age	2	0	2	3	PC
5	CSC112	Foundations of Computing Systems	3	0	0	3	IC
6	CSL111	Computer Programming Lab	0	0	2	1	IC
7	ECL111	Electronic Devices and Digital Logic Design Lab	0	0	2	1	IC
		Total	14	3	6	20	

Semester II

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC122	Modern Algebra	3	0	0	3	PC
2	MAC123	Real and Complex Analysis	3	1	0	4	PC
3	MAC124	Vector Calculus and Applications	3	1	0	4	PC
4	CSC121	Data Structures	3	1	0	4	IC
5	HMC121	Personality Development and Soft Skills	3	0	0	3	PC
6	CSC122	Computer Organization and Architecture	3	1	0	4	IC
7	CSL121	Data Structures lab (C, C++)	0	0	2	1	IC
		Total	18	4	2	23	

Semester III

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC211	Probability, Statistics and Random Processes	3	1	0	4	PC
2	MAC212	Graphs and Networks in Computing Systems	3	1	0	4	PC
3	HMC211	Engineering Management and Entrepreneurship	2	0	0	2	PC
4	CSC211	Computer Networks	2	1	0	3	IC
5	CSC212	Design and Analysis of Algorithms	2	1	0	3	IC
6	CSC213	Theory of Computation	3	1	0	4	IC
7	CSL211	Computer Networks Lab	0	0	2	1	IC
8	CSL212	Algorithm Design Lab	0	0	2	1	IC
		Total	16	4	4	22	

Semester IV

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC221	Differential Equations and Transforms	3	1	0	4	PC
2	MAC222	Numerical Linear Algebra	3	1	0	4	PC
3	HMC221	Engineering Economics, Finance and Business Analytics	2	0	0	2	PC
4	CAC221	Machine Learning and Data Analysis	2	1	0	3	IC
5	CSC223	Database Management Systems	2	1	0	3	IC
6	CSC224	Advanced Data Structures	2	1	0	3	IC
7	CAL221	Machine Learning and Data Analysis Lab	0	0	2	1	IC
8	CSL223	Database Management Systems Lab	0	0	2	1	IC
		Total	14	5	4	21	

Semester V

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC311	Soft Computing	3	0	0	3	PC
2	MAC312	Optimization Techniques for Data Science	3	0	0	3	PC
3	OE	OE-1	2	1	0	3	OE
4	CAC312	Artificial Intelligence	3	0	0	3	IC
5	CAC313	Advanced Machine Learning	3	0	0	3	IC
6	CSC313	Cryptography and Network Security	3	1	0	4	IC
7	CAL313	Advanced Machine Learning Lab	0	0	2	1	IC
8	CSL313	Cryptography and Network Security Lab	0	0	2	1	IC
		Total	17	3	2	21	

Semester VI

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC321	Linear Programming and Combinatorial Optimization	3	0	0	3	PC
2	MAC322	Numerical Methods and Scientific Computing	3	1	0	4	PC
3	MAC323	Topological Data Analysis	3	1	0	4	PC
4	OE	Foreign Language	2	0	0	2	OE
5	CAO003	Design Thinking and Innovation	3	0	0	3	OE
6	MAC324	Applied Functional Analysis	3	1	0	4	PC
7	MAE321	Elective I	3	0	0	3	PE
8	Honours	Advanced Course I	3	0	0	3	
		Total	20	3	0	23	

Semester VII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAC411	Measure and Integration	3	0	0	3	PC
2	MAE411	Elective II	3	0	0	3	PE
3	MAE412	Elective III	3	0	0	3	PE
4	MAE413	Elective IV	3	0	0	3	PE
5	MAE414	Elective V	3	0	0	3	PE
6	MAE415	Elective VI	3	0	0	3	PE
7	MAP411	BTP I	0	0	6	3	P
8	Honours	Advanced Course II	3	0	0	3	
9	Honours	Honours Project Phase I	0	0	6	3	P
		Total	18	0	6	21	

Semester VIII

Sl.	Code	Course Title	L	T	P	Credits	Category
1	MAP421	BTP II	0	0	18	9	P
2	IHP421	Honours Project II	0	0	22	11	P
		Total	0	0	18	9	

DETAILED SYLLABUS

MAC111 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To study the basic topological properties of the real numbers.
- To develop knowledge of sequences of real numbers and their convergence.
- To study the notion of continuous functions and their properties.
- To gain an understanding of systems of linear equations.
- To introduce the fundamental concepts of vector spaces.
- To impart the basics of linear transformations, orthogonalization, basis, dimension, and eigenvalues.
- To provide the knowledge required to apply concepts of linear algebra in engineering applications.

Course Outcomes

- CO1: Understand the concepts of limits, continuity, differentiability, and classical theorems (Rolle's, Mean Value, Fundamental Theorem of Calculus) to model and solve real-world problems in science and engineering.
- CO2: Analyze convergence of sequences and series, and optimize multivariable functions using Hessian matrices to address practical problems in data modelling, optimization and artificial intelligence.
- CO3: Formulate and solve systems of linear equations using matrix methods, and interpret solutions in applied contexts such as network flows, resource allocation, and computational simulations.
- CO4: Apply vector space concepts, orthogonalization techniques, and inner product structures to design efficient representations in signal processing, machine learning, and numerical computation.
- CO5: Compute eigenvalues and eigenvectors, perform matrix diagonalization, and apply these techniques in stability analysis, principal component analysis, engineering design etc.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Reimann Integration, fundamental theorem of calculus. Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence; Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Textbooks / References

1. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th Edition, Wiley, 2011.
2. T. M. Apostol, *Calculus*, Volume I, 2nd Edition, Wiley, 2007.
3. G. Strang, *Linear Algebra and Its Applications*, 5th Edition, Wellesley–Cambridge Press/SIAM, 2016.
4. K. Hoffman and R. Kunze, *Linear Algebra*, 2nd Edition, PHI, 2009.
5. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
6. Thomas and Finney, *Calculus*, 9th Edition, Addison Wesley Publishing, 1998.
7. Anton, Biven, *Calculus*, 11th Edition, Wiley, 2015
8. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press, 2019.

CSC111 Computer Programming [3-1-0-4]

Course Objectives

- To introduce problem solving techniques and their representation using algorithms, flowcharts, and pseudocode.
- To teach core concepts of C programming including data types, operators, control flow constructs, and input/output mechanisms.
- To familiarize students with arrays, character arrays, and user-defined functions with parameter passing mechanisms in C.
- To introduce the concepts of recursion and pointer-based programming in C.
- To familiarize students with derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Gain knowledge of problem solving techniques using algorithmic thinking, flowcharts, and pseudocode.
- CO2: Students learn to apply core C programming concepts including data types, control flow, arrays, functions, recursion, and pointers in problem solving.
- CO3: Students learn to use derived data types and structured programming constructs for developing efficient solutions in C.

Syllabus

Introduction to C Programming: Introduction to algorithm, flowchart, pseudocode. Introduction to C programming - Character set of C, basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and compiling simple C programs.

Control Flow Constructs: Conditional statements, if-else and switch constructs, nested conditions. Entry tested loops, exit tested loops, nested loop programs.

Arrays and Functions: Single-dimensional arrays, multidimensional arrays, character arrays. User-defined functions, function declaration, definition, call, return types, actual and formal parameters, parameter passing by value.

Recursion and Pointers: Recursive problem definitions, base case and recursive case. Example programs: factorial using recursion, Fibonacci series, Printing patterns. Introduction to pointers, address and dereferencing, pass-by-reference using pointers.

Derived Data Types: Use of typedef for user-defined type names, enumeration, unions and structures in C. Defining and accessing union members and structure members.

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, *Programming in ANSI C*, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, *C: How to Program*, Ninth Edition, Pearson Education, 2022.
4. V. Rajaraman, *Computer Programming in C*, Second Edition, Prentice Hall India, 2004.
5. R. G. Dromey, *How to Solve It by Computer*, Pearson Education India, 2007.
6. Reema Thareja, *Computer Fundamentals and Programming in C*, Second Edition, Oxford University Press, 2016.

ECC111 Electronic Devices and Digital Logic Design [3-1-0-4]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To analyse the operation of fundamental diode circuits and transistor switches.
- To develop a strong foundation in number systems, Boolean algebra, logic gates, and simplification techniques used in digital electronic systems.
- To analyze and design combinational and sequential logic circuits using standard digital building blocks such as multiplexers, counters, flip-flops, and shift registers.

Course Outcomes

- CO1: Analyse pn-junction behaviour under various operating conditions and design diode circuits
- CO2: Understand the working of MOSFETs as switching devices
- CO3: Apply number system conversions, Boolean algebra, Karnaugh maps, and logic gate principles to analyze and simplify digital circuits.
- CO4: Design and evaluate combinational and sequential circuits including adders, multiplexers, counters, flip-flops, and shift registers with appropriate timing analysis.

Syllabus

Introduction to Semiconductors: Intrinsic and extrinsic semiconductors, pn junction diode, diode equation (derivation not required), diode characteristics, ideal and practical diode (Silicon), diode applications: clipping and clamping circuits. Field Effect Transistors: voltage-current relation in MOSFET (derivation not required), MOSFETs as switches.

Number Systems and base conversion, Binary Arithmetic, Representation of Negative Numbers: sign and magnitude, 2's complement, 1's complement forms, shifts, and overflow, Binary Coded Decimal (BCD). Boolean algebra, De Morgan's laws, Basic Theorems, truth tables, SOP/POS. Minterm and Maxterm Expansions, Incompletely Specified Functions, don't–care conditions, Karnaugh map, Karnaugh map SOP and POS minimizations (Two, Three, Four and Five-variable Karnaugh maps), NAND/NOR realization.

Combinational Logic Circuits–Half and Full Adders, Parallel Binary Adders, Binary Subtractor, Magnitude Comparators, Decoders, Encoders, Code Converters, Multiplexers, Demultiplexers.

Sequential Logic Circuits – Latches and Flip–Flops, SR Latch, Gated D Latch, Edge–Triggered D Flip-Flop, SR Flip–Flop, JK Flip-Flop, T Flip–Flop, Flip-Flops with Asynchronous Clear and Preset inputs, Master-Slave Flip-Flops, Excitation table, Flip-Flop conversions, Counters – Asynchronous Counters, Synchronous Counters, Up/Down Synchronous Counters, Design of Synchronous Counters. Shift Registers – data movements in shift registers, SISO, SIPO, PISO, PIPO shift registers, Design of sequence detector circuits.

Textbooks / References

1. Sedra A. and Smith K. C., Microelectronic Circuits, Oxford University Press, Seventh Edition, 2015.
2. Robert L. Boylestad and Louis Nashelsky, Electronic Devices and Circuit Theory, Pearson Education, Eleventh Edition, 2015.
3. Ben G. Streetman and Sanjay Kumar Banerjee, Solid State Electronic Devices, Seventh Edition, Pearson Education, 2016.
4. M. Morris Mano and Michael D. Ciletti, Digital Design, 6th Edition, Pearson Education, 2018.
5. Thomas L. Floyd, Digital Fundamentals, 11th Edition, Pearson Education, 2015.
6. Charles H. Roth Jr. and Larry L. Kinney, Fundamentals of Logic Design, 7th Edition, Cengage Learning, 2013.

HMC111 Communication Skills for the Digital Age [2-0-2-3]

Course Objectives

- Understand the fundamentals of communication and technical communication.
- Develop effective listening, speaking, reading, and writing skills for academic and professional contexts.
- Apply grammatical structures and vocabulary appropriately in oral and written communication.
- Analyze communication barriers, audience needs, and communication styles in different contexts.
- Create professional documents, presentations, and technical content using appropriate language and format.
- Understand the impact of Artificial Intelligence and Digital Humanities on contemporary communication practices.
- Practice ethical, responsible, and inclusive communication in digital and multilingual environments.

Course Outcomes

- CO1: Apply effective listening, speaking, reading, and writing strategies in academic and professional contexts.
- CO2: Analyze communication barriers, audience needs, and communication styles in different contexts.
- CO3: Create professional documents, presentations, and technical content using appropriate language and format.
- CO4: Evaluate the role of AI and digital technologies in communication with ethical awareness.
- CO5: Demonstrate responsible and effective communication in multilingual and digital environments.

Syllabus

Fundamentals of Communication: Process and types of communication - Technical communication and its importance - Barriers to communication - Non-verbal communication - Communication in academic and professional contexts - Shifts in communication in the digital age

Listening and Speaking Skills: Listening process and active listening - Barriers to listening and strategies to overcome them - Fluency and accuracy in speech - Self-expression and tonal variations - Group discussion and interpersonal communication - Public speaking and presentation skills - Interviews and professional interaction - Responsible use of AI in communication

Reading, Writing and Digital Communication: Reading comprehension and critical reading - Elements of effective writing - Professional emails and formal letters - Technical reports and presentations - Introduction to academic writing - AI in multilingual communication and translation - Ethical and legal considerations in AI-mediated communication - Introduction to Digital Humanities and digital communication practices

Grammar and Vocabulary for Communication: Parts of speech and sentence structures - Tenses and Subject–Verb Agreement - Reported Speech and Active–Passive Voice - Common errors in English usage - Punctuation and mechanics of writing - Vocabulary enhancement in context

Practical / Language Laboratory Component: Listening to recorded talks and comprehension activities - Active listening exercises - Group discussion and interpersonal conversation - Pronunciation and tonal variation practice - Extempore speech and presentation activities - Mock interviews and persuasive speaking exercises - Speech assessment and peer feedback - Technology-assisted listening and speaking practice.

Textbooks / References

1. Anderson, Paul V., and Kerry Surman. *Technical Communication: A Reader-Centered Approach*. Thomson/ Wadsworth, 2007.
2. Berry, David M. "Introduction: Understanding the Digital Humanities." In *Understanding Digital Humanities*. Palgrave Macmillan UK, 2012.
3. Davis, Marjorie T. "Assessing Technical Communication within Engineering Contexts Tutorial." *IEEE Transactions on Professional Communication*, 53(1), 2010.
4. Duin, Ann Hill, and Isabel Pedersen. *Augmentation Technologies and Artificial Intelligence in Technical Communication: Designing Ethical Futures*. Routledge, 2023.
5. Raman, Meenakshi, and Sangeeta Sharma. *Technical Communication: Principles and Practice*. Oxford University Press, 2004.
6. Verma, Shalini. *Technical Communication for Engineers*. Vikas Publishing House, 2015.
7. Yule, George. *Oxford Practice Grammar Advanced*. Oxford University Press, 2015.

CSC112 Foundations of Computing Systems [3-0-0-3]

Course Objectives

- Cultivate a systematic problem-solving mindset using computational thinking principles (decomposition, abstraction, and algorithmic design).
- Establish the mathematical and logical fundamentals governing digital data representation, reliability, and physical hardware execution.
- Provide a unified view of computer systems by bridging the structural hierarchy from physical logic gates up through software layers and networking frameworks.
- Introduce the mechanics of distributed execution, concurrency, and real-world system modeling.
- Evaluate the physical and mathematical limits of classical computing while exploring the architecture of emerging, next-generation paradigms.

Course Outcomes

- CO1: Apply computational thinking and modular decomposition to design structured algorithms while selecting optimal data types and number systems for digital representation.
- CO2:Trace how digital data encoding, binary arithmetic, and overflow constraints directly dictate the design of physical logic circuits and processor registers.
- CO3: Map the abstraction layers of a computing system, linking high-level software execution down to machine code, processor cycles, and physical hardware switches.
- CO4: Analyze data flow and structural topologies by mapping algorithmic design strategies (such as graphs or recursion) onto packet routing mechanisms and network layered frameworks.
- CO5: Construct conceptual models for distributed applications, evaluating how concurrency, message passing, and algorithmic constraints influence deadlocks and systemic performance.
- CO6: Formulate simulation workflows for complex, real-world processes while appraising their theoretical computational complexity (P vs. NP) and adaptability to non-classical hardware.

Syllabus

Introduction to computational thinking as a foundational problem-solving methodology. Decomposition: breaking complex tasks down into manageable, modular components. Pattern Recognition and Abstraction: identifying structural similarities and filtering out unnecessary details. Algorithmic Design: developing procedures using pseudocode and flowcharts, featuring basic strategies (searching, insertion sort, and recursion). Introduction to Linear Data Structures: conceptual overview of storing relationships using lists, matrices, and basic graphs.
The mathematical foundations of digital data storage and transmission. Number Systems: positional notation, binary, octal, decimal, and hexadecimal bases; standard conversion techniques. Binary Arithmetic: signed and unsigned integer representation, Two's complement systems, and arithmetic overflow. Data Encoding & Multimedia: text representation (ASCII, Unicode), and the digitization of continuous media (image pixels, color depth, audio sampling). Introduction to Compression & Reliability: conceptual difference between lossy and lossless compression; basic error detection (parity bits).
The basic structural hierarchy of physical computer hardware and system software. The Hardware Layer: transistors as binary switches, basic logic gates (AND, OR, NOT, XOR), and simple Boolean expressions. Digital Logic Circuits: introductory overview of combinational logic (adders) and sequential logic components (registers). Processor Architecture: the classic Von Neumann model framework (ALU, Control Unit, Memory) and a conceptual overview of the Fetch-Decode-Execute instruction cycle. Software Layer Abstraction: high-level languages vs. machine code, and the basic role of an Operating System.
Introduction to interconnected processing units and network communication. Networking Foundations: physical topologies (star, mesh), and layered abstraction frameworks (basic concepts of the OSI and TCP/IP models). The Internet Stack: packet routing, logical addressing (IP), and physical addressing (MAC). Distributed Systems & Concurrency: conceptual introduction to client-server and cloud computing models; high-level awareness of concurrency, message passing/Remote Procedure Calls (RPC), and deadlocks.
Computational approaches to mimicking reality and future paradigms. System Modeling & Simulation: introduction to using computers to simulate real-world processes (stochastic vs. deterministic context, queuing systems). Limits of Classical Computing: brief conceptual overview of what makes a problem computationally hard (P vs. NP mapping). Emerging Paradigms: high-level qualitative overview of Quantum Computing (qubits, superposition), neuromorphic architectures, and the Internet of Things (IoT).

Textbooks / References

1. Brookshear, J. G., & Brylow, D. Computer Science: An Overview. 13th Edition, Pearson, 2019.
2. Patterson, D. A., & Hennessy, J. L. Computer Organization and Design: The Hardware/Software Interface. 6th Edition, Morgan Kaufmann, 2020.
3. Kurose, J. F., & Ross, K. W. Computer Networking: A Top-Down Approach. 8th Edition, Pearson, 2020.
4. Patt, Yale N., and Sanjay J. Patel. Introduction to Computing Systems: From Bits & Gates to C/C++ & Beyond. 3rd ed., McGraw-Hill Education, 2020.
5. Tanenbaum, A. S., & Austin, T. Structured Computer Organization. 6th Edition, Prentice Hall, 2012.
6. George Coulouris, Jean Dollimore, and Tim Kindberg. (2011). Distributed Systems: Concepts and Design (5th ed.). Addison-Wesley.
7. Sipser, Michael. Introduction to the Theory of Computation. 3rd ed., Cengage Learning, 2012.
8. Zeigler, Bernard P., et al. Theory of Modeling and Simulation: Discrete Event & Iterative System Computational Foundations. 3rd ed., Academic Press, 2018.

CSL111 Computer Programming Lab [0-0-2-1]

Course Objectives

- To enable students to write, compile, and execute C programs systematically.
- To familiarize students with core C programming concepts including data types, operators, control flow constructs, and input/output mechanisms.
- To develop programming skills using arrays, character arrays, and user-defined functions with parameter passing mechanisms.

- To introduce recursive and pointer-based programming through hands-on implementation in C.
- To enable students to implement programs using derived data types such as typedef, enumerations, unions, and structures in C.

Course Outcomes

- CO1: Students learn to translate programming concepts into C programs using structured programming techniques.
- CO2: Students learn to develop recursive and pointer-based solutions for computational problems.
- CO3: Students learn to use C programs for solving real world problems.

Syllabus

Introduction to C Programming: Basic data types, operators, scope of variables, formatted input and output using scanf and printf. Writing and executing simple C programs.

Control Flow Constructs: Implementation of programs using conditional statements and loop constructs.

Arrays and Functions: Implementation of programs using arrays and character arrays. Programs involving user-defined functions with parameter passing by value.

Recursion and Pointers: Implementation of recursive programs. Implementation of programs using pointers and pass-by-reference mechanisms.

Derived Data Types: Implementation of programs using typedef, enumerations, unions, and structures in C. Defining and accessing union members and structure members.

Textbooks / References

1. Brian W. Kernighan and Dennis M. Ritchie, *The C Programming Language*, Second Edition, Prentice Hall, 1988.
2. E. Balagurusamy, *Programming in ANSI C*, Ninth Edition, McGraw Hill, 2024.
3. Paul Deitel and Harvey Deitel, *C: How to Program*, Ninth Edition, Pearson Education, 2022.
4. Ashok N. Kamthane, *Programming in C*, Third Edition, Pearson India, 2015.

ECL111 Electronic Devices and Digital Logic Design Lab [0-0-2-1]

Course Objectives

- Identify and safely use basic electronic laboratory equipment, instruments, and electronic components for circuit implementation and testing.
- To implement diode-based clipper and clamper circuits
- To implement combinational circuits using standard ICs
- Explain the behaviour of flip-flops, and analyse their characteristic equations and timing
- To design and implement counters and shift registers

Course Outcomes

- CO1: Design, implement, and evaluate diode-based clippers and clampers
- CO2: Identify and verify the operation of logic gates, and design combinational circuits
- CO3: Design and implement synchronous and asynchronous sequential circuits, counters, using state diagrams and excitation tables.
- CO4: Configure and shift registers and apply them in serial data transfer applications.

Experiments

- Familiarization of basic electronic lab equipment and components
- Diode Clippers
- Diode Clampers
- Design of Combinational Logic Circuits - Adders, Multiplexers, Demultiplexers
- Familiarization of Flip-Flops and Latches. SR, D and JK
- Design of asynchronous counters
- Design of synchronous counters
- Shift registers

MAC122 Modern Algebra [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of abstract algebra.
- To introduce the principles of propositional and predicate logic, rules of inference, and formal methods of proof as foundational tools for computer science.
- To build a strong mathematical foundation for understanding modern cryptographic techniques and protocols.
- To formally connect algebraic structures with notions of security, authentication, and privacy.
- To enable students to apply abstract algebraic principles in solving problems related to cryptography and information security.
- To provide a rigorous and integrated perspective of abstract algebra as a pillar of cryptography and an emerging key area of Computer Science and Engineering.

Course Outcomes

- CO1: Apply various methods of proof to validate properties of mathematical structures using propositions, predicates, and quantifiers, and verify validity using truth table, rules of inference.
- CO2: Analyze algebraic structures such as of groups, rings, fields and finite fields to establish connections with security primitives.
- CO3: Demonstrate proficiency in solving cryptographic problems using algebraic tools such as modular arithmetic, factorization, and polynomial rings.

- CO4: Critically evaluate the security of cryptographic protocols by applying algebraic principles to authentication, encryption, and digital signature schemes.
- CO5: Integrate abstract algebraic theory with modern cryptographic techniques to design and assess secure systems, bridging mathematical rigor with real-world applications in computer science and engineering.

Syllabus

Propositions, negation, disjunction and conjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof

Binary operation and its properties, Definition of a group. Examples, and basic properties. Subgroups, Coset of a subgroup, Lagrange's theorem.

Cyclic groups, Order of a group. Normal subgroups, Quotient group. Homomorphisms, Kernel Image of a homomorphism, Isomorphism theorems.

Permutation groups, Cayley's theorems.

Basic functions of cryptology, encryption, signature, and identification problems.

Direct product of groups. Structure of finite abelian groups. Applications, Private and public key cryptography, some nontrivial examples.

Rings: definition, examples, and basic properties. Zero divisors, Integral domains, Fields, Characteristic of a ring, Quotient field of an integral domain.

Subrings, Ideals, Quotient rings, Isomorphism theorems. Ring of polynomials.

Prime, Irreducible elements and their properties, UFD, PID and Euclidean domains. Prime ideal, maximum ideals.

Extension fields, Algebraic extensions, Finite fields.

Application to RSA encryption, elliptic curve cryptography, and advanced cryptographic standards.

Textbooks / References

1. J. A. Gallian, *Contemporary Abstract Algebra*, 7th Edition, Brooks/Cole, 2009.
2. J. B. Fraleigh, *A First Course in Abstract Algebra*, Narosa Publishing House, 2003.
3. O. Goldreich, *Foundations of Cryptography*, Volumes I & II, Cambridge University Press, 2001, 2004.
4. J. Katz and Y. Lindell, *Introduction to Modern Cryptography*, Chapman & Hall/CRC, USA, 2007.
5. W. B. Mao, *Modern Cryptography: Theory and Practice*, Prentice Hall, USA, 2003.

MAC123 Real and Complex Analysis [3-1-0-4]

Course Objectives

- To develop a rigorous foundation of the real number system and abstract analytical structures.
- To study convergence and completeness in metric and normed spaces.
- To build theoretical understanding of continuity, differentiability, and fixed point methods beyond computational calculus.
- To introduce analytic functions and the structural theory of complex analysis.
- To demonstrate the role of analysis in optimization, numerical methods, machine learning, and scientific computing.
- To cultivate proof-writing skills and mathematical rigor essential for higher mathematics and theoretical computing.

Course Outcomes

- CO1: Apply completeness, order properties, and metric space concepts to construct rigorous mathematical arguments and analyze convergence.
- CO2: Analyze topological, metric, and normed space structures that underpin optimization methods, numerical algorithms, and data analysis techniques.
- CO3: Apply fixed point theorems and fundamental results of real analysis to study convergence and stability of iterative computational methods.
- CO4: Understand complex differentiability, analyticity, and harmonic functions using rigorous mathematical reasoning.
- CO5: Apply core theorems of complex analysis, including Cauchy's Theorem, Cauchy's Integral Formula, and the Residue Theorem, to problems arising in engineering, signal processing, and scientific computing.
- CO6: Demonstrate mathematical maturity through proof construction, abstract reasoning, and the ability to connect analytical theory with applications in computing, optimization, and data science.

Syllabus

Real Analysis: Structure of the real number system: field and order axioms, completeness axiom, supremum and infimum, Archimedean property; countable and uncountable sets. Metric spaces, open and closed sets, limit points, compactness and connectedness in R; sequences in metric spaces, convergence, Cauchy sequences, completeness; Bolzano–Weierstrass theorem; series of real numbers and absolute convergence. Continuity as a topological property; uniform continuity; contraction mappings and Banach Fixed Point Theorem; applications of fixed point methods in iterative numerical algorithms. Introduction to normed spaces, vector norms and distance measures, convergence in normed spaces, error estimation and approximation.

Applications to Computing and Data Science: Distance metrics in clustering and classification, convergence analysis of optimization algorithms, fixed point methods in machine learning, nearest neighbor search, recommendation systems, and numerical computing.

Complex Analysis: Algebra and geometry of complex numbers; complex sequences and series; limits and continuity of complex functions; complex differentiability. Cauchy Riemann equations and their consequences; analytic and harmonic functions. Complex integration along paths; Cauchy's Integral Theorem and Cauchy's Integral Formula; Liouville's Theorem; Fundamental Theorem of Algebra. Isolated singularities; Laurent series; residues and the Residue Theorem.

Applications to Computing and Engineering: Fourier transforms and signal processing, stability analysis of dynamical systems, conformal mapping and image processing, applications in electro magnetics and fluid flow, contour integration methods in scientific computing and engineering mathematics.

Textbooks / References

1. R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, Wiley.
2. T. Tao, *Analysis I*, 4th Edition, Texts and Readings in Mathematics, Vol. 37, Hindustan Book Agency.
3. T. Tao, *Analysis II*, 4th Edition, Texts and Readings in Mathematics, Vol. 38, Hindustan Book Agency.
4. K. A. Ross, *Elementary Analysis: The Theory of Calculus*, Springer.
5. T. M. Apostol, *Mathematical Analysis*, Addison–Wesley.
6. W. Rudin, *Principles of Mathematical Analysis*, McGraw–Hill.
7. S. Kumaresan, *Topology of Metric Spaces*, Narosa.
8. Lars Ahlfors, *Complex Analysis*, 3rd Edition.
9. S. Ponnusamy, *Foundations of Complex Analysis*.
10. Erwin Kreyszig, *Introductory Functional Analysis with Applications*, Wiley.
11. Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press.
12. Lloyd N. Trefethen and David Bau III, *Numerical Linear Algebra*, SIAM.
13. Gilbert Strang, *Linear Algebra and Learning from Data*, Wellesley-Cambridge Press.
14. Trevor Hastie, Robert Tibshirani and Jerome Friedman, *The Elements of Statistical Learning*, Springer.
15. David S. Watkins, *Fundamentals of Matrix Computations*, Wiley.

MAC124 Vector Calculus and Applications [3-1-0-4]

Course Objectives

- To understand the basic concepts of vector calculus in space.
- To learn how to compute line, surface, and multiple integrals.
- To study gradient, divergence, curl, and important integral theorems.
- To understand the Jacobian, Hessian, Taylor series, and Laplacian for multivariable functions.
- To learn the basics of discrete vector calculus and finite difference approximations.

Course Outcomes

- CO1: Compute arc length, line integrals, surface integrals, and apply Green's, Stokes', and Gauss' theorems.
- CO2: Find gradient, directional derivatives, divergence, and curl of vector fields.
- CO3: Use polar, cylindrical, and spherical coordinates in solving integrals.
- CO4: Compute and interpret Jacobian matrices, Hessian matrices, multivariable Taylor expansions, and the Laplacian operator.
- CO5: Apply finite difference methods to approximate gradients and the discrete Laplacian.

Syllabus

Parameterised curves in space, Arc length, Tangent and normal vectors, Line integral, Gradient, Directional derivatives, Tangent plane and normal vector, Vector field, Divergence, Curl.

Parameterised surface, Double and triple integrals,Change of variables formula, Applications to area and volume, Surface integral, Integral theorems: Green's Theorem, Stokes' theorem, Gauss' divergence theorem. Coordinate system: Polar, spherical, cylindrical.

The Jacobian Matrix (and its geometric interpretation). The Hessian Matrix and Multivariable Taylor Series. The Laplacian Operator and its physical significance. Conservative Vector Fields, Path Independence.

Discrete Vector Calculus: Finite difference approximations for gradients and the discrete Laplacian.

Textbooks / References

1. M. R. Spiegel, *Vector Analysis*, Schaum's Outline Series, McGraw-Hill, 1959.
2. E. Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2011.
3. J. Stewart, *Calculus: Early Transcendentals*, 8th Edition, Cengage Learning, 2015.
4. G. Strang, *Computational Science and Engineering*, Wellesley-Cambridge Press, 2007.
5. H. Anton, I. C. Bivens, and S. Davis, *Calculus: Early Transcendentals*, 12th Edition, John Wiley & Sons, 2021.

CSC121 Data Structures [3-1-0-4]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs.
- Design and specify algorithms for searching and sorting, and those associated with the above data structures .
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences.
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms.

Course Outcomes

- CO1: Students learn to define operations of data structures like arrays, linked lists, trees and graphs.
- CO2: Students learn to design and specify algorithms involving above types of data structures.
- CO3: Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis.
- CO4: Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - List, Sorted Lists, Single and Double Linked Lists, Skip list. Dynamic Arrays, Circular Linked Lists, Applications of Lists in Browser History and Text Editors.

Stack & Queue with applications.

Binary Trees- Insertion and Deletion of nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues. Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting -Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching. Hashing- Application problems on hashing.

Graphs- Representations, types and properties of graphs, Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

HMC121 Personality Development and Soft Skills [3-0-0-3]

Course Objectives

- To understand the significance of soft skills, professional etiquette, self-awareness, and ethical behaviour in personal and professional development
- To help students understand and explain individual behaviour and personality in interpersonal and professional contexts.
- To apply communication, leadership, teamwork, presentation, and interpersonal skills in academic and workplace contexts
- To create awareness regarding common psychological concerns and the importance of psychological wellbeing in academic and personal life.
- To analyze workplace situations and develop problem-solving, emotional intelligence, adaptability, and stress management skills for professional
- To familiarize students with strategies for emotional regulation, coping, and healthy personality development.

Course Outcomes

- CO1: Apply self-assessment and goal-setting techniques-for personal and professional growth
- CO2: Understand one's own personality and that of others, appreciate individual differences, and adapt effectively to people and situations.
- CO3:Analyze workplace situations using communication, leadership, and emotional intelligence skills
- CO4: Develop the ability to cope with personal, academic, and professional challenges through an understanding of human behaviour.
- CO5: Demonstrate effective speaking, presentation, and interpersonal communication skills
- CO6: Evaluate professional challenges related to stress, ethics, teamwork, and work–life balance

Syllabus

Foundations of soft skills and personality development: Introduction to soft skills - Importance and aspects of soft skills - Self-esteem and self- awareness - Professional etiquette and social etiquette - Professional ethics and workplace behavior - Professional body language - SMART goals and personal goal setting - SWOT analysis for personality development.

From Campus to Corporate: Leadership skills and teamwork - Group discussion and meeting management - Adaptability and work ethics - Time management and stress management - Advanced speaking skills - Oral presentations, speeches, and debates - Combating nervousness and stage fear - Planning and preparation of presentations - Making effective presentations - Speeches for various occasions - Interviews and facing job interviews - Emotional intelligence and critical thinking -Work– life balance - Applied grammar for professional communication.

Understanding Personality and Behaviour: Personality: Meaning and Assessment. Psychoanalytic and Neo Psychoanalytic Approach; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality; Indian perspective on personality.

Mental Health, Institutions, and Social Contexts: Mental health in social, cultural, and institutional contexts; campus and workplace mental health. Institutional violence, bullying, harassment, and discrimination based on caste, religion, gender, sexuality, disability, body image, and other social locations. Structural inequalities, stigma, inclusion, and access to mental health support. Student mental health and suicide prevention, including institutional accountability and recommendations of the Supreme Court's National Task Force on Student Suicides. Critical perspectives on positive psychology and mainstream mental health interventions; indigenous and community-based approaches to well being. Help-seeking, collective care, peer support, and professional mental health services.

Textbooks / References

1. Carnegie, Dale, *How to Win Friends and Influence People*, John Wiley & Sons, 2026.
2. Covey, Stephen R., *The 7 Habits of Highly Effective People*, Personal Finance Magazine, 2024.
3. Ghai, A., *Rethinking Disability in India*, Routledge, 2015.
4. Guru, G., and Sarukkai, S., *The Cracked Mirror: An Indian Debate on Experience and Theory*, Oxford University Press, 2012.
5. Goleman, Daniel, *Emotional Intelligence*, Bloomsbury Publishing, 2020.
6. Kumar, N., *Mental Health and Well-Being: An Indian Psychology Perspective*, Routledge, 2023.
7. Medden, Stephanie, et al., "From Public Speaking to Multimodal Communication: A Reevaluation of the Basic Communication Course," *Basic Communication Course Annual*, 37(1), 2025.
8. Patel, V., *Where There Is No Psychiatrist: A Mental Health Care Manual*, 2nd Edition, Royal College of Psychiatrists, 2014.
9. Patel, V., Saxena, S., Lund, C., Thornicroft, G., Baingana, F., Bolton, P., et al., "The Lancet Commission on Global Mental Health and Sustainable Development," *The Lancet*, 392(10157), 2018, pp. 1553–1598.
10. Pease, Barbara, and Allan Pease, *The Definitive Book of Body Language: The Hidden Meaning Behind People's Gestures and Expressions*, Bantam, 2008.
11. Rao, Manchanahalli Satyaranayana, *Soft Skills–Enhancing Employability: Connecting Campus with Corporate*, IK International Pvt. Ltd., 2010.
12. Tracy, Brian, *Eat That Frog!: 21 Great Ways to Stop Procrastinating and Get More Done in Less Time*, Berrett-Koehler Publishers, 2025.
13. Nevid, J. S., Rathus, S. A., and Greene, B., *Abnormal Psychology in a Changing World*, 11th Edition, Pearson, 2022.
14. Snyder, C. R., and Lopez, S. J. (Eds.), *Positive Psychology: The Scientific and Practical Explorations of Human Strengths*, 5th Edition, 2021.
15. Carr, A., *Positive Psychology: The Science of Happiness and Human Strengths*, 4th Edition, 2022.
16. Kabat-Zinn, J., *Mindfulness for Beginners: Reclaiming the Present Moment—and Your Life*, 2018.
17. Ryckman, R. M., *Theories of Personality*, 10th Edition, 2017.
18. Schultz, D. P., and Schultz, S. E., *Theories of Personality*, 12th Edition, 2020.

CSC122 Computer Organization and Architecture [3-1-0-4]

Course Objectives

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.
- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Course Outcomes

- CO1: Understand the basics of computer hardware and how software interacts with computer hardware.
- CO2: Analyse and evaluate the performance of computers.
- CO3: Understand basics of Instruction Set Architectures and their impact on processor design.
- CO4: Understand how computers represent and manipulate data.
- CO5: Understand how computers perform arithmetic operations, how they are optimized and made to run faster.
- CO6: Design a pipeline for consistent execution of instructions with minimum hazards.
- CO7: Understand how the memory management takes place in a computer system.
- CO8: Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic Unit, control unit, and data, instruction and address flow.

Syllabus

Basic principles, hardware components Measuring performance: evaluating, comparing and summarizing performance.

Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multicycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies

Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors.

Textbooks / References

1. D. A. Pattersen and J. L. Hennesy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, ZvonkoVranesic and SafwatZaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

CSL121 Data Structures lab (C, C++) [0-0-2-1]

Course Objectives

- Develop proficiency in implementing fundamental data structures and algorithms.
- Analyze the time and space complexity of data structure operations.
- Apply appropriate data structures to solve computational problems efficiently.
- Gain practical experience with tree, graph, sorting, searching, and hashing techniques.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate linear and non-linear data structures.
- CO2: Analyze the performance of data structure operations and associated algorithms.
- CO3: Apply trees, graphs, hashing, sorting, and searching techniques to solve computational problems.
- CO4: Design efficient solutions using suitable data structures and justify their selection.

Syllabus

Linear Data Structures: Implementation and analysis of Lists, Sorted Lists, Singly Linked Lists, Doubly Linked Lists, Circular Linked Lists, Dynamic Arrays, and Skip Lists. Applications of lists in Browser History Management and Text Editors.

Stacks and Queues: Implementation using arrays and linked lists. Applications including Expression Evaluation, Celebrity Problem, Largest Rectangular Area in a Histogram, and Queue-based Scheduling Problems.

Trees: Binary Tree Operations, Insertion and Deletion of Nodes, Tree Traversals, Polish Notations, AVL Trees, Heaps, Priority Queues, Optimal Binary Search Trees, and Application Problems on Optimal Binary Search Trees.

Sorting and Searching: Implementation and performance analysis of Bubble Sort, Selection Sort, Insertion Sort, Merge Sort, Quick Sort, Radix Sort, Heap Sort, Linear Search, and Binary Search.

Hashing: Hash Functions, Collision Resolution Techniques, and Application Problems on Hashing.

Graphs: Graph Representations, Types and Properties of Graphs, Breadth First Search (BFS), Depth First Search (DFS), Shortest Path Algorithms, and Minimum Spanning Tree Algorithms.

Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of a data structure-based solution to a real-world problem.

Textbooks / References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005.

MAC211 Probability, Statistics and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.

- To develop the ability to perform statistical inference and data analysis using estimation, confidence intervals, and hypothesis testing techniques.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness-of-fit tests.
- To introduce the concept of random processes, Markov chains, and Brownian Motion.
- To introduce stochastic modeling techniques and their applications in data-driven systems.

Course Outcomes

- CO1: Define and apply the concepts of probability and conditional probability.
- CO2: Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions.
- CO3: Understand the concepts and significance of estimation theory and hypothesis testing.
- CO4: Apply estimation and hypothesis testing techniques to construct confidence intervals for means and variances and conduct goodness-of-fit tests for statistical inference and data-driven decision making.
- CO5: Analyze random processes, Markov chains, and Brownian motion and explain their applications in stochastic modelling and computing systems.

Syllabus

Basics of Probability: Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulas.

Random Variables and Distributions: Univariate, Bivariate and multivariate random variables, cumulative and marginal distribution functions, conditional and multivariate distributions, functions of random variables: sum, product, ratio, change of variables, mathematical expectations, moments, moment generating function, characteristic functions

Special Probability Distributions: discrete and continuous distributions, Limit theorems, Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central Limit Theorem, Tchebeychev's inequality, Law of Large Numbers.

Estimation Theory: Bias of estimates, confidence intervals, minimum variance unbiased estimation, Bayes' estimators, moment estimators, maximum likelihood estimators, Chi-square distribution, confidence intervals for parameters of normal distribution.

Hypothesis Testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Random Processes: Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Textbooks / References

1. S. Ross, *Introduction to Probability and Statistics for Engineers and Scientists*, 3rd ed., Elsevier, 2004.
2. P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Probability Theory*, Universal Book Stall, 2000.
3. S. M. Ross, *Introductory Statistics*, 2nd ed., Academic Press, 2009.
4. J. Medhi, *Stochastic Processes*, 3rd ed., New Age International, 2009.
5. V. K. Rohatgi and A. K. Saleh, *An Introduction to Probability and Statistics*, 3rd ed., Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, *Probability and Random Processes*, Oxford Univ. Press, 2001.
7. W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.
8. S. M. Ross, *Stochastic Processes*, 2nd ed., Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, *Introduction to Probability*, 2nd ed., Universities Press India, 2009.
10. S. Ross, *A First Course in Probability*, 10th ed., Pearson Education, Delhi, 2018.
11. K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.
12. G. Casella and R. L. Berger, *Statistical Inference*, Chapman and Hall/CRC, 2024.
13. S. Ross, *Introduction to Probability Models*, 13th ed., Elsevier, 2023.
14. R. V. Hogg, J. W. McKean, and A. T. Craig, *Introduction to Mathematical Statistics*, 8th ed., Pearson, 2019.

MAC212 Graphs and Networks in Computing Systems [3-1-0-4]

Course Objectives

- Formulate the fundamental concepts of graph theory and explore its wide range of computational applications.
- Develop the ability to model and analyse complex systems using graphs, evaluating their key structural and topological properties.
- Provide a deep understanding of algorithmic approaches to solve network problems.
- Introduce the foundational characteristics and structural dynamics of emerging complex networks.
- Equip students with the ability to apply graph-theoretic methods and data structures to practical engineering, science, and data analysis problems.

Course Outcomes

- CO1: Explain foundational graph theory terminologies, structural properties, and physical network topologies.
- CO2: Analyze the connectivity, independent sets, and structural bounds of graphs using mathematical reasoning and core theorems.
- CO3: Formulate and apply classic optimization strategies to solve routing, matching, and traveling/postman problems.
- CO4: Apply and implement fundamental traversal and routing algorithms to determine shortest paths and network connectivity.
- CO5: Evaluate large-scale real-world complex networks by calculating distance-based and degree-based centrality metrics to identify influential nodes.
- CO6: Model the structural dynamics of emerging complex networks and assess their fault tolerance and resilience against disruptions.

Syllabus

Introduction to Graphs and Applications, Paths, Cycles, and Trails, Connectivity and Bipartite Graphs, Eulerian Circuits, Vertex Degrees and Counting, Degree-Sum Formula, Matrix Representation, Chinese Postman Problem, Network Topologies: Mesh, Star, Ring, Bus, and Hybrid topologies

Trees, Properties of Trees, Spanning Trees, Cayley's Formula, Matchings and Covers, Hall's Condition, Independent Sets and Covers, Cuts and Connectivity, k-Connected Graphs, Menger's Theorem, Euler Graph, Hamiltonian graph, Shortest path, Graph coloring, Brooks' Theorem, BFS and DFS Search, Dijkstra's Algorithm.

Introduction to area of complex networks, Application examples such as social network, Inet, WWW, protein-protein, Description of basic notions and phenomena such as scale-free, small-world, community structure. Network centrality generalizing vertex degree, distinguishing local and global centrality, Centralities based on distances generalizing eccentricity such as closeness centrality, Betweenness centrality and its combinatorial properties, Fault tolerance and network resilience.

Textbooks / References

1. Chartrand, G., and Zhang, P., *A First Course in Graph Theory*, Dover Publications, 2012.
2. Cormen, T. H., Leiserson, C. E., Rivest, R. L., and Stein, C., *Introduction to Algorithms*, 3rd Edition, MIT Press, 2009.
3. Deo, N., *Graph Theory with Applications to Engineering and Computer Science*, Prentice-Hall, 1974.
4. Newman, M. E. J., *Networks: An Introduction*, Oxford University Press, 2010.
5. Tanenbaum, A. S., Feamster, N., and Wetherall, D. J., *Computer Networks*, 6th Edition, Pearson Education, 2021.
6. van Steen, M., *Graph Theory and Complex Networks: An Introduction*, Maarten van Steen, 2010.
7. van Steen, M., and Tanenbaum, A. S., *Distributed Systems*, 4th Edition, distributed-systems.net, 2023.
8. West, D. B., *Introduction to Graph Theory*, 2nd Edition, Prentice Hall, 2001.

HMC211 Engineering Management and Entrepreneurship [2-0-0-2]

Course Objectives

- To introduce the principles and practices of engineering management and enable students to understand managerial functions in technology-oriented organizations.
- To provide students with tools and techniques for planning, organizing, leading, and controlling engineering projects and resources.
- To develop an entrepreneurial mindset by exposing students to innovation, opportunity identification, startup ecosystems, and venture creation.
- To equip students with the knowledge required to evaluate business opportunities and prepare sustainable technology based business models and ventures.

Course Outcomes

- Understand the functions, responsibilities, and practices of managers in engineering and technology organizations.
- Apply basic management, project management, and decision-making tools for effective management of engineering projects and teams.
- Analyze entrepreneurial opportunities, innovation processes, and startup ecosystems for technology-driven ventures.
- Develop a preliminary business plan incorporating market, financial, legal, and sustainability considerations.

Syllabus

Engineering Management: Meaning, definition, scope and significance of management in engineering organizations; Evolution of management thought; Contributions of F.W. Taylor, Henry Fayol, Elton Mayo and Behavioral School; Functions of management; Levels of management; Managerial roles and skills; Strategic thinking; Management lessons from Indian ethos and philosophy; Teamwork, productivity, effectiveness and efficiency.

Planning and Organizing: Planning process, objectives, policies, strategies and decision making; Management by Objectives (MBO); Organizing principles; Organizational structures and design; Departmentation; Centralization and decentralization; Delegation of authority; Span of control; Staffing and human resource management; Team building; Communication and coordination in engineering organizations.

Leadership, Operations and Project Management: Leadership concepts, leadership styles and motivation theories; Directing and controlling functions; Operations management fundamentals; Productivity and quality management; Lean concepts and continuous improvement; Engineering project management; Project life cycle; Work Breakdown Structure (WBS); PERT and CPM techniques; Resource allocation; Cost estimation; Risk management; Project monitoring and control.

Entrepreneurship and Venture Creation: Entrepreneurship concepts and characteristics; Entrepreneurial competencies; Creativity, innovation and design thinking; Opportunity identification and feasibility analysis; Startup ecosystem in India; Technology entrepreneurship; Incubation and acceleration; Business Model Canvas; Customer discovery and value proposition; Marketing fundamentals for startups; Sources of finance including bootstrapping, angel investment, venture capital and crowdfunding; Intellectual Property Rights (IPR); Business planning; Sustainable and social entrepreneurship; Case studies of successful technology ventures.

Textbooks / References

1. P. K. Wright, *Engineering Management*, Addison-Wesley, 1998.
2. H. Koontz and H. Weihrich, *Essentials of Management*, Tata McGraw-Hill, 1998.
3. S. P. Robbins and M. A. Coulter, *Management*, 6th Edition, Pearson Education, 2004.
4. S. P. Robbins and M. A. Coulter, *Management*, 15th Edition, Pearson Education, 2022.
5. P. C. Tripathi and P. N. Reddy, *Principles of Management*, Tata McGraw-Hill, 2017.
6. Y. K. Bhushan, *Fundamentals of Business Organisation and Management*, Sultan Chand & Co., 2019.
7. R. D. Hisrich, M. P. Peters, and D. A. Shepherd, *Entrepreneurship*, McGraw-Hill Education, 2023.
8. C. B. Gupta and N. P. Srinivasan, *Entrepreneurship Development*, S. Chand Publishing, 2022.
9. A. Osterwalder and Y. Pigneur, *Business Model Generation*, Wiley, 2010.
10. T. H. Byers, R. C. Dorf, and A. J. Nelson, *Technology Ventures: From Idea to Enterprise*, McGraw-Hill Education, 2020.
11. P. F. Drucker, *Innovation and Entrepreneurship*, Harper Business, 2006.
12. E. Ries, *The Lean Startup*, Crown Business, 2011.

CSC211 Computer Networks [2-1-0-3]

Course Objectives

- To understand the principles, architecture, and operation of modern computer networks.
- To study networking protocols and services across different layers of the TCP/IP model.
- To understand routing, switching, and transport mechanisms in computer networks.
- To explore wireless networking technologies and emerging networking paradigms.
- To analyze network performance, reliability, and security mechanisms.
- To provide the foundation required for network design, administration, and advanced networking studies.

Course Outcomes

- CO1: Explain networking architectures, communication models, and protocol layering principles.
- CO2: Analyze data transmission, error control, flow control, and medium access mechanisms in communication networks.
- CO3: Apply Internet addressing, routing, and transport protocols for network communication and management.
- CO4: Explain the operation of common application-layer protocols and network services.
- CO5: Assess wireless networking technologies, network security threats, and emerging networking paradigms.

Syllabus

Introduction to Computer Networks: Evolution of computer networks, network components and network types, network topologies, layered network architecture, OSI reference model, TCP/IP protocol suite, Internet standards and RFCs.

Data Communication and Data Link Layer: Data transmission fundamentals, transmission media, encoding techniques, framing, error detection and correction, flow control, sliding window protocols, HDLC and PPP protocols, medium access control protocols, switching concepts.

Network Layer and Routing: Internet Protocol (IP), addressing and subnetting, ARP, ICMP, DHCP, CIDR and NAT, routing principles, distance vector and link state routing, RIP, OSPF and BGP, introduction to Software Defined Networking (SDN).

Transport and Application Layers: UDP and TCP, reliable data transfer, flow control, congestion control, Quality of Service (QoS), DNS, HTTP and HTTPS, electronic mail protocols, multimedia networking basics.

Wireless, Network Security and Emerging Networks: Wireless LAN and Wi-Fi standards, network security fundamentals, firewalls and intrusion detection concepts, Internet of Things (IoT) networking, edge computing, and future Internet architectures.

Textbooks / References

1. Kurose, J. F. and Ross, K. W., *Computer Networking: A Top-Down Approach*, 8th Edition, Pearson, 2021.
2. Tanenbaum, A. S. and Wetherall, D. J., *Computer Networks*, 6th Edition, Pearson, 2021.
3. Peterson, L. L. and Davie, B. S., *Computer Networks: A Systems Approach*, 6th Edition, Morgan Kaufmann, 2021.

4. Forouzan, B. A., *Data Communications and Networking*, 5th Edition, McGraw-Hill Education, 2013.
5. Comer, D. E., *Internetworking with TCP/IP: Principles, Protocols, and Architecture*, 6th Edition, Pearson, 2013.

CSC212 Design and Analysis of Algorithms [2-1-0-3]

Course Objectives

- Understand the principles of algorithm design and analysis.
- Analyze the correctness and efficiency of algorithms using appropriate mathematical techniques.
- Understand fundamental algorithmic paradigms such as divide-and-conquer, greedy methods, and dynamic programming to solve computational problems.
- Develop solutions using graph algorithms.
- Understand computational complexity, NP-completeness, and approaches for solving intractable problems.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Analyze the time and space complexity of algorithms using asymptotic and amortized analysis techniques.
- CO2: Design and implement algorithms using divide-and-conquer, greedy, and dynamic programming paradigms.
- CO3: Apply graph algorithms to solve real-world computational problems.
- CO4: Classify computational problems based on complexity classes and explain concepts of NP-completeness and approximation algorithms.

Syllabus

Introduction to algorithm design paradigms: Branch and Bound, Backtracking.
Divide and Conquer: Analysis by Recurrence Relations Methods for solving recurrence: Recursion Tree, Substitution Method, Master Theorem. Applications of the Master Theorem.
Greedy Algorithms: Optimization problems, optimal substructures, greedy choice, Applications of advanced data structures - Union-Find, Fibonacci heap.
Dynamic Programming: Reusing work across sub computations, optimal substructures, overlapping subproblems, ordering of subproblems and memoization. **Asymptotic Analysis of algorithms**: Asymptotic notations, Insertion Sort- Proof of Correctness and Complexity.
Amortized Analysis of algorithms: Aggregate Method, Accounting Method, Potential Method, Dynamic Tables–Balanced Trees.
Graph Algorithms: Topological Sorting, Strongly Connected Components, Minimum Spanning Tree, Shortest Path. **Intractable Problems**: Polynomial Time– class P, Polynomial Time Verifiable Algorithms– class NP, NP completeness and reducibility, NP Hard Problems, NP completeness proofs.
Introduction to Approximation Algorithms.

Textbooks / References

1. Introduction to Algorithms, T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, 3rd ed. Cambridge, MA: MIT Press, 2009.
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.
3. The Design and Analysis of Algorithms, Dexter C. Kozen, Monographs in Computer Science, Springer New York, NY, 2011.
4. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

CSC213 Theory of Computation [3-1-0-4]

Course Objectives

The primary objectives of this course are to enable students to:

- Understand the abstract mathematical models of computation and how they relate to modern computer hardware and software design (such as compilers).
- Analyze and Design various types of automata, including Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Pushdown Automata (PDA), and Turing Machines (TM) for specified languages.
- Master the relationship between formal languages, grammars, and abstract machines across the Chomsky hierarchy.
- Apply Mathematical Tools, such as the Pumping Lemma, to rigorously prove the limitations of specific computational models.
- Distinguish between decidable and undecidable problems, and gain a foundational understanding of computational complexity classes.

Course Outcomes

By the end of this course, learners will be able to:

- CO1: Design, minimize, and interconvert Finite State Machines, regular expressions, and regular grammars.
- CO2: Apply formal mathematical tools, such as the Pumping Lemma, to prove the limitations of regular and context-free languages.
- CO3: Construct, simplify, and transform Context-Free Grammars and design equivalent Pushdown Automata models.
- CO4: Synthesize standard and variant Turing Machines to compute functions and classify complex languages within the Chomsky Hierarchy.
- CO5: Evaluate the decidability of computational problems using reductions and analyze foundational principles of Complexity Theory.

Syllabus

Introduction: Notion of formal language- Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automaton model, acceptance of strings and languages.

Finite Automata – deterministic and nondeterministic, Regular operations, Regular Expression, equivalence between DFA, NFA and REs, Interconversions of DFA, NFA and REs, minimization of FSM, Non regular languages and pumping lemma, Moore and Mealy machines. Regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, inter conversion between RE and RG.

Context free grammars: Derivation, parse trees. Language generated by a CFG. Eliminating useless symbols, productions, and unit productions, Chomsky Normal Form, Non CFLs and pumping lemma for CFLs.

Pushdown automata: Definition, instantaneous description as a snapshot of PDA computation, notion of acceptance for PDAs, Equivalence of PDA and CFG, Properties of CFLs, DCFLs.

Turing machine: definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TM's, RAM machine, Configuration graph, closure properties of decidable languages, decidability properties of regular languages and CFLs.

Undecidability and classes of problems: Reductions, Chomsky hierarchy of languages, Rice's Theorem, Universal Turing Machine, un-decidability of post's correspondence problem, introduction to complexity theory.

Textbooks / References

1. M. Sipser, *Introduction to the Theory of Computation*, Second Edition, Thomson, 2005.
2. H. P. Lewis and C. H. Papadimitriou, *Elements of the Theory of Computation*, Fourth Edition, Prentice Hall of India, 2007.
3. S. Arora and B. Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press, 2009.
4. C. H. Papadimitriou, *Computational Complexity*, Addison-Wesley Publishing Company, 1994.
5. D. C. Kozen, *Theory of Computation*, Springer, 2006.
6. M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP-Completeness*, Freeman, New York, 1979.
7. J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, Third Edition, Pearson, 2008.

CSL211 Computer Networks Lab [0-0-2-1]

Course Objectives

- To provide practical exposure to networking protocols, services, and communication mechanisms.
- To develop skills in network programming, protocol analysis, and network simulation.
- To enable students to analyze, configure, and troubleshoot networked systems using modern networking tools.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Use networking tools to analyze network traffic and protocol behavior.
- CO2: Develop client-server applications using socket programming.
- CO3: Simulate and evaluate networking protocols using network simulators.
- CO4: Configure, troubleshoot, and evaluate network communication scenarios.

Syllabus

Network Programming

1. Introduction to Linux Networking Commands
2. Socket Programming – TCP Client/Server
3. Socket Programming – UDP Client/Server
4. Multi-client Communication Applications
5. Remote Procedure Call (RPC) Concepts
6. Implementation of Routing Algorithms

Network Analysis

7. Packet Capture and Analysis using Wireshark
8. Measurement of Network Performance Metrics

Network Simulation

10. Introduction to NS3 Simulator
11. Point-to-Point Network Simulation
12. Routing Protocol Simulation
13. Mini Project on Network Design and Performance Evaluation

Textbooks / References

1. Panwar, S., Mao, S., Ryoo, J.-D., and Li, Y., *Computer Networking Fundamentals*, Cambridge University Press, 2024.
2. Stevens, W. R., *UNIX Network Programming*.
3. Wireshark Documentation.
4. NS-3 Documentation.

CSL212 Algorithm Design Lab [0-0-2-1]

Course Objectives

- Analyze and compare the performance of algorithms through experimentation.
- Apply algorithm design paradigms to solve computational problems.

- Gain experience with graph algorithms.
- Strengthen problem-solving and programming skills through algorithmic projects and case studies.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Implement and evaluate the performance of fundamental algorithms and data structures.
- CO2: Apply divide-and-conquer, greedy, dynamic programming, and graph-based techniques to solve computational problems.
- CO3: Design efficient algorithmic solutions and analyze their correctness and complexity.

Syllabus

Implementation and Analysis of Basic Algorithms: Insertion Sort, Merge Sort, Binary Search,

Asymptotic and Recurrence Analysis: Experimental verification of asymptotic growth, Solving recurrence relations and complexity comparison

Graph Algorithms like Topological Sorting, Strongly Connected Components, Minimum Spanning Tree (Prim's and Kruskal's Algorithms), Shortest Path Algorithms

Greedy Algorithms like Interval Scheduling, Fractional Knapsack, Huffman Coding, Minimum Spanning Tree Applications

Dynamic Programming algorithms like Rod Cutting Problem, Matrix Chain Multiplication, Longest Common Subsequence, Machine Scheduling Problems, Bellman-Ford and Floyd-Warshall Algorithms

Complexity and Computational Problems: Problem classification (P, NP, NP-Complete, NP-Hard), Simple reductions and complexity analysis

Approximation: Implementation of basic approximation algorithms Mini Project / Case Study (optional): Design, implementation, analysis, and presentation of an algorithmic solution to a real-world problem.

Textbooks / References

1. The algorithm design manual, Skiena, S. S. (2nd ed.). Springer (2008).
2. Algorithms, 1st Edition, Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, 2024.
3. Algorithm Design, Kleinberg, J., & Tardos, E., Pearson/ Addison-Wesley, 2006.

MAC221 Differential Equations and Transforms [3-1-0-4]

Course Objectives

- Develop a comprehensive understanding of Fourier Series, Fourier Transform, and Z-Transform concepts with their fundamental properties and applications.
- Provide knowledge of analytical techniques for solving ordinary and partial differential equations arising in engineering and scientific applications.
- Enable students to apply transform methods for analyzing continuous and discrete systems and solving mathematical models.
- Introduce the concepts of convergence, harmonic analysis, and solution techniques for differential equations using appropriate mathematical approaches.
- Improve students' mathematical modelling and problem-solving capabilities by applying transform techniques and differential equation methods to real-world applications.

Course Outcomes

- CO1: Apply Fourier Series concepts, convergence criteria, and harmonic analysis techniques for representing periodic functions.
- CO2: Utilize Fourier Transform and Z-Transform techniques to analyze signals, systems, and solve engineering problems.
- CO3: Solve ordinary differential equations using appropriate analytical methods including variation of parameters and special forms of equations.
- CO4: Formulate and solve partial differential equations using standard methods and Fourier series-based solution techniques.
- CO5: Apply transformation methods and differential equation models to analyze and solve problems in science and engineering disciplines.

Syllabus

Fourier Series: Dirichlet's conditions - General Fourier series - Odd and even functions - Half-range sine and cosine series - Complex form of Fourier series - Parseval's identity - Harmonic analysis. Convergence of Fourier series, differentiation and integration of Fourier series.

Fourier Transform: Fourier Integral Theorem - Fourier transform pair - Sine and cosine transforms - Properties - Transform of elementary functions - Convolution theorem - Parseval's identity.

Z-transforms: Linearity Property, Damping rule, Shifting property, Multiplication by n, Initial and Final value theorem, Inverse Z- transforms, Convolution theorem, Evaluation of Inverse Z-transform, Application to difference equations.

Ordinary Differential Equations: Method of variation of parameters - Method of undetermined coefficients - Homogeneous equations of Euler's and Legendre's type - System of simultaneous linear differential equations with constant coefficients.

Partial Differential Equations: Formation - Solutions of first-order equations - Standard types and equations reducible to standard types - Singular solutions - Lagrange's linear equation - Integral surface passing through a given curve - Classification of partial differential equations - Solution of linear equations of higher order with constant coefficients - Linear non-homogeneous equations.

Fourier Series Solutions of Partial Differential Equations: Method of separation of variables - Solutions of one-dimensional wave equation and one-dimensional heat equation - Steady-state solution of two-dimensional heat equation - Fourier series solutions in Cartesian coordinates.

Textbooks / References

1. C. Edwards and D. Penney, *Elementary Differential Equations with Boundary Value Problems*, 6th Edition, Pearson, 2003.
2. W. E. Boyce and R. C. DiPrima, *Elementary Differential Equations*, 7th Edition, John Wiley & Sons, 2002.
3. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th Edition, Wiley, 2015.
4. Tyn Myint-U and L. Debnath, *Linear Partial Differential Equations for Scientists and Engineers*, 4th Edition, Birkhäuser, 2007.
5. B. S. Grewal, *Higher Engineering Mathematics*, 44th Edition, Khanna Publishers, 2021.
6. K. A. Stroud and Dexter J. Booth, *Engineering Mathematics*, 7th Edition, Palgrave Macmillan, 2013.

MAC222 Numerical Linear Algebra [3-1-0-4]

Course Objectives

- To introduce the theoretical foundations and computational aspects of numerical linear algebra.
- To analyze stability, conditioning, and floating-point effects in matrix computations.
- To study direct and iterative methods for solving large-scale linear systems.
- To develop understanding of eigenvalue problems and singular value decomposition.
- To connect numerical linear algebra with modern applications in data science and machine learning.

Course Outcomes

- CO1: Analyze computational complexity of matrix algorithms and assess numerical stability.
- CO2: Apply LU, Cholesky, and QR factorizations to solve linear systems and least-squares problems.
- CO3: Compute eigenvalues and singular values using numerical algorithms.
- CO4: Implement and analyze iterative methods such as Conjugate Gradient and GMRES.
- CO5: Apply matrix decomposition techniques to problems in data science, dimensionality reduction, and graph analytics.
- CO6: Evaluate conditioning and sensitivity of numerical problems arising in scientific computing.

Syllabus

Matrix–vector multiplication and computational cost; vector and matrix norms; orthogonality and projections; floating-point arithmetic, rounding errors, and machine precision; conditioning of linear systems and sensitivity analysis. Gaussian elimination and LU factorization; pivoting strategies and numerical stability; Cholesky factorization for symmetric positive definite matrices; applications to solving linear systems in data fitting and regression. Overdetermined systems and least-squares problems; orthogonal projections and normal equations; QR decomposition via Gram–Schmidt, Modified Gram–Schmidt, and Householder transformations; reduced least-squares problems and linear regression.

Eigenvalue Problems, Iterative Methods, and Data Applications: Eigenvalues and eigenvectors; power method and inverse iteration; Schur and Hessenberg reductions; tridiagonal and bidiagonal forms; QR algorithm with and without shifts; Singular Value Decomposition (SVD), low-rank approximation, and numerical computation. Krylov subspace methods including Arnoldi iteration; Conjugate Gradient method for symmetric systems; GMRES for non-symmetric systems; preconditioning techniques and convergence analysis. Applications in data science including PCA, Linear Discriminant Analysis (LDA), matrix completion, Non-negative Matrix Factorization (NMF), sparse matrices and sparse solutions, spectral graph methods and PageRank, and introductory tensor decompositions.

Textbooks / References

1. V. Sundarapandian, *Numerical Linear Algebra*, PHI, 2008.
2. L. N. Trefethen and David Bau III, *Numerical Linear Algebra*, SIAM, 1997.
3. D. S. Watkins, *Fundamentals of Matrix Computation*, 2nd Edition, Wiley, 2002.
4. Biswa Nath Dutta, *Numerical Linear Algebra and Applications*, SIAM, 2010.
5. Roger A. Horn and Charles R. Johnson, *Matrix Analysis*, Cambridge University Press, 1994.
6. William Ford, *Numerical Linear Algebra with Applications*, Academic Press, 2014.

HMC221 Engineering Economics, Finance and Business Analytics [2-0-0-2]

Course Objectives

- To introduce the fundamental concepts of engineering economics and their application in engineering decision-making.
- To provide students with knowledge of financial management principles, financial markets, and investment evaluation techniques.
- To familiarize students with business analytics concepts, data-driven decision making, and analytical tools used in modern organizations.
- To develop the ability to evaluate economic, financial, and business alternatives for effective managerial and entrepreneurial decision making.

Course Outcomes

- Understand the principles of engineering economics, cost analysis, and economic decision-making relevant to engineering and technology sectors.
- Apply financial concepts and techniques for budgeting, investment appraisal, and financial planning.
- Analyze business data using descriptive, predictive, and prescriptive analytics approaches to support managerial decisions.
- Evaluate business opportunities and organizational performance using economic, financial, and analytical tools.

Syllabus

Engineering Economics: Nature and scope of engineering economics; Economic decision making in engineering; Demand, supply and market equilibrium; Cost concepts and cost analysis; Fixed and variable costs; Break-even analysis; Time value of money; Simple and compound interest; Cash flow diagrams; Economic comparison of alternatives; Present worth, annual worth and future worth methods; Depreciation concepts and methods; Inflation and its impact on economic decisions; Replacement and maintenance analysis.

Financial Management and Financial Markets: Introduction to finance and financial management; Objectives and functions of financial management; Financial statements and financial analysis; Ratio analysis; Working capital management; Capital budgeting; Sources of finance; Cost of capital; Risk and return concepts; Financial planning and budgeting; Overview of financial markets and institutions; Equity, debt and mutual funds; Basics of investment management and portfolio concepts; Venture capital and startup financing.

Business Analytics Fundamentals: Introduction to business analytics; Role of analytics in business decision making; Types of analytics—descriptive, diagnostic, predictive and prescriptive analytics; Data collection and data visualization; Measures of central tendency and dispersion; Probability concepts; Hypothesis Testing, Correlation and regression analysis; Forecasting techniques; Data-driven decision making; Introduction to spreadsheet-based analytics and business intelligence tools.

Applications of Analytics in Business and Engineering: Analytics for operations, marketing, finance and human resources; Key Performance Indicators (KPIs); Dashboard reporting; Decision trees and risk analysis; Data ethics and governance; Artificial Intelligence and analytics in business; Case studies in engineering economics, financial decision making and business analytics; Applications in startups, technology ventures and digital enterprises.

Textbooks / References

1. W. G. Sullivan, E. M. Wicks, and C. P. Koelling, *Engineering Economy*, McGraw-Hill Education, 2018.
2. L. Blank and A. Tarquin, *Engineering Economy*, Pearson Education, 2020.
3. J. A. White, K. E. Case, and D. B. Pratt, *Principles of Engineering Economic Analysis*, Wiley, 2012.
4. E. F. Brigham and J. F. Houston, *Fundamentals of Financial Management*, Cengage Learning, 2021.
5. I. M. Pandey, *Financial Management*, Vikas Publishing House, 2021.
6. E. F. Brigham and M. C. Ehrhardt, *Financial Management: Theory and Practice*, McGraw-Hill Education, 2023.
7. J. Evans, *Business Analytics*, Pearson Education, 2020.
8. S. C. Albright and W. L. Winston, *Business Analytics: Data Analysis and Decision Making*, Cengage Learning, 2019.
9. F. Provost and T. Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
10. R. Sharda, D. Delen, and E. Turban, *Business Intelligence, Analytics, and Data Science*, Pearson Education, 2018.
11. U. D. Kumar and D. Kumar, *Business Analytics: The Science of Data-Driven Decision Making*, Wiley India, 2017.
12. C. R. Thomas and S. C. Maurice, *Managerial Economics*, Oxford University Press, 2020.
13. M. Hirschey, *Managerial Economics*, McGraw-Hill Education, 2022.
14. D. R. Anderson, D. J. Sweeney, T. A. Williams, J. D. Camm, and J. J. Cochran, *Statistics for Business and Economics*, Pearson Education, 2020.
15. T. H. Davenport and J. G. Harris, *Competing on Analytics*, Harvard Business Review Press, 2017.

CAC221 Machine Learning and Data Analysis [2-1-0-3]

Course Objectives

- To introduce the mathematical and statistical foundations required for machine learning and data analysis.
- To develop skills in exploratory data analysis, preprocessing, and visualization techniques.
- To provide knowledge of supervised, unsupervised, and reinforcement learning methods.
- To enable students to design, implement, and evaluate machine learning models.
- To develop analytical and problem-solving skills for intelligent data-driven applications.

Course Outcomes

- CO1: Understand the mathematical and statistical foundations of machine learning.
- CO2: Perform exploratory data analysis and preprocessing on datasets.
- CO3: Apply supervised and unsupervised learning algorithms for predictive analytics.

- CO4: Evaluate and optimize machine learning models using appropriate metrics.
- CO5:Develop intelligent data-driven solutions for real-world applications.

Syllabus

Mathematical Foundations and Exploratory Data Analysis Overview of Linear Algebra: Scalars, vectors, and matrices, Matrix operations, Systems of linear equations, Eigenvalues and eigenvectors, Matrix decomposition. Overview of Probability and Statistics: Random variables, Probability distributions, Descriptive statistics, Measures of central tendency, Measures of dispersion, Sampling techniques. Overview of Optimization Techniques: Optimization fundamentals, Objective functions and constraints, Gradient and Hessian matrix, Least squares optimization, Gradient descent methods. Exploratory Data Analysis (EDA): Data collection and preprocessing, Handling missing values, Data transformation, Outlier detection, Correlation analysis, Data visualization techniques.
Statistical Learning and Model Selection Introduction to Machine Learning: Definitions and goals of machine learning, History and applications of machine learning, Types of machine learning. Statistical Learning: Linear classification, Classification errors, Regression techniques, Predictive analytics concepts. Model Selection Techniques: Empirical Risk Minimization (ERM), Structural Risk Minimization (SRM), Cross-validation techniques, Regularization methods, Bias-variance tradeoff, Performance evaluation metrics.
Mining Frequent Patterns, Associations, and Correlations Association Rule Mining: Frequent itemset mining, Apriori algorithm, FP-growth concepts, Association rules, Support and confidence measures, Market basket analysis applications.
Supervised Learning Fundamentals: Supervised learning concepts, Generative and discriminative learning, Parametric and non-parametric learning. Regression Techniques: Linear regression, Model fitting and prediction. Classification Techniques: Decision trees, k-Nearest Neighbors (KNN), Naïve Bayes classifier with Laplace smoothing, Logistic regression with Maximum Likelihood Estimation Model Evaluation:Regression Metrics and Confusion matrix, Precision, Recall, F1-score, ROC curve.
Unsupervised Learning: Unsupervised learning concepts, Clustering techniques, K-Means clustering, Hierarchical clustering. Dimensionality Reduction: Principal Component Analysis (PCA), Feature selection and feature extraction techniques.

Textbooks / References

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer.
2. Tom M. Mitchell, Machine Learning, McGraw-Hill.
3. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow, O'Reilly Media.
4. Gareth James et al., An Introduction to Statistical Learning, Springer.
5. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements of Statistical Learning, Springer.
6. Gilbert Strang, Introduction to Linear Algebra, Wellesley-Cambridge Press.
7. 3. Stephen Boyd and Lieven Vandenberghe, Convex Optimization, Cambridge University Press.

CSC223 Database Management Systems [2-1-0-3]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced SQL features.
- To apply normalization techniques for efficient database design.
- To understand indexing, query processing, transaction management, and database security mechanisms.
- To introduce modern database technologies including NoSQL, cloud databases, and vector databases.

Course Outcomes

- CO1: Explain database concepts, architectures, and data modeling techniques.
- CO2: Design relational database schemas and write SQL queries for data management and retrieval.
- CO3: Apply normalization techniques to develop efficient database designs.
- CO4: Analyze indexing, query processing, transaction management, and security mechanisms in database systems.
- CO5: Identify and apply modern database technologies such as NoSQL, cloud databases, and vector databases for emerging applications.

Syllabus

Database Modelling: Database system concepts and architecture, Data models, Schemas and instances, Database system environment, Data modelling using Entity Relationship (ER) model, Enhanced ER model, Specialization, Generalization.

Relational Databases: Relational Model, Relational database design using ER to relational mapping, Constraints, Relational algebra, Relational calculus, SQL, Advanced SQL (Views, Stored Procedures, Triggers, CTEs, Window Functions), PL/SQL.

Database Design: Database design theory and methodology, Functional dependencies and normalization of relations, Normal Forms, Properties of relational decomposition, Algorithms for relational database schema design, Denormalization and scalable schema design.

Indexing and Query Processing in DBMS: Single level and multi-level indexing, Dynamic Multi-level Indexing using B Trees and B+ Trees, Hash-Based Indexing (Static and Dynamic Hashing), Query processing, Query evaluation plans, Query optimization.

Database Transactions and Security: Transaction processing concepts, ACID properties, Schedules and serializability, Concurrency control, Two Phase Locking Techniques, Deadlock, Database recovery concepts and techniques. Database Security: Introduction to database security, Authentication and authorization, Access control mechanisms, Privileges using GRANT and REVOKE.

Advanced Database Systems and Emerging Database Technologies: Big Data overview and NoSQL database concepts, NewSQL Databases, Cloud Databases, Vector Databases for AI Applications, Emerging trends in database technologies.

Textbooks / References

1. R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th Edition, Pearson, 2022.
2. R. Ramakrishnan and J. Gehrke, *Database Management Systems*, 3rd Edition, McGraw-Hill Higher Education, 2003.
3. M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st Edition, O'Reilly Media, 2017.

CSC224 Advanced Data Structures [2-1-0-3]

Course Objectives

The course aims to:

- Master Advanced Data Structures.
- Optimize Information Retrieval.
- Solve Network Connectivity.
- Design Probabilistic Solutions.

Course Outcomes

After successful completion of the course, students will be able to:

- CO1: Analyze and Implement Self-Balancing Search Structures: Demonstrate the ability to implement and evaluate balanced search trees to optimize disk access and dynamic data retrieval.
- CO2: Evaluate Advanced Heap Architectures: Compare the structural and amortized performance of Binomial and Fibonacci Heaps to optimize priority queue operations in network and graph algorithms.
- CO3: Design Efficient Pattern Matching and String Processing Systems: Apply advanced string algorithms (KMP, Rabin-Karp) and prefix/suffix structures (Tries, Suffix Trees/Arrays) to solve complex text-processing and biological data problems.
- CO4: Formulate Graph Connectivity Solutions: Deploy advanced graph algorithms to identify strongly connected components, detect cycles, and map topology in complex networks.
- CO5: Synthesize and Apply Probabilistic Algorithmic Strategies.

Syllabus

Self-Balancing Trees & Extensions but not limited to: Red-Black Trees, B-Trees & B+ Trees.

Advanced Heaps & Priority Queues but not limited to: Fibonacci Heaps , Binomial Heaps.

String and Prefix Processing but not limited to:Tries (Prefix Trees), Suffix Trees & Suffix Arrays, KMP Algorithm, Rabin Karp algorithm.

Algorithms on Graph connectivity but not limited to: Tarjan's algorithm and Kosaraju's algorithm for detection of strongly connected components, Detect cycles in an undirected graph.

Network Flow: Max Flow min cut theorem, Ford Fulkerson algorithm, Bipartite matching using flows.

Introduction to Randomized Algorithms but not limited to: Las Vegas Algorithms -Quick Sort, Monte Carlo Algorithms- Miller-Rabin primality test.

Textbooks / References

1. Introduction to Algorithms (4th Edition) by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein.
2. Algorithms (4th Edition) by Robert Sedgewick and Kevin Wayne.
3. Advanced Data Structures by Peter Brass.
4. Randomized Algorithms by Rajeev Motwani and Prabhakar Raghavan.

CAL221 Machine Learning and Data Analysis Lab [0-0-2-1]

Course Objectives

- Recall the core terms, concepts, and mathematical foundations used in data analysis and machine learning.
- Interpret and understand statistical methods, learning paradigms, and visualization principles.
- Apply and analyse data preprocessing, statistical, mining, and machine learning techniques on datasets.
- Synthesize and create a complete data analysis and machine learning workflow for a real problem.

Course Outcomes

- CO1: Identify the essential concepts, tools, and mathematical ideas used in data analysis and machine learning.
- CO2: Explain statistical procedures, learning methods, and visualization principles in practical contexts.
- CO3: Implement and evaluate preprocessing, prediction, clustering, mining, and visualization techniques on datasets.
- CO4: Design and develop an end-to-end analytical solution using machine learning and visualization tools.

Syllabus

1. Exploring matrix operations, eigenvalues and eigenvectors, gradients and probability distributions.

2. Understanding data cleaning, transformation, normalization, feature selection, and dimensionality reduction techniques.
3. Applying the measures of central tendency and dispersion, along with exploratory analysis using python.
4. Practicing simple and multiple regression for prediction, with basic error analysis and evaluation.
5. Understanding model Selection and Generalization by learning cross-validation, regularization, and bias–variance trade-off.
6. Creating supervised Learning models with parametric/non-parametric methods such as decision trees, Naïve Bayes, and k-NN.
7. Synthesising Unsupervised Learning models using Clustering, dimensionality reduction and outlier analysis techniques.
8. Mining association Rules using techniques such as Apriori, FP-growth etc.
9. Generating meaningful plots such as line, bar, scatter, histogram, box, heat map, and treemap using Matplotlib, Seaborn, and/or Plotly.

Textbooks / References

1. Hastie, T., Tibshirani, R., and Friedman, J. H. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, 2nd Edition, Springer Series in Statistics, Springer, New York, USA, 2009. ISBN: 978-0-387-84857-0 (Hardcover), ISBN: 978-0-387-84858-7 (eBook), 745 pages.
2. Bishop, C. M. Pattern Recognition and Machine Learning, Information Science and Statistics Series, Springer, New York, USA, 2006. ISBN: 978-0-387-31073-2 (Hardcover), ISBN: 978-0-387-45528-0 (eBook), 738 pages.
3. Han, J., Kamber, M., and Pei, J. Data Mining: Concepts and Techniques, 3rd Edition, Morgan Kaufmann (Elsevier), Waltham, MA, USA, 2011. ISBN: 978-0-12-381479-1, 744 pages.
4. Murphy, K. P. Machine Learning: A Probabilistic Perspective, Adaptive Computation and Machine Learning Series, The MIT Press, Cambridge, MA, USA, 2012. ISBN: 978-0-262-01802-9 (Hardcover), ISBN: 978-0-262-30432-0 (eBook), 1104 pages.
5. Géron, A. Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems, 1st Edition, O'Reilly Media, Sebastopol, CA, USA, 2025. ISBN: 979-8341607989.

CSL223 Database Management Systems Lab [0-0-2-1]

Course Objectives

- To understand the fundamentals of database systems and data modeling techniques.
- To design and implement relational databases using SQL and advanced database programming constructs.
- To apply normalization and schema design principles for efficient database development.
- To understand query processing, indexing, transaction management, recovery, and database security.
- To explore modern database technologies such as NoSQL, cloud databases, and vector databases for contemporary applications.

Course Outcomes

- CO1: Design conceptual and logical database schemas using ER and Enhanced ER models.
- CO2: Implement relational databases and develop SQL/PL-SQL programs for data management.
- CO3: Apply normalization techniques and evaluate database schema quality.
- CO4: Analyze indexing methods, query execution plans, and optimization strategies.
- CO5: Implement transaction management, concurrency control, security mechanisms, and modern database applications using NoSQL and emerging database technologies.

Syllabus

1. Database Design using ER/EER Models: Design ER and Enhanced ER diagrams for real-world applications.
2. Relational Database Design: Convert ER models into relational schemas and implement constraints.
3. SQL Fundamentals: Practice DDL, DML, DCL, and TCL commands.
4. SQL Queries and Joins: Write simple, nested, and join queries for data retrieval.
5. Advanced SQL: Implement views, set operators, CTEs, and window functions.
6. PL/SQL Programming: Develop programs using control structures, cursors, and exception handling.
7. Stored Procedures and Functions: Create and execute procedures and user-defined functions.
8. Database Triggers: Implement triggers for data validation and auditing.
9. Normalization and Schema Design: Apply functional dependencies and normalization techniques.
10. Indexing and Query Optimization: Create indexes and analyze query execution plans.
11. Transaction Management and Security: Perform transaction control, concurrency control, and privilege management.
12. NoSQL Database Operations: Implement CRUD operations and aggregation using MongoDB.
13. Modern Database Technologies: Explore Cloud Databases, NewSQL Databases, and Vector Databases.
14. Mini Project: Design and implement a complete database application for a real-world problem.

Textbooks / References

1. Silberschatz, A., Korth, H. F., & Sudarshan, S., *Database System Concepts*, 7th ed., McGraw-Hill Education, 2019.
2. Elmasri, R., & Navathe, S. B., *Fundamentals of Database Systems*, 7th ed., Pearson Education, 2016.
3. Viescas, J. L., & Hernandez, M. J., *SQL Queries for Mere Mortals: A Hands-On Guide to Data Manipu-*

lation in SQL, 5th ed., Addison-Wesley Professional, 2023.
4. Beaulieu, A., *Learning SQL: Generate, Manipulate, and Retrieve Data*, 3rd ed., O'Reilly Media, 2020.
5. Kleppmann, M., *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, 1st ed., O'Reilly Media, 2017.
6. Bradshaw, S., Brazil, E., & Chodorow, K., *MongoDB: The Definitive Guide: Powerful and Scalable Data Storage*, 3rd ed., O'Reilly Media, 2019.
7. Thusoo, A., & Sen Sarma, J., *Vector Search: Building Intelligent Search Systems with Embeddings*, 1st ed., O'Reilly Media, 2025.

MAC311 Soft Computing [3-0-0-3]

Course Objectives

- To summarize basic learning laws and architectures of neural networks.
- To describe supervised and unsupervised learning laws of neural networks.
- To introduce fuzzy logic, fuzzy relations, and fuzzy mathematics for designing a fuzzy logic controller.
- To discuss neuro-fuzzy approaches like ANFIS and CANFIS.

Course Outcomes

- CO1: Translate biological motivations into various characteristics of neural networks.
- CO2: Comprehend and analyse basic learning laws of neural networks and activation functions used.
- CO3: Interpret associative memories for storing and recalling input patterns.
- CO4: Learn and implement supervised and unsupervised learning laws for various applications.
- CO5: Decide on fuzzification and defuzzification methods for fuzzy inference systems.
- CO6: Apply and integrate various neuro-fuzzy techniques for designing intelligent systems using ANFIS and CANFIS.
- CO7: Design models using neural networks and fuzzy logic for various applications.

Syllabus

Artificial Neural Network: Introduction, characteristics, learning methods, taxonomy, evolution of neural networks, basic models, important technologies, applications. Fuzzy Logic: Introduction, crisp sets, fuzzy sets, crisp relations and fuzzy relations: Cartesian product of relation, classical relation, fuzzy relations, tolerance and equivalence relations, non-iterative fuzzy sets. Genetic Algorithm: Introduction, biological background, traditional optimization and search techniques, genetic basic concepts.

Neural Networks: McCulloch-Pitts neuron, linear separability, Hebb network, supervised learning network: perceptron networks, adaptive linear neuron, multiple adaptive linear neuron, BPN, RBF, TDNN; associative memory network: auto-associative memory network, hetero-associative memory network, BAM, Hopfield networks, iterative auto-associative memory network and iterative associative memory network; unsupervised learning networks: Kohonen Self-Organizing Feature Maps, LVQ, CP networks, ART network.

Fuzzy Logic: Membership functions: features, fuzzification, methods of membership value assignments; Defuzzification: lambda cuts, methods; fuzzy arithmetic and fuzzy measures: fuzzy arithmetic, extension principle, fuzzy measures, measures of fuzziness, fuzzy integrals; fuzzy rule base and approximate reasoning: truth values and tables, fuzzy propositions, formation of rules, decomposition of rules, aggregation of fuzzy rules, fuzzy reasoning, fuzzy inference systems, overview of fuzzy expert systems, fuzzy decision making.

Neuro-Fuzzy Modelling: Adaptive Neuro-Fuzzy Inference Systems (ANFIS) - Introduction, Architecture, Hybrid Learning Algorithm, ANFIS as a Universal Approximator. Applications – Printed Character Recognition, Inverse Kinematics Problem, Automobile LPG Prediction.

Textbooks / References

1. S. R. Jang, C. T. Sun and E. Mizutani, *Neuro-Fuzzy and Soft Computing*, PHI/Pearson Education, 2004.
2. S. N. Sivanandam and S. N. Deepa, *Principles of Soft Computing*, Wiley India Pvt. Ltd., 2018.
3. S. Rajasekaran and G. A. Vijayalakshmi Pai, *Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis and Applications*, PHI Learning Pvt. Ltd., 2017.
4. Kwang H. Lee, *First Course on Fuzzy Theory and Applications*, Springer, 2005.
5. George J. Klir and Bo Yuan, *Fuzzy Sets and Fuzzy Logic: Theory and Applications*, Prentice Hall, 2015.
6. James A. Freeman and David M. Skapura, *Neural Networks: Algorithms, Applications, and Programming Techniques*, Addison-Wesley, 2003.

MAC312 Optimization Techniques for Data Science [3-0-0-3]

Course Prerequisite

- Preliminary knowledge of linear algebra is required for the formulation of the concept.
- Knowledge of multivariate calculus is needed.
- Knowledge of the basic concept of probability is required.
- Knowledge of the programming language is essential for implementation.

Course Objectives

- To introduce the mathematical foundations of optimization and their role in data science and machine learning.

- To develop the ability to formulate data science and machine learning problems as optimization problems.
- To provide a comprehensive understanding of convex optimization, first-order methods, second-order methods, and constrained optimization techniques.
- To enable the selection and application of appropriate optimization algorithms for data-driven optimization problems.
- To introduce evolutionary optimization techniques for solving combinatorial optimization problems.

Course Outcomes

- CO1: Apply the mathematical foundations of optimization to analyze and solve optimization problems arising in data science.
- CO2: Formulate machine learning and data science problems as continuous or discrete optimization problems, and as unconstrained or constrained optimization problems.
- CO3: Analyze the properties of convex and non-convex optimization problems, including the role of regularization and sparsity.
- CO4: Apply first-order and second-order optimization algorithms to machine learning models such as linear regression, logistic regression, neural networks, and support vector machines.
- CO5: Utilize constrained optimization, duality principles, and evolutionary optimization techniques to solve real-world data science and combinatorial optimization problems.

Syllabus

Module 1: Mathematical Foundations for Optimization:
Vectors and matrices, vector and matrix norms, eigenvalues and eigenvectors, positive definite matrices, multivariate calculus, partial and directional derivatives, gradient, Jacobian and Hessian matrices, Taylor series expansion, local and global minima and maxima, first-order and second-order necessary and sufficient optimality conditions. Introduction to optimization problems and objective functions.

Module 2: Convexity, Regularization, and Machine Learning Optimization Landscapes:
Convex sets and convex functions, first-order and second-order characterizations of convexity, strong convexity and smoothness, convex and non-convex optimization problems, subgradients and non-smooth optimization. Machine learning models including linear regression, logistic regression, and neural networks; loss functions including mean squared error, cross-entropy, and hinge loss; convex and non-convex loss landscapes. L1 and L2 regularization, Ridge regression, LASSO regression, sparsity, and regularized learning models.

Module 3: First-Order Optimization Methods:
Gradient descent, batch gradient descent, stochastic gradient descent, mini-batch gradient descent, subgradient descent, convergence analysis of first-order methods, momentum method, Nesterov accelerated gradient, adaptive optimization methods including AdaGrad, RMSProp, and Adam, line search methods, backtracking line search, Armijo sufficient decrease condition, Wolfe conditions, and Goldstein conditions. Applications to optimization of linear regression, logistic regression, LASSO regression, and neural networks through backpropagation.

Module 4: Second-Order Optimization Methods:
Newton's method, convergence properties of Newton's method, quasi-Newton methods, BFGS method, convergence properties of BFGS, limited-memory BFGS (L-BFGS), and large-scale optimization.

Module 5: Constrained Optimization and Duality:
Equality and inequality constrained optimization, Lagrange multipliers, Karush-Kuhn-Tucker (KKT) conditions, weak and strong duality, primal and dual optimization problems, and quadratic optimization problems. Applications to support vector machines and their optimization formulations.

Module 6: Evolutionary Optimization:
Introduction to genetic algorithm and its applications to combinatorial optimization problems (such as feature selection, TSP)

Textbooks / References

1. Bertsekas, D. P., *Nonlinear Programming*, 2nd Edition, Athena Scientific, Belmont, MA, 1999.
2. Bertsekas, D., *Convex Optimization Algorithms*, Athena Scientific, 2015.
3. Bertsekas, D. P., *Constrained Optimization and Lagrange Multiplier Methods*, Academic Press, 2014.
4. Boyd, S. P., and Vandenberghe, L., *Convex Optimization*, Cambridge University Press, 2004.
5. Nocedal, J., and Wright, S. J., *Numerical Optimization*, Springer, New York, 1999.
6. Kochenderfer, M. J., and Wheeler, T. A., *Algorithms for Optimization*, MIT Press, 2019.

CAC312 Artificial Intelligence [3-0-0-3]

Course Objectives

- To understand the foundations of Artificial Intelligence and intelligent agents, environments, and rational decision-making processes in AI systems.
- To apply problem-solving techniques in uninformed, informed, adversarial, and constraint search space.
- To understand knowledge representation, reasoning and planning in applications.
- To explore advanced AI concepts including natural language processing, expert systems, fuzzy logic, and hybrid intelligent systems.

Course Outcomes

- CO1: Explain the core principles and architectures of Artificial Intelligence systems.
- CO2: Implement search algorithms and reasoning techniques to solve AI problems.
- CO3: Apply knowledge representation and probabilistic reasoning methods in intelligent applications.
- CO4: Analyze the applicability of advanced AI techniques such as NLP, fuzzy systems, expert systems in real-world applications.

Syllabus

Module 1: Artificial Intelligence & Problem Solving

Introduction to AI: What is AI? Foundation, History, State of the art, Risks and Benefits.

Intelligent agent: Agents and environments, concept of rationality, nature of environment, structure of agents.

Problem solving: Search space problem, uninformed search strategies, informed search strategies, local search algorithms, constraint satisfaction problems, adversarial search and games.

Module 2: Knowledge, Reasoning and Planning

Logical agents: Knowledge-Based Agents, Propositional logic, propositional theorem proving.

First order logic: Syntax and semantics, knowledge engineering, inference: unification, forward and backward chaining, resolution.

Knowledge representation: Ontology, categories, objects, events.

Planning: Classical planning, heuristics of planning, hierarchical planning.

Module 3: Uncertain Knowledge and Reasoning

Quantifying uncertainty: Acting under uncertainty, Bayes' rule, Naïve Bayes model, semantics of Bayes networks, inference in temporal models, combining beliefs and desires under uncertainty, decision networks, sequential decision problems, relational probability models.

Fuzzy logic: Crisp logic, Fuzzy logic, Membership function, Membership function, Fuzzy logic Applications.

Module 4: Expert Systems and Advanced AI

Expert Systems: What is Expert system, Conventional systems vs. Expert systems, Basic Concepts, Human Expert Behaviors, Knowledge Types, Inferencing, Rules, Structure of Expert Systems, ES Components, Knowledge Engineer, Expert Systems Working, Problem Areas Addressed by Expert Systems, benefits limitations, Applications of expert systems.

Advanced AI: Natural language processing, Hybrid intelligent systems, Introduction to: Ethics of AI, Responsible AI, Generative AI, Explainable AI, Agentic AI.

Textbooks / References

1. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, Fourth Edition, Pearson Education, 2021.
2. Elaine Rich and Kevin Knight, *Introduction to Artificial Intelligence*, McGraw Hill, Third Edition, 2017.
3. Michael Negnevitsky, *Artificial Intelligence: A Guide to Intelligent Systems*, Addison Wesley, Third Edition, 2017.
4. David L. Poole and Alan K. Mackworth, *Artificial Intelligence: Foundations of Computational Agents*, Third Edition, 2023.
5. Deepak Khemani, *A First Course in Artificial Intelligence*, McGraw Hill Education (India), 2013.
6. Denis Rothman, *Artificial Intelligence by Example*, Packt, Second Edition, 2020.
7. Michael Munn and David Pitman, *Explainable AI for Practitioners*, O'Reilly Media, 2022. ISBN: 9781098119 133.

Web Resources

1. Stanford Encyclopedia of Artificial Intelligence.
2. IBM AI Learning Hub.
3. NPTEL Artificial Intelligence Course.

CAC313 Advanced Machine Learning [3-0-0-3]

Course Objectives

- To provide a strong foundation in advanced supervised learning, ensemble learning, and probabilistic modeling techniques.
- To develop an understanding of deep neural networks, optimization algorithms, and regularization methods used in modern deep learning systems.
- To introduce advanced unsupervised and representation learning techniques for clustering, dimensionality reduction, anomaly detection, and feature learning.
- To familiarize students with emerging machine learning paradigms including reinforcement learning, semi-supervised learning, self-supervised learning, and federated learning.
- To enable learners to analyze, design, and implement machine learning models for real-world intelligent applications.

Course Outcomes

- CO1: Analyze and apply advanced supervised learning techniques including SVMs, ensemble learning methods, and probabilistic models for classification and prediction tasks.
- CO2: Design and optimize deep neural network models using appropriate activation functions, optimization algorithms, and regularization techniques.
- CO3: Implement advanced unsupervised and representation learning methods for clustering, dimensionality reduction, anomaly detection, and high-dimensional data analysis.
- CO4: Apply reinforcement learning, semi-supervised learning, self-supervised learning, and federated learning concepts to solve modern intelligent system problems.

- CO5: Evaluate machine learning models using suitable performance metrics and select appropriate algorithms for real-world applications.

Syllabus

Advanced Supervised and Ensemble Learning: Overview of Machine Learning and Learning Theory , Basic concept, Probably Approximately Correct (PAC) Learning, Version Space, Computational Complexity of Learning, Introduction to VC Dimension. Margin-Based Models: Support Vector Machines (SVM), Margin maximization, Soft-margin classifiers, Kernel methods, Kernel trick, Linear and non-linear classification techniques. Ensemble Learning: Ensemble learning fundamentals, Bagging techniques, Random Forests, Boosting methods, AdaBoost, Gradient Boosting, XGBoost concepts, Ensemble model evaluation and applications. Probabilistic Models: Fundamentals of probabilistic learning, Bayesian learning and inference, Bayesian Belief Networks, Hidden Markov Models (HMM), Monte Carlo Methods, Probabilistic graphical models and applications.

Deep Neural Networks and Optimization Techniques: Deep Neural Networks: Fundamentals of deep learning, Artificial Neural Networks (ANN), Perceptrons, Feedforward and multilayer neural networks, Activation functions, Loss functions, Backpropagation algorithm, Weight initialization techniques. Optimization Methods: Gradient Descent and its variants, Stochastic Gradient Descent (SGD), Momentum optimization, RMSProp, Adam optimizer, Learning rate scheduling techniques. Regularization Techniques: Overfitting and underfitting problems, L1 and L2 regularization, Dropout techniques, Early stopping, Batch normalization.

Advanced Unsupervised and Representation Learning: Clustering and Dimensionality Reduction: Density-based clustering (DBSCAN), Linear Discriminant Analysis (LDA), t-SNE and manifold learning concepts. Representation Learning: Feature extraction and feature selection, Latent feature representation, Autoencoders, Sparse autoencoders, Variational Autoencoders (VAE) concepts, Representation learning for high-dimensional data. Anomaly and Pattern Detection: Outlier detection techniques, Anomaly detection models, Pattern mining concepts, Applications in fraud detection and healthcare analytics.

Advanced Machine Learning Applications and Emerging Trends: Advanced Learning Paradigms: Reinforcement Learning fundamentals, Markov Decision Process (MDP), Q-Learning concepts, Policy-based learning methods, Exploration and exploitation strategies, Semi-supervised learning techniques, Self-supervised learning concepts, Federated learning architecture and applications.

Textbooks / References

1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
2. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.
3. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, The Elements of Statistical Learning, Springer, 2nd Edition.
4. Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT Press, 2012.
5. Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction, MIT Press, 2nd Edition.
6. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media.
7. François Chollet, Deep Learning with Python, Manning Publications.
8. Sebastian Raschka and Vahid Mirjalili, Python Machine Learning, Packt Publishing.
9. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer.
10. Ethem Alpaydin, Introduction to Machine Learning, MIT Press.

CSC313 Cryptography and Network Security [3-1-0-4]

Course Objectives

- To understand the mathematical foundations and security principles underlying modern cryptographic systems.
- To study symmetric and public-key cryptographic techniques, authentication mechanisms, digital signatures, and key management schemes.
- To analyse secure communication protocols, network security architectures, and defence mechanisms against cyber attacks.
- To examine vulnerabilities in computer networks, web applications, and wireless systems, and evaluate appropriate security countermeasures.
- To develop the ability to analyse, design, and evaluate secure computing and communication systems using contemporary cryptographic and network security techniques.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain the principles of modern cryptography, number-theoretic foundations, and security services used in secure communication systems.
- CO2: Apply symmetric and public-key cryptographic techniques, authentication mechanisms, and digital signatures to secure information systems.
- CO3: Analyse network security protocols, secure communication mechanisms, and defence strategies for protecting computer networks.
- CO4: Evaluate vulnerabilities, attacks, and security risks in networked systems and recommend appropriate protection mechanisms.
- CO5: Design secure computing and communication solutions by integrating cryptographic techniques, net-

work security protocols, and contemporary cybersecurity practices.

Syllabus

Introduction to information security, Security goals and services, CIA triad, Threats, vulnerabilities, attacks, and risk assessment, Security models and design principles, Malware: viruses, worms, trojan horses, ransomware, botnets, firewalls - Cryptographic fundamentals, Mathematical foundations of cryptography: divisibility, prime numbers, greatest common divisor, Euclidean algorithm, modular arithmetic, congruence relations, Euler's theorem, Fermat's Little Theorem, Classical cryptographic techniques: Caesar cipher, Playfair cipher, Hill cipher, Vigenère cipher, Cryptanalysis and Kerckhoffs' principle.

Symmetric-key cryptography, Block and stream ciphers, DES, Triple DES, AES, Modes of operation, Public-key cryptography, RSA cryptosystem, Diffie–Hellman key exchange, Elliptic Curve Cryptography (ECC), Cryptographic hash functions, SHA-2 and SHA-3, Message Authentication Codes (MAC), Digital signatures, Public Key Infrastructure (PKI), Digital certificates, Key management and key distribution, Kerberos authentication protocol.

Principles of network security, Secure communication protocols, Transport Layer Security (TLS), Secure Sockets Layer (SSL), IP Security (IPSec), Virtual Private Networks (VPNs), Secure Shell (SSH), Electronic mail security, Pretty Good Privacy (PGP), S/MIME, Wireless security principles, IEEE 802.11 security, WPA2 and WPA3, Network access control, Secure network architecture.

Textbooks / References

1. Stallings, W., Cryptography and Network Security: Principles and Practice, 8th Edition, Pearson, 2023.
2. Forouzan, B. A. and Mukhopadhyay, D., Cryptography and Network Security, 4th Edition, McGraw-Hill Education, 2022.
3. Paar, C. and Pelzl, J., Understanding Cryptography: A Textbook for Students and Practitioners, Springer, 2010.
4. Bishop, M., Computer Security: Art and Science, 2nd Edition, Addison-Wesley, 2018.
5. Kizza, J. M., Guide to Computer Network Security, 6th Edition, Springer, 2020.
6. Anderson, R., Security Engineering: A Guide to Building Dependable Distributed Systems, 3rd Edition, Wiley, 2020.
7. Easttom, C., Network Defense and Countermeasures: Principles and Practices, 3rd Edition, Pearson, 2022.
8. NIST, The NIST Cybersecurity Framework (CSF) 2.0, National Institute of Standards and Technology, 2024.

CAL313 Advanced Machine Learning Lab [0-0-2-1]

Course Objectives

- Explore the concepts, terminology, and working principles of advanced machine learning techniques such as linear models, ensembles, neural networks, GANs, CNNs, RNNs, transfer learning, and reinforcement learning.
- Interpret and understand the role of model selection, optimization, and learning strategies in building effective machine learning systems.
- Apply and analyse advanced learning techniques on different types of data, including tabular, image, sequential, and unlabeled datasets.
- Create suitable machine learning workflows by selecting, tuning, and integrating advanced models for real-world problem solving.

Course Outcomes

- CO1: Identify the core concepts, terminology, and techniques used in advanced machine learning models and workflows.
- CO2: Explain the principles and functioning of linear models, ensemble methods, neural networks, GANs, CNNs, RNNs, transfer learning, reinforcement learning, and unsupervised learning methods.
- CO3: Implement and evaluate advanced machine learning techniques on structured, image-based, sequential, and unlabeled datasets using appropriate optimization and tuning methods.
- CO4: Design and develop an end-to-end advanced machine learning solution by selecting and integrating suitable learning techniques for a given application.

Syllabus

1. Implement and analyze non-linear regression and classification models to study decision boundaries, margin maximization, and model performance on complex datasets.
2. Develop ensemble learning models using techniques such as bagging, random forests, AdaBoost, gradient boosting, and perform comparative performance analysis.
3. Design and train artificial neural networks including perceptrons and multilayer feedforward networks using backpropagation and optimization algorithms.
4. Construct probabilistic and deep learning models for data generation and representation learning, including adversarial training and synthetic data evaluation.
5. Apply convolution-based deep learning architectures for feature extraction, spatial pattern analysis, and image classification tasks.
6. Develop recurrent neural network models for sequence learning, temporal pattern recognition, and prediction of sequential data.
7. Perform clustering and dimensionality reduction using advanced unsupervised learning techniques such as DBSCAN, LDA, and manifold learning methods.
8. Investigate the impact of hyperparameter optimization, learning-rate scheduling, regularization, and model refinement strategies on predictive performance.

9. Implement transfer learning approaches using pre-trained models for feature extraction, fine-tuning, knowledge distillation, and domain adaptation applications.
10. Develop reinforcement learning solutions involving agent – environment interaction, Markov Decision Processes, Q-learning, policy-based learning, and comprehensive model evaluation using appropriate performance metrics.

Textbooks / References

1. Raschka, S., Liu, Y. (Hayden), and Mirjalili, V. Machine Learning with PyTorch and Scikit-Learn: Develop Machine Learning and Deep Learning Models with Python, 1st Edition, Packt Publishing Ltd., Birmingham, UK, 2022. ISBN: 978-1801819312, 774 pages.
2. Goodfellow, I., Bengio, Y., and Courville, A. Deep Learning, Adaptive Computation and Machine Learning Series, MIT Press, Cambridge, MA, USA, 2016. ISBN: 978-0262035613 (Hardcover), ISBN: 978-0262337373 (eBook), 800 pages.
3. Rojas, R. Neural Networks: A Systematic Introduction, 1st Edition, Springer-Verlag Berlin Heidelberg, Germany, 1996. ISBN: 978-3-540-60505-8 (Softcover), ISBN: 978-3-642-61068-4 (eBook), DOI: 10.1007/978-3-642-61068-4, 502 pages.
4. Krohn, J. Deep Reinforcement Learning and GANs: Advanced Topics in Deep Learning, O'Reilly Media, Sebastopol, CA, USA, 2020. ISBN: 978-1098114831.
5. Sarkar, D., Bali, R., and Ghosh, T. Hands-On Transfer Learning with Python: Implement Advanced Deep Learning and Neural Network Models Using TensorFlow and Keras, Packt Publishing Ltd., Birmingham, UK, 2018. ISBN: 978-1788831307.
6. Foster, D. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play, 2nd Edition, O'Reilly Media, Sebastopol, CA, USA, 2023. ISBN: 978-1098134181.

CSL313 Cryptography and Network Security Lab [0-0-2-1]

Course Objectives

- To provide hands-on experience in implementing classical and modern cryptographic algorithms.
- To develop practical skills in secure communication, authentication, and network security protocols.
- To familiarise students with network security tools for vulnerability assessment, traffic analysis, and intrusion detection.
- To analyse common security attacks and evaluate appropriate defensive mechanisms in controlled environments.
- To develop the ability to design, implement, and evaluate secure communication systems through practical exercises.

Course Outcomes

- CO1: Implement classical and modern cryptographic algorithms for secure communication.
- CO2: Configure and analyse authentication mechanisms and secure communication protocols.
- CO3: Perform network traffic analysis, vulnerability assessment, and security monitoring using standard security tools.
- CO4: Evaluate network attacks and implement suitable countermeasures in controlled laboratory environments.
- CO5: Develop secure communication solutions by integrating cryptographic techniques and network security mechanisms.

Syllabus

1. Implement classical cryptographic algorithms including Caesar, Playfair, Hill, and Vigenère ciphers.
2. Implement symmetric key cryptographic algorithms using AES and compare different modes of operation.
3. Implement public key cryptographic algorithms including RSA for encryption and digital signatures.
4. Generate and verify cryptographic hash values using SHA-2/SHA-3 and implement message authentication techniques.
5. Generate public-private key pairs, digital certificates, and perform secure key exchange using OpenSSL.
6. Configure and analyse Transport Layer Security (TLS) for secure client-server communication.
7. Capture and analyse network traffic using Wireshark to identify network protocols and security vulnerabilities.
8. Perform network discovery, port scanning, and service enumeration using Nmap in a controlled environment.
9. Configure and evaluate firewall rules for traffic filtering and network access control.
10. Install and evaluate an Intrusion Detection/Prevention System (IDS/IPS) using suitable open-source tools.
11. Analyse common web security vulnerabilities including SQL Injection and Cross-Site Scripting (XSS) in a controlled environment.
12. Configure secure remote access using SSH and Virtual Private Networks (VPNs).
13. Perform vulnerability assessment and basic penetration testing using standard security assessment tools.
14. Mini Project: Design and implement a secure communication system by integrating cryptographic techniques, authentication mechanisms, and network security tools.

Textbooks / References

1. Stallings, W., Cryptography and Network Security: Principles and Practice, 8th Edition, Pearson, 2023.
2. Orebaugh, A., Ramirez, G., and Burke, J., Wireshark & Ethereal Network Protocol Analyzer Toolkit, Syngress.
3. OpenSSL Project Documentation. OpenSSL Documentation. https://www.openssl.org/docs/
4. Nmap Project. Nmap Network Scanning Guide. https://nmap.org/

MAC321 Linear Programming and Combinatorial Optimization [3-0-0-3]

Course Objectives

- Provide students with a rigorous foundation in linear and integer programming.
- Introduce the geometry of polyhedra and its connection to optimization.
- Develop an algorithmic understanding of the solution of LPs (Simplex method, Ellipsoid method overview).
- Teach duality theory and its applications, including game-theoretic insights.
- Apply optimization to combinatorial problems such as shortest paths, flows, and matchings.

Course Outcomes

- CO1: Formulate real-world problems as Linear Programs (LPs) and Integer Linear Programs (ILPs).
- CO2: Analyze feasible regions using polyhedral geometry and convexity concepts.
- CO3: Apply the Simplex method, interpret results, and understand infeasibility and unboundedness certificates.
- CO4: Use duality theory (weak duality, strong duality, complementary slackness, and Farkas Lemma) in optimization analysis.
- CO5: Apply primal-dual techniques to combinatorial optimization problems such as shortest paths, flows, and matchings.
- CO6: Recognize advanced concepts such as the Ellipsoid method, totally unimodular matrices, and integrality of LP relaxations.

Syllabus

Module 1: Linear and Integer Programming Foundations:
Linear algebra concepts: systems of equations, matrix representation, Gaussian elimination, rank-nullity, column space-row space duality. Formulating real-world problems as Linear Programs (LPs) and Integer Linear Programs (ILPs). Examples of scheduling, logistics, and combinatorial problems.

Module 2: Geometry of Linear Programming:
Feasible regions of LPs as polyhedra, Convexity and convex hulls, extremal points, vertices, faces and facets of polyhedra, Relationship between feasible region geometry and LP solutions.

Module 3: Algorithms for Linear Programming:
Canonical forms of Linear Programs (LPs), bases and basic feasible solutions, degeneracy, characterization of LP outcomes (optimality, infeasibility, and unboundedness), certificates of LP outcomes, the Simplex method including Big-M and Two-Phase methods, geometric interpretation of the Simplex method, and an introduction to the Ellipsoid method.

Module 4: Duality Theory:
Dual linear programs, weak duality, strong duality, complementary slackness, Farkas Lemma, applications of duality in optimization, the Minimax theorem, and an overview of algorithmic game theory.

Module 5: Combinatorial Optimization:
Primal-dual methods and their applications to combinatorial optimization problems including shortest paths, minimum spanning trees, maximum flows, the Max-Flow Min-Cut Theorem, and minimum-cost perfect matchings; totally unimodular matrices and integrality of LP relaxations; semidefinite matrices and semidefinite programming.

Textbooks / References

1. Bertsimas, D., and Tsitsiklis, J. N., *Introduction to Linear Optimization*, Athena Scientific, 1997.
2. Karloff, H., *Linear Programming*, Birkhäuser Boston, Springer Science & Business Media, 2008.
3. Matoušek, J., and Gärtner, B., *Understanding and Using Linear Programming*, Springer, 2007.
4. Papadimitriou, C. H., and Steiglitz, K., *Combinatorial Optimization: Algorithms and Complexity*, Dover Publications, 1998.
5. Schrijver, A., *Combinatorial Optimization: Polyhedra and Efficiency*, Springer, 2003.
6. Grötschel, M., Lovász, L., and Schrijver, A., *Geometric Algorithms and Combinatorial Optimization*, 2nd Edition, Springer Science & Business Media, 2012.
7. Luenberger, D. G., and Ye, Y., *Linear and Nonlinear Programming*, 3rd Edition, Springer, 2008.
8. Nocedal, J., and Wright, S. J., *Numerical Optimization*, 2nd Edition, Springer, New York, 2006.

MAC322 Numerical Methods and Scientific Computing [3-1-0-4]

Course Objectives

- To impart knowledge about numerical methods for solving nonlinear equations, interpolation, differentiation, and integration.
- To enable students to implement direct and iterative methods for solving systems of equations.
- To enable students to solve initial value and boundary value problems numerically.
- To introduce finite difference methods for solving differential equations.
- To introduce scientific computing and high-performance computing techniques for solving large-scale computational problems.

Course Outcomes

- CO1: Obtain numerical solutions of nonlinear equations using iterative methods and analyze their convergence properties.

- CO2: Apply direct and iterative numerical techniques for solving systems of linear equations arising in scientific and engineering problems.
- CO3: Construct interpolation and approximation models and perform numerical differentiation and integration.
- CO4: Solve initial value and boundary value problems using appropriate numerical methods.
- CO5: Develop numerical schemes based on finite difference and weighted residual methods for differential equations.
- CO6: Analyze the accuracy, stability, efficiency, and computational complexity of numerical algorithms.
- CO7: Apply scientific computing and high-performance computing concepts for solving large-scale computational problems.

Syllabus

Numerical Methods

Solutions of Linear Systems: Gaussian elimination, Jacobi method,
Gauss-Seidel method, LU decomposition, convergence and error analysis. Iterative Methods for Nonlinear Equations: Fixed-point iteration, Newton's method, Regula-Falsi method, convergence analysis and error estimation. Interpolation and Approximation: Lagrange interpolation, Newton divided differences, Hermite interpolation, cubic spline interpolation, numerical extrapolation. Numerical Differentiation and Integration: Richardson extrapolation, Newton-Cotes formulas, composite numerical integration, Romberg integration, adaptive quadrature, Gaussian quadrature. Initial Value Problems (IVP): Euler method, Modified Euler method, Runge-Kutta methods, local and global truncation errors. Boundary Value Problems (BVP): Finite difference method, collocation method, Galerkin method.

Scientific Computing

Introduction to scientific computing and large-scale numerical simulation. Evolution of high-performance computing systems and applications in scientific computing. Single-processor performance, memory hierarchy, cache memory, pipelining, and performance considerations. Introduction to Message Passing Interface (MPI); basic MPI communication concepts and numerical applications.

Introduction to GPU computing and the CUDA programming model; CUDA architecture, threads, blocks, and grids; memory hierarchy in CUDA; and basic CUDA kernels for numerical computations. Performance evaluation: speedup, efficiency, and scalability. Applications of HPC in numerical linear algebra, differential equations, optimization, machine learning, scientific simulations, and data analytics.

Textbooks / References

1. S. R. K. Iyengar and R. K. Jain, *Numerical Methods for Scientific and Engineering Computation*, 6th Edition, New Age International.
2. S. S. Sastry, *Introductory Methods of Numerical Analysis*, 5th Edition, PHI Learning.
3. Richard L. Burden and J. Douglas Faires, *Numerical Analysis*, 11th Edition, Cengage Learning.
4. Kendall E. Atkinson, *An Introduction to Numerical Analysis*, 2nd Edition, Wiley.
5. Amos Gilat and Vish Subramaniam, *Numerical Methods for Engineers and Scientists*, Wiley.
6. C. F. Gerald and P. O. Wheatley, *Applied Numerical Analysis*, Pearson.
7. Victor Eijkhout, *Introduction to High Performance Scientific Computing*, Lulu Press.
8. Georg Hager and Gerhard Wellein, *Introduction to High Performance Computing for Scientists and Engineers*, CRC Press.
9. David B. Kirk and Wen-mei W. Hwu, *Programming Massively Parallel Processors: A Hands-on Approach*, Morgan Kaufmann.
10. William Gropp, Ewing Lusk and Anthony Skjellum, *Using MPI: Portable Parallel Programming with the Message-Passing Interface*, MIT Press.
11. Michael J. Quinn, *Parallel Programming in C with MPI and OpenMP*, McGraw-Hill.
12. Lloyd N. Trefethen and David Bau III, *Numerical Linear Algebra*, SIAM.

MAC323 Topological Data Analysis [3-1-0-4]

Course Objectives

- Establish a rigorous mathematical foundation in point-set topology and combinatorial structures tailored for discrete data spaces.
- Equip students with the theoretical knowledge to understand how continuous shapes are translated into computable, algebraic structures.
- Develop the practical ability to compute, visualize, and interpret persistent homology to distinguish topological signals from geometric noise.
- Enable students to integrate topological summaries into modern machine learning pipelines to uncover robust, large-scale patterns in complex, high-dimensional datasets.
- Analyze and interpret the "shape" of complex, high-dimensional data, going beyond traditional statistical methods to uncover robust, large-scale structures in datasets.

Course Outcomes

- CO1: Explain fundamental topological concepts—such as metric spaces, homeomorphisms, and manifolds—and relate them to the representation of point cloud data.
- CO2: Construct simplicial complexes from raw datasets to model underlying geometric structures.
- CO3: Compute Betti numbers and simplicial homology by formulating boundary operators as binary matrices and applying linear algebra operations over Z_2.

- CO4: Analyze persistence diagrams and barcodes to evaluate the stability of topological features across multiple spatial scales.
- CO5: Design automated TDA pipelines that vectorize topological summaries (using Persistence Images or Landscapes) for seamless integration into machine learning models.
- CO6: Apply the Mapper algorithm and persistence frameworks to extract meaningful, non-linear insights from real-world applications such as time-series analysis or network science.

Syllabus

Foundations of Point-Set Topology:
Metric Spaces: Distance functions, open and closed balls, ϵ-neighborhoods. The concept of a point cloud as a finite metric space. Topological Spaces: Formal definition of a topology, open sets, closed sets. Basis for a topology. Subspace topology. Continuity and Homeomorphisms: Continuous functions between topological spaces. Homeomorphism as topological equivalence Product topologies. Quotient topology and gluing (constructing cylinders, Möbius strips, and tori from flat planes) to build geometric intuition.

Topological Properties for Data:
Connectedness: Connected spaces, path-connected spaces, and connected components. Theoretical foundation for 0-dimensional homology and the Betti number β_0. Compactness: Open covers, compact spaces, and compactness in metric spaces. Heine–Borel theorem. Separation Axioms: Hausdorff (T_2) spaces and their significance in topological data analysis. Manifolds: Introduction to topological manifolds and the Manifold Hypothesis.

Combinatorial Topology and Simplicial Complexes:
Simplicial Complexes: Simplices, abstract simplicial complexes, and geometric realizations. Complexes from Data: Constructing C each complexes, Vietoris-Rips complexes, and Alpha complexes from point clouds. The concept of a filtration. Simplicial Homology (via Linear Algebra): Chain groups, cycle groups, and boundary operators over the field $\mathbb{Z}_2$. Formulating the boundary operator as a matrix. Betti Numbers: Computing β_0 (connected components), β_1 (loops), and β_2 (voids) using matrix null space and column space.

Persistent Homology and Topological Summaries: Persistent Homology, Visualizing persistence using barcodes and persistence diagrams. The Bottleneck and Wasserstein distances between persistence diagrams. The Stability Theorem, Vectorizing persistence diagrams using Persistence Landscapes and Persistence Images for downstream ML tasks.

Algorithms, Software, and Applications: The Mapper Algorithm: Filter functions, open covers, clustering, and constructing the Reeb graph/Mapper complex. Computational Frameworks: Implementing TDA pipelines using Python libraries (Gudhi, Ripser, scikit-tda). Real-World Applications: Time Series Analysis, Biology & amp; Medicine, Network Science

Textbooks / References

1. J. R. Munkres, *Topology*, 2nd Edition, Prentice Hall, 2000.
2. W. A. Sutherland, *Introduction to Metric and Topological Spaces*, 2nd Edition, Oxford University Press, 2009.
3. B. Mendelson, *Introduction to Topology*, 3rd Edition, Dover Publications, 1990.
4. H. Edelsbrunner and J. Harer, *Computational Topology: An Introduction*, American Mathematical Society, 2010.
5. R. W. Ghrist, *Elementary Applied Topology*, Createspace, 2014.
6. S. Y. Oudot, *Persistence Theory: From Quiver Representations to Data Analysis*, American Mathematical Society, 2015.
7. G. Carlsson, "Topology and Data," *Bulletin of the American Mathematical Society*, vol. 46, no. 2, pp. 255–308, 2009.
8. H. Adams *et al.*, "Persistence Images: A Stable Vector Representation of Persistent Homology," *Journal of Machine Learning Research*, vol. 18, no. 1, pp. 218–252, 2017.
9. T. K. Dey and Y. Wang, *Computational Topology for Data Analysis*, Cambridge University Press, 2022.
10. G. Carlsson and M. Vejdemo-Johansson, *Topological Data Analysis with Applications*, Cambridge University Press, 2021.
11. C. Farrelly and Y. U. Gaba, *The Shape of Data: Network Science, Geometry-based Machine Learning, and Topological Data Analysis*, No Starch Press, 2023.

OE Foreign Language [2-0-0-2]

Course Objectives

- To introduce the basics of French language and grammar to the students.
- To develop the four language skills at the initial level. It covers the fundamentals of French language, such as French alphabets and phonetics, essential grammar, and simple vocabulary.
- To provide a fundamental understanding of the French language and culture.
- To develop knowledge in core competencies such as listening, reading, speaking, and writing.

Course Outcomes

- CO1: Use French to describe, narrate, and ask/answer questions in the present tense on topics related to family, daily activities, and surroundings, and communi-

cate effectively in the present, near future, and recent past.
- CO2: Make short statements and ask/answer simple questions in the past; comprehend short conversations; read and understand simple literary and non-literary texts; and write sentences and short paragraphs on familiar topics.
- CO3: Develop an awareness of French and Francophone cultures and understand the basic functioning of French as a language.

Syllabus

Salutations, Alphabets, Presentation (to introduce someone), Conjugation of First and Second Group Verbs, Interrogative Sentences, Numbers from 0 to 100.

Definite and Indefinite Articles, French Vocabulary, French Foods, Costumes, etc., Conjugation of Third Group and Irregular Verbs, Negation.

Prepositions of Place, Possessive Adjectives, The Family, About House, Activities and Hobbies, Reflexive Verbs, Pronom Toniques, Conjunctions, Article Contracté, Partitive Articles, Possessive Form, Future Tense.

Simple Prepositions and Negation, Seasons, Adjectives, Irregular Adjectives, Recent Past, Demonstrative Adjectives, Time.

Textbooks / References

1. *Apprenons le Français 1 – Méthode de Français 1 & Cahier d'Exercices*, Mahitha Ranjith and Monica Singh.
2. *Saison 1 – Méthode de Français 1 & Cahier d'Exercices*, Didier.
3. *Écho A1 – Méthode*, Version Numérique DVD, CLE International.
4. *La Tendance* – Conversation Videos.

MAE321 Applied Functional Analysis [3-1-0-4]

Course Prerequisites

MAC111 Calculus and Linear Algebra

Course Objectives

- To introduce the fundamental concepts of metric spaces, normed spaces, Banach spaces, and Hilbert spaces that form the mathematical foundation of modern computational methods.
- To study linear operators, boundedness, continuity, operator norms, and dual spaces, with applications in mathematical modelling and scientific computing.
- To develop analytical skills for understanding functional spaces and operator theory used in optimization, machine learning, signal processing, and numerical analysis.
- To understand fundamental results of functional analysis including the Hahn–Banach Theorem, Banach–Steinhaus Theorem, Open Mapping Theorem, and Closed Graph Theorem.
- To apply Hilbert space methods, orthogonality, orthogonal projections, and bounded linear functionals in solving computational and engineering problems.

Course Outcomes

- CO1: Explain the structure and properties of metric spaces, normed spaces, Banach spaces, and Hilbert spaces, and relate them to mathematical and computational problems.
- CO2: Analyze bounded linear operators, operator norms, dual spaces, and linear functionals arising in mathematical modelling and scientific computing.
- CO3: Apply the Hahn–Banach theorem and related concepts to extend and characterize bounded linear functionals.
- CO4: Use major results of functional analysis such as the Banach–Steinhaus Theorem, Open Mapping Theorem, Closed Graph Theorem, and Bounded Inverse Theorem to study operator behaviour.
- CO5: Apply orthogonality principles, orthonormal bases, projections, and representation theorems in Hilbert spaces to problems in approximation theory, signal processing, optimization, and data analysis.

Syllabus

Metric spaces and normed linear spaces, Examples from function spaces and sequence spaces, Product and quotient spaces, Banach spaces, Hilbert spaces, Non-compactness of the unit ball in infinite-dimensional normed spaces.

Linear maps between normed spaces, Boundedness and continuity, Spaces of bounded linear operators, Dual spaces, Linear isometries, Linear functionals, Examples from approximation theory and computational mathematics.

Hahn–Banach theorem and applications, Banach Steinhaus theorem (Uniform Boundedness Principle), Open Mapping Theorem, Closed Graph Theorem, Bounded Inverse Theorem, Spectrum of bounded operators.

Orthogonality and Gram–Schmidt orthogonalization, Bessel's inequality, Riesz–Fischer theorem. Orthonormal bases, Parseval's identity, Orthogonal projections and decomposition, Riesz Representation Theorem and bounded linear functionals on Hilbert spaces, Applications in least-squares approximation and signal representation.

Textbooks / References

1. B. V. Limaye, *Functional Analysis*, 2nd ed., New Age International, New Delhi, 1996.
2. J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Graduate Texts in Mathematics, Vol. 96, Springer, 1990.

3. Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
4. Peter D. Lax, *Functional Analysis*, Wiley-Interscience, 2002.
5. Béla Bollobás, *Linear Analysis: An Introductory Course*, Cambridge Mathematical Textbooks, 1990.
6. S. Kesavan, *Functional Analysis*, 2nd Edition, Springer, 2023.
7. M. Thamban Nair, *Functional Analysis: A First Course*, Prentice Hall of India, 2008.
8. Walter Rudin, *Functional Analysis*, 2nd ed., McGraw-Hill, 2006.
9. Jan van Neerven, *Functional Analysis, Cambridge Studies in Advanced Mathematics*, Cambridge University Press, 2024.
10. J. Tinsley Oden and Leszek Demkowicz, *Applied Functional Analysis*, 3rd Edition, CRC Press, 2023.

MAC411 Measure and Integration [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of measurable spaces, measurable sets, and measurable functions.
- To develop a rigorous understanding of measures, outer measures, and the generation of measures, including Lebesgue measure.
- To explore advanced integration concepts such as functions of bounded variation, absolute continuity, and the Fundamental Theorem of Calculus in the Lebesgue framework.
- To develop knowledge of product measures, Fubini's theorem, and their applications.
- To introduce signed measures, Hahn and Jordan decompositions, Radon–Nikodym theorem, and L^p-spaces.

Course Outcomes

- CO1: Demonstrate understanding of Lebesgue measure, its properties, and recognize the existence of non-measurable sets.
- CO2: Apply Lebesgue integration theory, including the Monotone and Dominated Convergence Theorems, to evaluate integrals.
- CO3: Compare Lebesgue and Riemann integrals, and explain their differences and advantages.
- CO4: Use the Radon–Nikodym theorem to represent absolutely continuous measures.
- CO5: Apply product measures and Fubini's theorem to multi-dimensional integration problems.

Syllabus

Measurable spaces, measurable sets, measurable functions, measure, outer measures, and generation of measures.

Sigma-algebra of measurable sets. Completion of a measure. Lebesgue measure and its properties. Non-measurable sets.

Basic integration theorem, Integration and Convergence Theorems. Lebesgue integral, functions of bounded variation, and absolutely continuous functions. Fundamental Theorem of Calculus for Lebesgue integrals, comparison of Lebesgue and Riemann integrals.

Signed measures, Hahn and Jordan decomposition theorems, the Radon–Nikodym theorem, product measures and Fubini's theorem, L^p-spaces, duals of L^p-spaces. Riesz Representation Theorem.

Textbooks / References

1. H. L. Royden, *Real Analysis*, Third Edition, Macmillan Publishing Company, New York, 1988.
2. Walter Rudin, *Real and Complex Analysis*, Third Edition, McGraw-Hill Book Co., New York, 1987.
3. G. de Barra, *Measure Theory and Integration*, 2nd Edition, Ellis Horwood Ltd./John Wiley & Sons, 2000.
4. Krishna B. Athreya and Soumendra N. Lahiri, *Measure Theory and Probability Theory*, Springer Texts in Statistics, Springer-Verlag, 2006.
5. Edwin Hewitt and Karl Stromberg, *Real and Abstract Analysis*, Springer, 1969.

PROGRAMME ELECTIVE COURSES

MAEXXX Applied Complex Analysis for Engineering [2-1-0-3]

Course Objectives

- To understand the fundamentals of complex variables and functions.
- To understand the continuity and differentiability of complex functions.
- To understand analytic functions, their properties, and the series expansion of analytic functions.
- To evaluate complex integrals using Cauchy's Integral Theorem and Cauchy's Integral Formula.
- To introduce conformal mappings and residue calculus.
- To apply complex analysis concepts in engineering problems.

Course Outcomes

- CO1: Understand and apply the fundamental concepts of complex variables and functions in engineering contexts.
- CO2: Analyze the properties of analytic functions using the Cauchy–Riemann equations and explore their implications in physical systems.
- CO3: Evaluate complex integrals using contour integration and apply Cauchy's Integral Theorem and Formula to solve engineering problems.
- CO4: Classify singularities and compute residues to solve real integrals.

- CO5: Apply complex analysis techniques to solve problems in control systems and related engineering applications.

Syllabus

Fundamentals of Complex Numbers, geometry and topology of the complex plane, representation on the extended complex plane (Riemann Sphere).

Complex Functions, limit, continuity, differentiability, Cauchy–Riemann equations, analytic functions and their properties, harmonic functions, harmonic conjugates, exponential, trigonometric, logarithmic functions, conformal mapping, fixed points.

Complex integration, line integrals in the complex plane, basic properties, contour integrals, Cauchy Integral Theorem on simply connected domains, Cauchy Integral Theorem on multiply connected domains, Cauchy Integral Formula, derivatives of an analytic function.

Power series, zeros, singularities, residue integration method, evaluation of real improper integrals, engineering applications in electrical circuit analysis, control theory, artificial intelligence, etc.

Textbooks / References

1. J. W. Brown and R. V. Churchill, *Complex Variables and Applications*, 9th Edition, McGraw-Hill Education, 2009.
2. S. Ponnusamy and H. Silverman, *Complex Variables with Applications*, Birkhäuser, 2006.
3. E. Kreyszig, H. Kreyszig, and E. J. Norminton, *Advanced Engineering Mathematics*, 10th Edition, Wiley, New Delhi, 2015.
4. John H. Mathews and Russell W. Howell, *Complex Analysis for Mathematics and Engineering*, 6th Edition, Jones and Bartlett Publishers, 2011.
5. Hoai An Le Thi, Hoai Minh Le, and Tao Pham Dinh (Eds.), *Optimization of Complex Systems: Theory, Models, Algorithms and Applications*, Springer International Publishing, 2019.
6. C. R. Wylie and L. C. Barrett, *Advanced Engineering Mathematics*, 6th Edition, McGraw-Hill, New Delhi, 1995.

MAEXXX Markov Decision Process and Reinforcement Learning [2-0-0-2]

Course Prerequisites

Basic knowledge of probability theory, stochastic processes, linear algebra, and optimization.

Course Objectives

- To introduce the mathematical foundations of decision-making under uncertainty using Markov Decision Processes (MDPs).
- To apply dynamic programming methods for solving sequential optimization problems.
- To develop an understanding of reinforcement learning (RL) as a data-driven counterpart to MDP theory.
- To explore model-free and deep reinforcement learning algorithms and their applications in control, robotics, and AI systems.

Course Outcomes

- CO1: Formulate real-world decision-making problems as Markov Decision Processes.
- CO2: Apply Bellman equations and dynamic programming principles for optimal policy determination.
- CO3: Analyze finite and infinite horizon models with discounted, average, and total cost criteria.
- CO4: Implement iterative algorithms such as policy iteration, value iteration, and Q-learning.
- CO5: Understand model-free reinforcement learning techniques including Monte Carlo and temporal difference methods.
- CO6: Apply deep reinforcement learning methods to engineering and AI applications.

Syllabus

Foundations of Stochastic Systems and Markov Decision Processes: Review of probability and stochastic processes, Introduction to Markov chains and Markov models for discrete-time dynamic systems, Rewards, policies, and policy evaluation, Markov Decision Processes: definitions, examples, and formulations. Finite Horizon and Dynamic Programming: Finite horizon models and classes of policies, Optimality of Markov policies, Bellman's optimality principle and optimality equations, Dynamic programming principle and algorithm. Infinite Horizon Models: Stationary models and adequacy of Markov policies, Discounted cost models: pure policies, optimality results, Policy iteration, value iteration, and modified policy iteration algorithms, Linear and convex programming formulations. Average and Total Cost Models: Average cost models: unichain and multichain cases, Iterative algorithms for average cost MDPs, Expected total cost models: positive and negative models, Mathematical programming formulations of optimal policies.

Reinforcement Learning Fundamentals: RL framework and connections to MDPs, Exploration vs. exploitation dilemma, Generalized Policy Iteration, Monte Carlo learning, Temporal Difference learning: TD(0), TD(λ). Model-Free RL: Q-learning and SARSA, Function approximation in RL, Actor-Critic methods, Policy Gradient methods, Stochastic approximation in RL. Deep Reinforcement Learning and Applications: Neural networks in reinforcement learning, Deep Q-Networks (DQN) and extensions, Policy gradient with deep architectures, Applications and case studies in machine learning, robotics, control, communication, and optimization.

Textbooks / References

1. Puterman, M. L., *Markov Decision Processes: Discrete Stochastic Dynamic Programming*, John Wiley & Sons.
2. D. P. Bertsekas, *Dynamic Programming and Optimal Control*, Volumes 1 & 2, Athena Scientific.
3. E. A. Feinberg and A. Shwartz (Eds.), *Handbook of Markov Decision Processes*, Kluwer Academic Publishers, 2002.
4. Eitan Altman, *Constrained Markov Decision Processes*, Routledge.
5. J. A. Filar and K. Vrieze, *Competitive Markov Decision Processes*, Springer, 1997.
6. Richard S. Sutton and Andrew G. Barto, *Reinforcement Learning: An Introduction*, 2nd Edition, MIT Press, 2018.
7. D. P. Bertsekas and J. N. Tsitsiklis, *Neuro-Dynamic Programming*, Athena Scientific, 1996.
8. Sheldon M. Ross, *Applied Probability Models with Optimization Applications*, Courier Corporation, 2013.
9. Sheldon M. Ross, *Introduction to Stochastic Dynamic Programming*, Academic Press, 2014.
10. V. G. Kulkarni, *Modeling and Analysis of Stochastic Systems*, CRC Press.

MAEXXX Computational Algebra [3-1-0-4]

Course Prerequisites

MAC122 Modern Algebra

Course Objectives

- To introduce fundamental algorithms for arithmetic operations, modular arithmetic, and classical number-theoretic methods such as Euclid's algorithm and the Chinese Remainder Theorem.
- To study the structure and properties of fields, finite fields, rational fields, and polynomial rings.
- To develop understanding of polynomial factorization techniques over finite fields, including Hensel lifting and related applications.
- To explore advanced algorithmic techniques such as the Shortest Vector Problem and the LLL algorithm, with emphasis on applications in computational algebra.
- To introduce Gröbner bases, term orderings, and Buchberger's algorithm as central tools in computational algebra and algebraic geometry.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Apply efficient algorithms for arithmetic operations and modular computations in algebraic settings.
- CO2: Use Euclid's algorithm and the Chinese Remainder Theorem to solve algebraic and number-theoretic problems.
- CO3: Factor polynomials over finite fields and apply Hensel lifting techniques in computational problems.
- CO4: Apply lattice-based methods such as the LLL algorithm to shortest vector and polynomial factorization problems.
- CO5: Construct Gröbner bases and use Buchberger's algorithm to solve systems of polynomial equations and ideal membership problems.

Syllabus

Basic algorithms for arithmetic, addition, multiplication, and modular arithmetic. Euclid's algorithm, Chinese Remainder Theorem, and applications to problems such as determinant computation and Gaussian elimination.

Fields, finite fields, field of rationals, polynomial rings. Polynomial factorization over finite fields, Hensel lifting and applications to bivariate factorization. Shortest Vector Problem (SVP) and the LLL algorithm. Applications to factorization of polynomials over the rationals.

Multivariate systems of polynomial equations. Basic operations with monomial ideals and modules. Term orders and leading terms. Gröbner bases of ideals and modules. Buchberger's algorithm and applications.

Textbooks / References

1. W. W. Adams and P. Loustaunau, *An Introduction to Gröbner Bases*, American Mathematical Society, 1994.
2. H. Cohen, *A Course in Computational Algebraic Number Theory*, Springer GTM 138, 1993.
3. D. A. Cox, J. Little, and D. O'Shea, *Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra*, Fourth Edition, Springer UTM, 2015.
4. J. von zur Gathen and J. Gerhard, *Modern Computer Algebra*, Third Edition, Cambridge University Press, 2013.
5. M. Kreuzer and L. Robbiano, *Computational Commutative Algebra I*, Springer, 2000.
6. M. Kreuzer and L. Robbiano, *Computational Commutative Algebra II*, Springer, 2005.
7. B. Mishra, *Algorithmic Algebra*, Springer, 1993.

MAEXXX Approximation Algorithms [3-1-0-4]

Course Objectives

- To identify the design techniques used in approximation algorithms for computationally hard optimization problems.
- To design and analyze approximation algorithms using combinatorial optimization techniques.
- To apply advanced techniques such as randomization and linear programming based methods to NP-hard optimization problems.
- To understand and apply reduction techniques for proving hardness of approximation.

- To develop the ability to evaluate approximation guarantees and performance bounds of algorithms.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Identify and explain the design techniques used in approximation algorithms for NP-hard optimization problems.
- CO2: Design and analyze approximation algorithms using combinatorial techniques such as greedy methods, local search, and primal-dual approaches.
- CO3: Apply linear programming relaxations, randomized algorithms, and rounding techniques to obtain approximation guarantees.
- CO4: Evaluate approximation ratios and compare different approximation strategies for optimization problems.
- CO5: Analyze hardness of approximation results and understand the limitations of efficient approximation algorithms.
- CO6: Apply approximation algorithm techniques to practical problems in scheduling, network design, routing, and resource allocation.

Syllabus

Combinatorial Approximation Techniques: Review of NP-completeness and polynomial-time reductions. Approximation algorithms and approximation factors. Lower bounds on optimal solutions. Vertex Cover approximation algorithms. Greedy algorithms for Set Cover. Minimum Spanning Tree heuristic. Metric Steiner Tree and Metric Traveling Salesman Problem (TSP). Layering techniques and approximation algorithms for Feedback Vertex Set.

Linear Programming and Randomized Techniques: Linear programming relaxations. Simple rounding techniques for Set Cover, Bin Packing, and Max-SAT. Random sampling and randomized algorithms for Max-SAT and Max-Cut. Randomized rounding methods for Set Cover and Multiway Cut. Primal-dual method and applications.

Hardness of Approximation: Approximation-preserving reductions. Techniques for proving hardness of approximation. PCP theorem (overview). Hardness results for Set Cover, Vertex Cover, and Traveling Salesman Problem. Limits of approximation algorithms for NP-hard optimization problems.

Applications: Approximation algorithms in scheduling, network design, facility location, routing, clustering, and resource allocation problems.

Textbooks / References

1. Vijay V. Vazirani, *Approximation Algorithms*, Springer, 2003.
2. David P. Williamson and David B. Shmoys, *The Design of Approximation Algorithms*, Cambridge University Press, 2011.
3. Dorit S. Hochbaum (Ed.), *Approximation Algorithms for NP-Hard Problems*, PWS Publishing Company, 1997.
4. Jon Kleinberg and Éva Tardos, *Algorithm Design*, Pearson, 2013.
5. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein, *Introduction to Algorithms*, MIT Press, 4th Edition, 2022.
6. Sanjeev Arora and Boaz Barak, *Computational Complexity: A Modern Approach*, Cambridge University Press, 2009.

MAEXXX Advanced Convex Optimization [3-1-0-4]

Course Objectives

- Develop a rigorous understanding of convex analysis and convex optimization theory.
- Study canonical convex programs and their unifying framework.
- Learn duality theory, KKT conditions, and their role in optimization.
- Explore computational algorithms for solving convex problems at scale.
- Explore real-world applications of convex optimization in control, signal processing, statistics, and machine learning.

Course Outcomes

- CO1: Analyze the structure and properties of convex sets and convex functions.
- CO2: Formulate real-world problems as canonical convex programs (LP, QP, SOCP, SDP).
- CO3: Apply duality theory and KKT conditions to study the optimality and sensitivity of convex problems.
- CO4: Implement and evaluate first-order, interior-point, and split methods for large-scale convex optimization.
- CO5: Handle non-smooth convex optimization using tools such as conjugate functions and Moreau's regularization.
- CO6: Apply convex optimization techniques to applications in machine learning, control systems, and signal processing.
- CO7: Critically assess the limitations of convex approaches and explore relaxations for nonconvex problems.

Syllabus

Convex Analysis Foundations: Convex sets, convex cones, affine sets, Convex and concave functions, Epigraphs, sub level sets, perspective functions, First- and second-order characterizations of convexity

Canonical Convex Programs: Linear Systems (LS) and Linear Programming (LP), Quadratic Programming (QP), Second-Order Cone Programming (SOCP), Semidefinite

Programming (SDP), Generalized Inequalities and Cone Programming, Linear Matrix Inequalities (LMIs), conic duality and conic duality theorem.

Computational Methods: Gradient and Sub-gradient methods, Proximal methods and operator splitting, Newton's method for convex problems, Interior point methods: Barrier functions, Primal-dual path-following (LP, SOCP, SDP), Augmented Lagrangian methods, Alternating Direction Method of Multipliers (ADMM), Proximal minimization algorithms, Mirror-descent algorithms (MDA), Non-smooth Convex Optimization: Conjugate functions, Moreau's regularization, Smooth approximations via conjugation

Applications of SDP: Control and systems, Polynomial optimization, Combinatorial and nonconvex problems, Machine learning: SVMs, Lasso, logistic regression, matrix completion, Signal Processing: compressed sensing, filter design

Textbooks/References

1. Ben-Tal, Aharon, and Arkadi Nemirovski. Lectures on modern convex optimization: analysis, algorithms, and engineering applications. Society for industrial and applied mathematics, 2001.
2. Nesterov, Yurii. Introductory lectures on convex optimization: A basic course. Vol. 87. Springer Science & Business Media, 2013.
3. Boyd, Stephen P., and Lieven Vandenberghe. Convex optimization. Cambridge university press, 2004.
4. Bazaraa, Mokhtar S., Hanif D. Sherali, and Chitharanjan M. Shetty. Nonlinear programming: theory and algorithms. John wiley & sons, 2006.
5. Dimitri P. Bertsekas. Nonlinear Programming: Athena Scientific, 2nd Edition, 1999
6. Bertsekas, Dimitri, Angelia Nedic, and Asuman Ozdaglar. Convex analysis and optimization. Vol. 1. Athena Scientific, 2003.

MAEXXX Algebraic Graph Theory [3-1-0-4]

Course Prerequisites

MAC212 Graphs and Networks in Computing Systems

Course Objectives

- To develop a rigorous understanding of algebraic and spectral techniques in graph theory through matrix representations, eigenvalues, and graph spectra.
- To apply fundamental algebraic results such as the Perron–Frobenius theorem and Cauchy's interlacing theorem in the analysis of graph structures and networks.
- To investigate important graph classes including strongly regular, distance-regular, distance-transitive, and Cayley graphs, and examine their algebraic and combinatorial properties.
- To analyze graph symmetries, automorphisms, transitivity properties, and their implications for graph classification.
- To employ spectral graph theoretic methods for solving problems related to connectivity, flows, network design, and applications in computing and data science.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Compute and interpret graph spectra using adjacency, incidence, and Laplacian matrices, and apply the Perron–Frobenius and Cauchy's interlacing theorems in graph analysis.
- CO2: Analyze strongly regular graphs, symmetric graphs, vertex-transitive graphs, and edge-transitive graphs using algebraic and spectral techniques.
- CO3: Characterize distance-regular and distance transitive graphs through intersection arrays, Bose–Mesner algebras, association schemes, and Krein parameters.
- CO4: Apply graph automorphisms, equitable partitions, line graph constructions, and Cayley graph theory to study graph symmetries and structural properties.
- CO5: Employ advanced spectral graph theory concepts to investigate graph connectivity, cuts, flows, graph energy, and network-related problems arising in computing and data science.

Syllabus

Foundations of Spectral Graphs: Basic definitions in graph theory, Matrices associated with graphs, eigenvalues and spectrum of graphs, incidence matrix, adjacency matrix, Laplacian matrix and their spectral properties, Perron-Frobenius theorem, Cauchy's interlacing theorem, Strongly regular graphs and their spectra, Laplacian matrix, cuts and flows.

Graph Symmetry: Automorphisms of graphs, Vertex-transitive and edge-transitive graphs, Symmetric graphs and their properties, Cayley graphs, examples and constructions, eigenvalues of Cayley graphs.

Regular Graphs: Distance-transitive graphs, distance-regular graphs and related spectra. Intersection arrays, Krein parameters, Bose–Mesner algebra, association schemes, equitable partitions and applications. Line graphs and spectral properties of line graphs. Generalized quadrangles, root systems, graphs with minimum eigenvalue at least -2.

Advanced Spectral Structures: Incidence graphs and duality. Paley graphs, Paley digraphs and their properties. Graph energy, bounds on graph energy, hyperenergetic and hypoenergetic graphs. Applications of spectral graph theory in networks and combinatorial optimization.

Textbooks / References

1. Norman Biggs, *Algebraic Graph Theory*, 2nd Edition, Cambridge University Press, 1993.
2. Chris Godsil and Gordon F. Royle, *Algebraic Graph Theory*, Graduate Texts in Mathematics 207, Springer, 2001.
3. Chris Godsil, *Algebraic Combinatorics*, Chapman & Hall, 1993.
4. Bogdan Nica, *A Brief Introduction to Spectral Graph Theory*, European Mathematical Society, 2018.
5. R. B. Bapat, *Graphs and Matrices*, Hindustan Book Agency, 2014.
6. Dragoš Cvetković, Michael Doob, and Horst Sachs, *Spectra of Graphs: Theory and Application*, Academic Press, 1980.
7. Brouwer, A. E., Cohen, A. M., and Neumaier, A., *Distance-Regular Graphs*, Springer, 1989.
8. Andries E. Brouwer and Willem H. Haemers, *Spectra of Graphs*, Universitext, Springer, New York, 2012.
9. Daniel A. Spielman, *Spectral and Algebraic Graph Theory*, Yale University, 2019.

MAEXXX Algorithmic Graph Theory [3-1-0-4]

Course Prerequisites

MAC212 Graphs and Networks in Computing Systems

Course Objectives

- To understand graph representations in computer systems and analyze the algorithmic complexity of graph algorithms.
- To study trees, their properties, and important algorithms such as BFS, DFS, sorting and minimum spanning tree.
- To explore classical graph search techniques (DFS, BFS) and their applications.
- To provide a foundation in algorithmic paradigms such as sorting and greedy algorithms within graph problems.
- To introduce the concepts of networks, flows, and cuts, and study max-flow min-cut theorem and related algorithms.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Design and implement classical graph algorithms (DFS, BFS, greedy, sorting, topological sorting) with efficiency considerations.
- CO2: Apply data structures and algorithmic complexity analysis to represent and process graphs efficiently.
- CO3: Analyze and solve matching and flow problems using well known theorems and max-flow min-cut principles.
- CO4: Evaluate domination parameters in graphs and establish bounds and inequalities for domination numbers.
- CO5: Integrate and apply graph theory concepts into real-world applications such as network analysis, scheduling, and optimization.

Syllabus

Basic definitions in graph theory, Sub-graphs, Degree sequences, graphical sequence, Havel Hakimi theorem, special graphs, digraphs.

Basic data structures in computer science, representation of graphs in computers, properties of trees, the minimum spanning tree problem, algorithmic complexity: Big-O, Big-omega and Theta notations, space and time complexity, search algorithms, topological sorting, sorting algorithms, greedy algorithms, depth first search algorithms, breadth first search algorithms.

Maximum matching, Berge's theorem, perfect matching, matching in bipartite graphs, Konig's theorem, Hall's marriage theorem, matching in general graphs, Tutte's theorem, weighted matching, flows and matching, max-flow min-cut theorem, factorizations, block designs.

Domination in graphs, dominating set, minimal dominating set, domination number, independent dominating set, connected domination number, total domination number, bounds and inequalities involving domination numbers.

Textbooks / References

1. T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, MIT Press, 2022.
2. R. Diestel, *Graph Theory*, Graduate Texts in Mathematics, 5th Edition, Springer, 2017.
3. T. Kloks and M. Xiao, *A Guide to Graph Algorithms*, Springer, 2022.
4. J. A. McHugh, *Algorithmic Graph Theory*, Prentice-Hall, 1990.
5. M. C. Golumbic, *Algorithmic Graph Theory and Perfect Graphs*, Academic Press, 2004.
6. G. Valiente, *Algorithms on Trees and Graphs: With Python Code*, 2nd Edition, Springer, 2002.

MAEXXX Fractional Calculus [3-0-0-3]

Pre-Requisites

Real Analysis

Course Objectives

- To introduce the fundamental concepts and methods of fractional calculus and distinguish them from classical calculus.

- To develop an understanding of special functions and transform techniques essential for fractional order analysis.
- To formulate and solve fractional differential equations analytically using appropriate mathematical tools.
- To explore applications of fractional calculus in modeling and solving problems in science and engineering.

Course Outcomes

- CO1: Understand the basic concepts of fractional calculus and appreciate the differences and similarities between classical derivatives and fractional derivatives.
- CO2: Analyse the existence and uniqueness of solutions of fractional differential equations.
- CO3: Apply analytical techniques such as Laplace transforms, Mellin transforms, and power series methods to solve linear fractional differential equations.
- CO4: Apply fractional calculus concepts to model, analyze, and solve real-world problems arising in science and engineering, including Abel's integral equation and fractional diffusion equations.

Syllabus

Gamma function and its properties, Beta function, Contour integral representation. Mittag-Leffler function and its relation with other functions, Laplace transform and derivative of Mittag-Leffler function, Review of basic definitions of integer-order derivatives and integrals and their geometric and physical interpretations. Motivation and history of fractional calculus.

Fractional derivatives: Grunwald-Letnikov, Riemann-Liouville and Caputo's fractional derivative (Left and right fractional derivatives), Riesz fractional integro-differentiation, Properties of fractional derivatives, Leibniz rule for fractional derivatives, Geometric and physical interpretation of fractional integration and fractional differentiation. Computation of these FDs for some basic functions like constant, ramp, exponential, sine, cosine, etc.

Laplace transforms of fractional derivatives. Fourier transforms and Mellin transforms of fractional derivatives. Linear fractional differential equations: Equation of a general form, existence and uniqueness theorem as a method of solution, dependence of a solution on initial conditions, Laplace transform method, standard fractional differential equations, some methods for solving fractional order equations: Mellin transform, power series, numerical evaluation of fractional derivatives.

Fractional Differential Equations: Introduction, Linearly independent solutions, Solutions of the homogeneous and non-homogeneous fractional differential equations, Reduction of fractional partial differential equations to ordinary differential equations, applications of fractional calculus: Abel's integral equation, fractional diffusion equation.

Textbooks / References

1. K. B. Oldham and J. Spanier, *The Fractional Calculus*, Academic Press, San Diego, 1974.
2. K. S. Miller and B. Ross, *An Introduction to the Fractional Calculus and Fractional Differential Equations*, Wiley, New York, 1993.
3. I. Podlubny, *Fractional Differential Equations*, Academic Press, San Diego, 1999.

MAEXXX Mathematical Physics [3-1-0-4]

Course Prerequisites

MAC111 Calculus and Linear Algebra, MAC123 Real and Complex Analysis

Course Objectives

- To provide a strong foundation in the mathematical techniques essential for understanding advanced topics in theoretical and applied physics.
- To develop problem-solving skills using linear algebra, vector and tensor analysis, differential equations, and special functions.
- To introduce complex analysis and integral transforms as tools for solving physical and engineering problems.
- To enhance analytical reasoning and apply mathematical methods to model and interpret physical phenomena.

Course Outcomes

After successful completion of this course, students will be able to:

- CO1: Apply vector, matrix, and tensor methods to describe physical systems and transformations.
- CO2: Analyze linear vector spaces and operators relevant to quantum mechanics and normal mode analysis.
- CO3: Solve differential equations using analytical and series techniques.
- CO4: Employ special functions and orthogonal polynomials in solving boundary value problems.
- CO5: Utilize Laplace transforms to address problems in potential theory, wave propagation, and signal analysis.

Syllabus

Vector, Matrix, and Tensor Analysis: Scalar and vector fields, coordinate systems, Matrices, Tensors: definition, rank, covariant and contravariant tensors, Kronecker delta, metric tensor, tensor operations, covariant differentiation, and geodesic equations. Linear Operators: Introduction to Inner-product Spaces, Hermitian, unitary, and self-adjoint operators, Diagonalization of Hermitian matrices, spectral theorem, Application to quantum mechanics and normal mode problems.

Special Functions: Exact and series solutions: Frobenius method, Sturm Liouville eigenvalue problems; Hermitian operators, Special functions: Bessel, Neumann, Hankel, Legendre, Laguerre, and Hermite functions-orthogonality, recurrence relations, and generating functions, Gamma and Beta functions: properties and applications, Quantum harmonic oscillator, hydrogen atom, vibrating membranes.

Laplace transform: inverse transform, and convolution theorem, Applications to boundary value problems, heat conduction, and electrical circuit analysis, Discrete Fourier transform (DFT).

Textbooks / References

1. G. B. Arfken and H. J. Weber, *Mathematical Methods for Physicists*, Elsevier Academic Press.
2. Mary L. Boas, *Mathematical Methods in the Physical Sciences*, Wiley.
3. K. F. Riley, M. P. Hobson, and S. J. Bence, *Mathematical Methods for Physics and Engineering*, Cambridge University Press.
4. T. M. Apostol, *Mathematical Analysis*, Addison-Wesley.
5. P. M. Morse and H. Feshbach, *Methods of Theoretical Physics*, McGraw–Hill.
6. F. W. Byron and R. W. Fuller, *Mathematics of Classical and Quantum Physics*, Dover.
7. E. Kreyszig, *Advanced Engineering Mathematics*, Wiley.
8. P. Dennery and A. Krzywicki, *Mathematics for Physicists*, Courier Corporation, 1996.
9. H. K. Dass, *Mathematical Physics*, S. Chand Publishing.
10. S. Prakash, *Mathematical Physics with Classical Mechanics*, Sultan Chand & Sons.

MAEXXX Stochastic Processes and Applications [3-0-0-3]

Course Objectives

- To provide a rigorous foundation in stochastic modeling, probabilistic reasoning, and their real-world applications.
- To study discrete-time and continuous-time stochastic processes and their long-term behavior.
- To introduce martingales, Brownian motion, stochastic calculus, and stochastic differential equations.
- To develop the ability to model uncertainty in engineering, finance, communication systems, and data science.
- To expose students to modern applications of stochastic processes in machine learning, reliability, and optimization.

Course Outcomes

- CO1: Apply probability spaces, random variables, expectations, and convergence concepts in stochastic modeling.
- CO2: Use conditional expectation and Bayesian reasoning to analyze uncertain systems.
- CO3: Model and analyze discrete-time and continuous-time stochastic processes, including Markov chains and Poisson processes.
- CO4: Apply martingale theory, Brownian motion, and stochastic calculus in the study of random dynamical systems.
- CO5: Formulate and solve stochastic differential equations arising in science, engineering, and finance.
- CO6: Apply stochastic process models to queueing systems, reliability analysis, communication networks, and machine learning applications.

Syllabus

Probability Foundations and Random Variables: Probability spaces and σ-algebras, conditional probability and conditional distributions, expectation, variance, covariance, higher moments, moment generating functions and characteristic functions, Law of Large Numbers, Central Limit Theorem, Monte Carlo estimation, reliability modeling, and empirical data distributions. Conditional Expectation and Convergence Concepts: Conditional expectation and its properties, tower property, law of total expectation, convergence in probability, convergence in distribution, almost sure convergence, applications in estimation theory and Bayesian inference, Bayesian Networks: structure, inference, and learning.

Discrete-Time Stochastic Processes: Definition and classification of stochastic processes, Markov property, transition probability matrices, Chapman–Kolmogorov equations, classification of states: recurrent, transient, and absorbing states, limiting distributions and stationary behavior, random walks and discrete-time martingales, applications to inventory systems, gambler's ruin, PageRank, queue length models, Hidden Markov Models (HMMs), filtering and prediction.

Continuous-Time Stochastic Processes: Poisson process, inter-arrival and waiting-time distributions, memoryless property, non-homogeneous Poisson processes, compound Poisson processes, continuous-time Markov chains (CTMCs), generator matrices, Kolmogorov forward and backward equations, birth-death processes, steady-state solutions, queueing models including M/M/1, M/M/c, M/G/1, and network queues, applications in communication systems, reliability engineering, and service systems.

Martingales and Stochastic Calculus: Discrete-time martingales, stopping times and optional stopping theorem, continuous-time martingales and quadratic variation, Brownian motion and Wiener processes, Itô integral, Itô's lemma, stochastic differential equations (SDEs), numerical methods for SDEs, stochastic control, applications in finance, signal processing, and machine learning.

Textbooks / References

1. S. M. Ross, *Introduction to Probability Models*, Academic Press.
2. P. G. Hoel, S. C. Port, and C. J. Stone, *Introduction to Stochastic Processes*, Houghton Mifflin.
3. V. K. Rohatgi and A. K. Md. Ehsanes Saleh, *An Introduction to Probability and Statistics*, Wiley.
4. G. Grimmett and D. Stirzaker, *Probability and Random Processes*, Oxford University Press.

5. R. Durrett, *Stochastic Calculus: A Practical Introduction*, CRC Press.
6. E. Çinlar, *Introduction to Stochastic Processes*, Prentice Hall.
7. C. W. Gardiner, *Handbook of Stochastic Methods*, Springer.
8. D. Barber, *Bayesian Reasoning and Machine Learning*, Cambridge University Press.
9. B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, Springer.
10. S. Karlin and H. M. Taylor, *A First Course in Stochastic Processes*, Academic Press.

MAEXXX Wavelet Theory and Applications [3-1-0-4]

Course Prerequisites

- MAC111 Calculus and Linear Algebra
- MAC211 Probability and Statistics.
- MAC213 Differential Equations and Integral Transforms

Course Objectives

- To provide a comprehensive introduction to wavelet theory and its mathematical foundations, emphasizing time-frequency analysis beyond classical Fourier methods.
- To develop an understanding of continuous and discrete wavelet transforms, multiresolution analysis, and wavelet-based signal representations.
- To study the construction and properties of important wavelet families and their implementation through filter banks and fast algorithms.
- To equip students with practical skills for applying wavelet techniques in signal processing, image compression, denoising, feature extraction, and data analysis.
- To enable students to implement and evaluate wavelet based methods using modern computational tools such as Python and MATLAB.

Course Outcomes

- CO1: Explain the limitations of the Fourier Transform for non-stationary signals and the motivation for time-frequency analysis.
- CO2: Formulate and compute the Continuous Wavelet Transform (CWT) and understand its properties.
- CO3: Describe the mathematical framework of Multiresolution Analysis (MRA) using scaling functions and wavelet functions.
- CO4: Implement the Discrete Wavelet Transform (DWT) and its inverse using filter banks and the Fast Wavelet Transform algorithm.
- CO5: Analyze and compare different wavelet families, such as Haar, Daubechies, and Coiflets, and select appropriate wavelets for specific tasks.
- CO6: Apply wavelet-based techniques to practical problems like signal denoising, data compression, and edge detection in images.

Syllabus

Review of Fourier Theory: The Fourier Transform and its properties. Limitations for analyzing non-stationary signals. Time-Frequency Analysis: The Short-Time Fourier Transform (STFT), the Gabor Transform, and the Heisenberg Uncertainty Principle. Introduction to Wavelets: The concept of a wavelet. Motivation for using variable-sized windows for signal analysis.

Definition of the CWT: Wavelet families, scaling, and translation. Admissibility Condition: The requirement for invertibility and the role of the zero-mean property. Properties and Interpretation: Linearity, covariance, and interpretation of the time-frequency plane (scalogram). Discretization of the CWT: Frame theory and wavelet series.

Multiresolution Analysis (MRA): Axioms of MRA. Nested approximation spaces. Scaling and Wavelet Functions: The scaling equation (dilation equation) and the wavelet equation. Filter Banks: Quadrature Mirror Filters (QMF), relationship between MRA and filter banks. The two-channel filter bank for DWT. The Discrete Wavelet Transform (DWT): The Fast Wavelet Transform (FWT) algorithm. Downsampling and upsampling. Perfect reconstruction. Wavelet Families: Construction and properties of Haar wavelets, Daubechies wavelets, Coiflets, and Symlets.

Image Compression: The 2D DWT. Application in the JPEG 2000 standard. Signal Denoising: Wavelet thresholding (hard and soft thresholding). Feature and Edge Detection: Identifying sharp changes and transients in signals and images.

Tutorial Sessions

The weekly 1-hour tutorial will focus on reinforcing concepts through problem-solving and guided coding exercises using Python (with libraries like PyWavelets and SciPy) or MATLAB.
Activities will include:

- Solving theoretical problems related to Fourier analysis, CWT, and MRA.
- Guided implementation of the Fast Wavelet Transform algorithm.
- Hands-on analysis of various signals using STFT and CWT.
- Comparing the performance and properties of different wavelet families through code.
- Step-by-step walkthroughs of applications like signal denoising and image compression.
- Discussion and progress tracking for a course mini-project where students apply wavelet techniques to a problem of their choice (e.g., analyzing financial data, compressing an audio signal, or detecting features in a biomedical signal).

Textbooks / References

1. Mallat, S., *A Wavelet Tour of Signal Processing: The Sparse Way*, 3rd Edition, Academic Press, 2008.
2. Daubechies, I., *Ten Lectures on Wavelets*, SIAM, 1992.
3. Strang, G., and Nguyen, T., *Wavelets and Filter Banks*, Wellesley-Cambridge Press, 1996.
4. Burrus, C. S., Gopinath, R. A., and Guo, H., *Introduction to Wavelets and Wavelet Transforms: A Primer*, Prentice Hall, 1998.
5. Chui, C. K., *An Introduction to Wavelets*, Vol. 1, Academic Press, 1992.
6. Hubbard, B. B., *The World According to Wavelets: The Story of a Mathematical Technique in the Making*, A K Peters/CRC Press, 1998.
7. Addison, P. S., *The Illustrated Wavelet Transform Handbook*, 2nd Edition, CRC Press, 2017.

MAEXXX Fuzzy Mathematics and Intelligent Systems [3-0-0-3]

Course Objectives

- To develop a strong foundation in fuzzy sets, fuzzy numbers, and fuzzy arithmetic for modelling uncertainty.
- To study ranking techniques and fuzzy decision-making methods for complex real-world problems.
- To introduce advanced fuzzy representations including intuitionistic, interval-valued intuitionistic, Type-2, hesitant, and circular fuzzy sets.
- To understand fuzzy optimization, fuzzy inference systems, and intelligent decision-support methodologies.
- To explore applications of fuzzy methodologies in artificial intelligence, machine learning, robotics, healthcare, finance, and data analytics.

Course Outcomes

- CO1: Explain fuzzy set concepts, fuzzy number representations, and arithmetic operations for modelling uncertainty.
- CO2: Analyze and compare ranking methods for fuzzy numbers and select suitable approaches for decision-making problems.
- CO3: Apply advanced fuzzy models including intuitionistic, interval-valued intuitionistic, Type-2, hesitant, and circular fuzzy sets.
- CO4: Apply fuzzy multi-criteria decision-making and fuzzy optimization techniques to solve complex decision problems.
- CO5: Analyze fuzzy inference systems, neuro-fuzzy systems, and intelligent decision-support frameworks.
- CO6: Evaluate real-world applications of fuzzy methodologies in artificial intelligence, machine learning, robotics, healthcare, finance, and data analytics.

Syllabus

Foundations of Fuzzy Mathematics: Fuzzy sets, membership functions, α-cuts, fuzzy numbers, triangular and trapezoidal fuzzy numbers, L–R fuzzy numbers, arithmetic operations on fuzzy numbers, fuzzy maximum and minimum operators, ordered fuzzy numbers, uncertainty modelling using fuzzy mathematics.

Ranking of Fuzzy Numbers: Need for ranking fuzzy numbers, classical ranking approaches, comparison function methods, ranking using left and right scores, centroid-based ranking methods, Chen's approach, Chen and Hwang's approach, Yager's centroid index, comparative analysis and selection criteria for ranking methods.

Advanced Fuzzy Models: Intuitionistic fuzzy sets, interval-valued intuitionistic fuzzy sets, Type-2 fuzzy sets, hesitant fuzzy sets, circular fuzzy sets, operations and representations of advanced fuzzy models, comparative study of fuzzy representations under uncertainty.

Fuzzy Decision Making and Optimization: Fuzzy multi-attribute decision making, fuzzy simple additive weighting methods, fuzzy AHP, fuzzy TOPSIS, fuzzy VIKOR, fuzzy DEMATEL, fuzzy linear programming, multi-objective fuzzy optimization, decision-support systems under uncertainty.

Intelligent Systems and Applications: Fuzzy inference systems, fuzzy rule-based systems, neuro-fuzzy systems, ANFIS, fuzzy clustering, explainable AI using fuzzy logic, applications in artificial intelligence, machine learning, robotics, healthcare analytics, financial decision making, recommender systems, and data analytics.

Textbooks / References

1. G. J. Klir and B. Yuan, *Fuzzy Sets and Fuzzy Logic: Theory and Applications*, Prentice Hall, 1995.
2. T. J. Ross, *Fuzzy Logic with Engineering Applications*, 3rd ed., Wiley, 2010.
3. C. Kahraman (Ed.), *Fuzzy Multi-Criteria Decision Making: Methods and Applications*, Springer, 2008.
4. H.J. Zimmermann, *Fuzzy Set Theory and Its Applications*, 4th ed., Springer, 2001.
5. S. J. Chen and C. L. Hwang, *Fuzzy Multiple Attribute Decision Making*, Springer-Verlag, 1992.
6. D. Dubois and H. Prade, *Fuzzy Sets and Systems: Theory and Applications*, Academic Press.
7. G. Beliakov, A. Pradera, and T. Calvo, *Aggregation Functions: A Guide for Practitioners*, Springer.
8. M. Delgado, J. L. Verdegay, and M. A. Vila, *Fuzzy Optimization: Recent Advances*, Physica-Verlag.
9. J. M. Mendel, *Uncertain Rule-Based Fuzzy Systems: Introduction and New Directions*, Springer.
10. W. Pedrycz and F. Gomide, *Fuzzy Systems Engineering*, Wiley.
11. A. Kandel and G. Langholz, *Fuzzy Control Systems*, CRC Press.
12. S. N. Sivanandam and S. N. Deepa, *Principles of Soft Computing*, Wiley.
13. J. S. R. Jang, C.T. Sun, and E. Mizutani, *Neuro-Fuzzy and Soft Computing*, Prentice Hall.

14. O. Castillo and P. Melin, *Type-2 Fuzzy Logic: Theory and Applications*, Springer.
15. J. Kacprzyk and W. Pedrycz (Eds.), *Springer Handbook of Computational Intelligence*, Springer.

MAEXXX Operations Research [3-0-0-3]

Course Prerequisite

- Prerequisite course MAC321 Linear Programming and Combinatorial Optimisation

Course Objectives

- To introduce the fundamental concepts, models, and methodologies of Operations Research for scientific decision making.
- To develop mathematical modeling skills for optimization problems arising under certainty and uncertainty.
- To apply classical and modern algorithms for linear, integer, dynamic, and stochastic optimization problems.
- To analyze structured optimization models such as transportation, assignment, queuing, and game-theoretic models.
- To demonstrate the application of Operations Research techniques in logistics, manufacturing, healthcare, finance, supply chains, and service systems.

Course Outcomes

- CO1: Formulate real-world decision-making problems as mathematical optimization models including linear, integer, and nonlinear programming formulations.
- CO2: Analyze the geometric and algebraic structure of optimization problems using concepts such as convexity, feasible regions, and polyhedra.
- CO3: Apply simplex-based methods, duality theory, and sensitivity analysis to solve and interpret linear programming problems.
- CO4: Implement algorithms for integer programming, dynamic programming, and combinatorial optimization problems.
- CO5: Solve practical problems involving transportation, assignment, queuing, and game theory.
- CO6: Evaluate and compare exact, approximation, and heuristic approaches for complex optimization and decision making problems.

Syllabus

Introduction to Operations Research: History and development of Operations Research, scope and applications, characteristics of OR models, phases of an OR study, decision making under certainty and uncertainty, mathematical modeling framework, optimization concepts, applications in industry, business, transportation, healthcare, and public systems.

Integer and Nonlinear Programming: Integer programming formulations, branch-and-bound method, cutting-plane methods, mixed-integer programming models, binary programming, applications of integer programming, introduction to nonlinear programming, unconstrained and constrained optimization problems, Kuhn–Tucker conditions, applications in operations research.

Transportation and Assignment Problems: Transportation problem formulation, balanced and unbalanced transportation problems, North-West Corner Method (NWCM), Least Cost Entry Method (LCEM), Vogel's Approximation Method (VAM), optimality tests using the u–v method, dual interpretation of transportation problems, assignment problem formulation, Hungarian method, variations and applications.

Dynamic Programming: Principle of optimality, recursive equations and Bellman formulation, deterministic and stochastic dynamic programming, resource allocation problems, shortest path problems, knapsack problems, inventory control models, stage-wise optimization and applications.

Introduction to game theory: Basic overview

Queueing Theory: Elements of queueing systems, arrival and service processes, Kendall's notation, steady-state analysis of Markovian queueing models, M/M/1 queue, M/M/1 with limited waiting space, M/M/c systems, M/M/c with limited capacity, M/G/1 models, performance measures, waiting-time distributions, applications in communication networks and service systems.

Textbooks / References

1. A. Ravindran, D. T. Phillips, and J. J. Solberg, *Operations Research: Principles and Practice*, 2nd Edition, John Wiley & Sons, 2012.
2. H. A. Taha, *Operations Research: An Introduction*, 10th Edition, Pearson Education, 2018.
3. M. S. Bazaraa, J. J. Jarvis, and H. D. Sherali, *Linear Programming and Network Flows*, 4th Edition, John Wiley & Sons, 2013.
4. F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, McGraw-Hill, 10th Edition.
5. B. E. Gillett, *Introduction to Operations Research: A Computer-Oriented Algorithmic Approach*, McGraw-Hill, 1989.
6. W. L. Winston, *Operations Research: Applications and Algorithms*, Cengage Learning.
7. M. S. Bazaraa, H. D. Sherali, and C. M. Shetty, *Nonlinear Programming: Theory and Algorithms*, Wiley.
8. D. P. Bertsekas, *Dynamic Programming and Optimal Control*, Athena Scientific.
9. R. W. Conway, W. L. Maxwell, and L. W. Miller, *Theory of Scheduling*, Dover Publications.
10. R. Panneerselvam, *Operations Research*, PHI Learning.

MAEXXX Introduction to Dynamical Systems Theory [3-0-0-3]

Course Objectives

- To introduce the fundamental concepts of nonlinear science and dynamical systems theory.
- To provide an understanding of nonlinear behavior in physical, biological, engineering, and computational systems.
- To develop the ability to analyze flows, fixed points, stability, and bifurcations in one- and two-dimensional dynamical systems.
- To explore mathematical tools such as phase space analysis, attractors, Lyapunov exponents, invariant manifolds, and bifurcation theory.
- To familiarize students with chaos theory, Hamiltonian systems, limit cycles, and discrete dynamical systems.

Course Outcomes

- CO1: Explain the fundamental concepts of dynamical systems, nonlinear phenomena, attractors, phase space, and deterministic versus stochastic dynamics.
- CO2: Analyze periodic points, stable manifolds, Lyapunov exponents, and characterize chaotic behavior in nonlinear dynamical systems.
- CO3: Apply linear stability analysis to determine the stability of fixed points in one-dimensional and two-dimensional systems.
- CO4: Identify, classify, and analyze local bifurcations including saddle-node, transcritical, pitchfork, and Hopf bifurcations.
- CO5: Understand Hamiltonian systems, conservative dynamics, area-preserving maps, and limit-cycle behavior in nonlinear systems.
- CO6: Model and analyze discrete dynamical systems, evaluate stability properties, and investigate the emergence of complex and chaotic dynamics.

Syllabus

Fundamentals of Nonlinear Science and Dynamical Systems: Dynamics and dynamical systems, classification of dynamical systems, nonlinearity and nonlinear phenomena, dissipative and conservative systems, deterministic versus stochastic systems, degrees of freedom, state space and phase space, trajectories, invariant sets, attractors, fixed points, periodic orbits, examples from physical, biological, and engineering systems.

Area Preservation, Stability, and Chaos: Periodic points and their classification, area-preserving maps, stable and unstable manifolds, Stable Manifold Theorem, homoclinic points and homoclinic intersections, construction of stable manifolds, Lyapunov exponents, random matrix interpretations, sensitivity to initial conditions, definitions and mathematical characterization of chaos, strange attractors, and chaotic dynamics.

One-Dimensional Flows and Hamiltonian Systems: One-dimensional autonomous systems, fixed points and their stability, linear stability analysis, bifurcation theory, saddle-node bifurcation, transcritical bifurcation, pitchfork bifurcation, flows on the circle, periodic solutions, Hamiltonian systems, phase portraits of Hamiltonian systems, conservation laws, and elements of Hamiltonian mechanics.

Two-Dimensional Flows and Discrete Dynamical Systems: Linear systems and phase-plane analysis, nonlinear autonomous systems, phase portraits, fixed points and linearization, conservative systems, index theory, limit cycles, Poincaré–Bendixson theorem, Liénard systems, local bifurcations including saddle-node, transcritical, pitchfork, and Hopf bifurcations, introduction to discrete dynamical systems, iterated maps, stability of fixed and periodic points, routes to chaos, and applications of discrete models.

Textbooks / References

1. S. Wiggins, *Introduction to Applied Nonlinear Dynamical Systems and Chaos*, Springer-Verlag, 2003.
2. K. Alligood, T. Sauer, and J. A. Yorke, *Chaos: An Introduction to Dynamical Systems*, Springer-Verlag, 1996.
3. M. W. Hirsch, S. Smale, and R. L. Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, Academic Press, 2012.
4. J. T. Sandefur, *Discrete Dynamical Systems: Theory and Applications*, Clarendon Press, 1990.
5. M. Han, *Bifurcation Theory and Methods of Dynamical Systems*, World Scientific, 1997.
6. F. Diacu and P. Holmes, *Celestial Encounters: The Origins of Chaos and Stability*, Princeton University Press, 2009.
7. B. Hasselblatt and A. Katok, *A First Course in Dynamics: With a Panorama of Recent Developments*, Cambridge University Press, 2003.
8. S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, CRC Press.
9. J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer.
10. E. Ott, *Chaos in Dynamical Systems*, Cambridge University Press.

MAEXXX Mathematical and Computational Finance [3-1-0-4]

Pre-Requisites

- Knowledge of multivariable calculus, differential equations, and basic probability theory.
- Familiarity with programming in MATLAB, Python, or C++.

Course Objectives

- To introduce the mathematical foundations of modern financial markets and derivative pricing.
- To develop understanding of probabilistic models and stochastic processes used in asset pricing.
- To equip students with computational and numerical techniques for solving finance-related PDEs and SDEs.
- To explore applications of mathematical models in portfolio optimization, option pricing, and risk management.

Course Outcomes

- CO1: Explain and apply the concepts of time value of money, arbitrage, and portfolio optimization.
- CO2: Formulate and analyze stochastic models governing asset prices, including Brownian motion and Itô calculus.
- CO3: Derive and implement the Black-Scholes-Merton (BSM) model and its extensions for pricing derivatives.
- CO4: Compute Greeks, implied volatility, and analyze option moneyness and spread strategies.
- CO5: Apply numerical methods such as Monte Carlo simulation, binomial/trinomial trees, and finite difference schemes for option valuation.
- CO6: Understand basics of risk measurement, credit risk modeling, and stochastic volatility models.

Syllabus

Introduction to Financial Markets and Portfolio Theory: Time value of money, arbitrage, and no-arbitrage principles, forwards, futures, swaps, and options, Introduction to portfolio theory, mean-variance optimization, efficient frontier.

Probability Models and Stochastic Processes: Random walk, martingale, and introduction to Lévy processes/jumps, Brownian motion, Itô's lemma, and stochastic differential equations (SDEs) for asset pricing.

Option Pricing Theory: Black-Scholes-Merton (BSM) model assumptions and derivations, European options, Greeks, and put-call parity, Implied volatility: definition, computation, and the volatility smile/skew. Option moneyness: At-the-money (ATM), in-the-money (ITM), and out-of-the-money (OTM) options; spreads and their role in trading/hedging strategies. Exotic options: Asian, barrier, American via Longstaff-Schwartz, term structure models, Vasicek, Cox-Ingersoll-Ross (CIR).

Numerical Methods in Finance: Binomial and trinomial tree models for European/American options, Monte Carlo simulation for pricing and Greeks, variance reduction techniques like Quasi-Monte Carlo. Generating sample paths, discretization of SDE, Finite difference methods for Black-Scholes PDEs, explicit/implicit schemes, Crank-Nicolson; overview of Fourier methods for option pricing.

Risk Management and Credit Risk: Hedging, replication, and dynamic programming basics, Risk measures: Value at Risk (VaR), Expected Shortfall; introduction to credit risk models, Greeks and sensitivity analysis; stochastic volatility, Heston model.

Textbooks / References

1. J. C. Hull, *Options, Futures and Other Derivatives*, Pearson.
2. S. Shreve, *Stochastic Calculus for Finance I & II*, Springer.
3. M. Baxter and A. Rennie, *Financial Calculus: An Introduction to Derivative Pricing*, Cambridge University Press.
4. P. Wilmott, S. Howison and J. Dewynne, *The Mathematics of Financial Derivatives: A Student Introduction*, Cambridge University Press, 1997.
5. A. Hirsa and S. N. Neftci, *An Introduction to the Mathematics of Financial Derivatives*, Academic Press, 2013.
6. P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer.
7. R. Seydel, *Tools for Computational Finance*, Springer.
8. D. J. Higham, *An Introduction to Financial Option Valuation: Mathematics, Stochastics and Computation*, Cambridge University Press.
9. B. Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, Springer.
10. Manfred Gilli, Dietmar Maringer, and Enrico Schumann, *Numerical Methods and Optimization in Finance*, Elsevier, 2011.

MAEXXX Computational Fluid Dynamics [3-1-0-4]

Pre-Requisites

- ODE and PDE.

Course Objectives

- To introduce the fundamental concepts of mass, momentum, and energy transfer with governing conservation equations in different coordinate systems.
- Familiarize students with mathematical modeling using the continuity, momentum, energy, Euler and Navier–Stokes equations.
- To develop knowledge of numerical solution methods, including central schemes, explicit, and implicit techniques.
- Introduce grid generation techniques (structured and unstructured) and discretization methods for fluid flow and heat transfer problems.
- Build computational skills for solving algebraic systems using direct and iterative methods and analyzing numerical errors and stability.

Course Outcomes

- CO1: Explain the principles of mass, momentum, and energy conservation and apply them to derive governing equations in different coordinate systems.
- CO2: Implement numerical schemes, including central, explicit, implicit, and PISO methods for discretized fluid flow equations.
- CO3: Construct and use structured and unstructured computational grids to solve transport problems.

- CO4: Apply finite difference, finite volume and finite element discretization methods for convection-diffusion and boundary-value problems.
- CO5: Model and solve linear and nonlinear systems of equations using direct and iterative numerical techniques.

Syllabus

Basics of mass and heat transfer, conservation equations for mass, momentum, energy, coordinate systems, dimensionless analysis, different types of fluid flows, Boussinesq approximation, Boundary layer approximation.

Control Volume Concept, Substantial derivative, continuous equation, Momentum Equation, Energy Equation, Navier–Stokes Equation, Euler Equation.

Central schemes with combined and independent space time discretisation, Explicit and implicit methods; Grid generation: Structured and unstructured grid generation methods, Discretization techniques using finite difference methods, Errors and Analysis of Stability.

Finite volume method basics, Control volume approach, structured and unstructured grids discretization technique, diffusion and convection discretisation, boundary conditions, Solution methodology for linear and non-linear problems.

Finite element basics, discretization, finite element model interpretation functions, time dependent problems, axisymmetric problems, numerical integration, Solution of algebraic systems (direct and iterative basics).

Textbooks / References

1. P. Wessling, *Principles of Computational Fluid Dynamics*, Springer, 1991.
2. John D. Anderson Jr., *Computational Fluid Dynamics: The Basics with Applications*, McGraw-Hill, 1995.
3. Alexandre J. Chorin and Jerrold E. Marsden, *A Mathematical Introduction to Fluid Mechanics*, Third Edition, Springer, 2013.
4. G. D. Smith, *Numerical Solution of Partial Differential Equations: Finite Difference Methods*, Oxford University Press, 1985.
5. J. H. Ferziger and M. Perić, *Computational Methods for Fluid Dynamics*, Second Edition, Springer, Berlin, 1999.
6. H. K. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, Second Edition, 2007.
7. J. N. Reddy and D. K. Gartling, *The Finite Element Method in Heat Transfer and Fluid Dynamics*, Third Edition, 2010.

MAEXXX Statistical Learning [3-1-0-4]

Course Prerequisites

MAC111 Calculus and Linear Algebra, MAC 123 Real and Complex Analysis

Course Objectives

- To review and reinforce the foundational concepts of probability and random variables essential for statistical learning.
- To introduce the concepts of bias, variance, and mean squared error, and to compare various estimation models through case studies.
- To develop a thorough understanding of linear and polynomial regression models, regularization techniques such as Ridge and Lasso, and classification methods including Logistic Regression and Softmax Regression.
- To introduce multivariate data structures, classical multivariable linear models, and the Gauss–Markov theorem with its applications.
- To familiarize students with kernel methods, support vector machines, dimensionality reduction techniques such as PCA and Kernel PCA, and ensemble learning approaches such as Boosting.

Course Outcomes

- CO1: Recall and apply fundamental concepts of probability, random variables, distributions, expectation, and moment generating functions.
- CO2: Analyze and compare the bias–variance tradeoff across competing estimation and regression models using real-life case studies.
- CO3: Formulate and solve linear, polynomial, ridge, and lasso regression problems, and apply logistic and softmax regression for classification tasks.
- CO4: Apply multivariate linear models, state and use the Gauss–Markov theorem, and implement kernel-based learning methods.
- CO5: Implement and interpret SVMs, perform dimensionality reduction using PCA and Kernel PCA, and understand the principles of boosting algorithms.

Syllabus

Introduction to statistical learning; Statistical learning with single population, Illustration with applications to reliability analysis, Statistical learning with more than one population, ANOVA techniques, PAC learning, Empirical risk minimization.

Bias, variance and mean squared error: decomposition of MSE into bias and variance components, real-life examples.

Regression models: linear regression, polynomial regression, ridge regression, lasso regression; least squares solution for classification; logistic regression and K-class softmax regression.

Kernel methods: positive semi-definite kernels, kernel ridge regression, support vector machines, and linear classification.

Dimensionality reduction: principal component analysis (PCA) and kernel PCA.

Textbooks / References

1. Rohatgi, V. K., and A. K. Saleh. An Introduction to Probability and Statistics, Third Edition. Wiley Student Edition, 2006.
2. Ross, S. Introduction to Probability and Statistics for Engineers and Scientists, Third Edition. Elsevier, 2004.
3. Grinstead, C. M., and J. L. Snell. Introduction to Probability, Second Edition. Universities Press India, 2009.
4. Bishop, Christopher M., and Nasser M. Nasrabadi. Pattern Recognition and Machine Learning, Vol. 4, No. 4. Springer, New York, 2006.
5. Alpaydin, Ethem. Introduction to Machine Learning. MIT Press, 2020.
6. Hastie, Trevor, Robert Tibshirani, and Jerome H. Friedman. The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Vol. 2. Springer, New York, 2009.
7. Hastie, Trevor, Robert Tibshirani, and Jerome Friedman. An Introduction to Statistical Learning. Springer, 2009.

MAEXXX Number Theory and Mathematical Theory of Coding [3-0-0-3]

Course Objectives

- To understand the basics of number theory, theory of congruences, and modular arithmetic and to apply the same in data security.
- To analyze various error correction codes.

Course Outcomes

- Apply the basics of number theory in different scenarios.
- Learn different error-correcting codes.

Syllabus:

Introduction to Number Theory: Theory of congruences and applications, Divisibility theory - division algorithm, the greatest common divisor, the Euclidean algorithm, the Diophantine equation, Primes and their distribution, The fundamental theorem of arithmetic (no proof) and applications.

The Theory of Congruences: Binary and decimal representation of integers, Congruences in one unknown, Chinese remainder theorem, Fermat's theorem, Pseudo primes, Fermat factorization.

Number Theoretic Functions: Numbers of special form, Number theoretic functions - the sum and number of divisors, the greatest integer function, an application to the calendar, Euler's phi function, Euler's theorem.

Numbers of Special Form: Perfect numbers, Fermat numbers, Legendre symbol, Jacobi symbol. Algebraic Structures: Finite fields, The order of integer modulo, Primitive roots, Discrete logarithms, Quadratic residue, Quadratic reciprocity law, Primality testing. Introduction to Groups, Rings, and Fields: Polynomial arithmetic, Finite fields.

Introduction to Coding Theory: Error-correcting codes, Basic concepts and definitions, Encoding and decoding, Bounds on general codes, Linear codes, Syndrome decoding, The dual code, Hamming codes, Cyclic codes, Reed-Solomon codes.

Textbooks/References

1. D. A. Burton, Elementary Number Theory, 6/e, Tata McGraw Hill, 2007.
2. Stallings W., Cryptography and Network Security: Principles and Practice, 4/e, Pearson Education Asia, 2006.
3. Wade Trappe, Lawrence C. Washington, Introduction to Cryptography with Coding Theory, Second Edition.
4. Niven I., Zuckerman H. S., and Montgomery H. L., An Introduction to Theory of Numbers, 5/e, John Wiley and Sons, 2004.
5. Joseph A. Gallian, Contemporary Abstract Algebra, Narosa, 1998.

MAEXXX Advanced Cryptography for Cyber Security [3-1-0-4]

Course Objectives

- To lay a foundation on Security in Networks, Classical Cryptosystem, and Block Cipher Modes of Operation.
- To analyse various Private and Public key Cryptosystems for encryption, key exchange, hashing, and authentication protocols.
- To acquire fundamental knowledge on applications of cryptography.

Course Outcomes

- CO1: Understand the fundamental concepts of Classical and modern Cryptosystems.
- CO2: Compare various private and public key Cryptosystems for encryption, key exchange, and authentication algorithms.
- CO3: Understand the different applications of cryptography.

Syllabus

Introduction: Cryptography, cryptanalysis, cryptology, Classical Cryptosystem- Shift cipher, Affine cipher, Vigenère cipher, Substitution, Transposition techniques.

Block Ciphers and Modes of Operations: Data Encryption Standard (DES)- Block cipher principles- Block cipher modes of operation, AES- Triple DES- Blowfish- RC5.

Public Key Cryptography: Public Key Cryptosystem, Key distribution, Diffie-Hellman Key Exchange, Man-in-the Middle (MITM) Attack, RSA, Random Number Generation, Elliptic Curve Cryptography (ECC), Key Management.

Hash Functions and Digital Signatures: Authentication requirement, Authentication function, Message Authentication Code (MAC), Hash functions, SHA, HMAC, Digital signature and authentication protocols.

Applications: Authentication– Kerberos, IP Security IPSec, Web Security- SSL, TLS, Blockchain, IoT Security.

Textbooks / References

1. William Stallings, *Cryptography and Network Security*, 6th Edition, Pearson Education.
2. Behrouz A. Forouzan and Debdeep Mukhopadhyay, *Cryptography and Network Security*, 5th Edition, McGraw Hill Education.
3. Rich Helton and Johennie Helton, *Mastering Java Security: Cryptography Algorithms and Practices*, John Wiley Publishers.
4. Charles P. Pfleeger, *Security in Computing*, 5th Edition, Pearson Education Asia.
5. William Stallings, *Network Security Essentials: Applications and Standards*, Pearson Education Asia.
6. Charlie Kaufman, Radia Perlman and Mike Speciner, *Network Security: Private Communication in a Public World*, 6th Edition, Prentice Hall India.

MAEXXX Computational Neuroscience [3-0-0-3]

Course Prerequisites

Basic biology, chemistry, and physics, Differential equations, linear algebra, MATLAB.

Course Objectives

- To introduce mathematical and computational models of neurons and neural circuits.
- To develop skills in dynamical systems theory for analyzing neuronal excitability and network behavior.
- To familiarize students with cable theory, compartmental modeling, and numerical integration methods for neural systems.
- To expose students to models of synaptic plasticity, learning, and brain systems, and their relationship to artificial neural networks.
- To build proficiency in simulation tools such as MATLAB, NEURON, and XPP for analyzing neural models.

Course Outcomes

- CO1: Formulate mathematical models of single neurons and neural circuits based on biophysical principles.
- CO2: Analyze neuronal dynamics using dynamical systems concepts (phase planes, bifurcations, stability).
- CO3: Apply cable theory and compartmental modeling techniques to simulate dendritic and axonal signal propagation.
- CO4: Implement and simulate Hodgkin-Huxley type and reduced neuron models using MATLAB, NEURON, and XPP.
- CO5: Evaluate models of synaptic plasticity, learning, and network synchronization.
- CO6: Design, implement, and present an independent computational neuroscience project based on a model from the literature.

Syllabus

Electrophysiology of Nerve Cells and the Hodgkin Huxley Model: Basics of neuronal electrophysiology, ion channels and membrane currents, Hodgkin-Huxley model for the action potential, voltage-clamp experiments, generation and propagation of action potentials. Neurons as Dynamical Systems: Analysis of model neurons using dynamical systems theory, phase plane analysis, bifurcations and excitability, conductance-based neural models, simplified (reduced) neural models such as integrate-and-fire and FitzHugh-Nagumo. Cable Theory and Compartmental Modeling: Passive and active compartmental modeling, Kirchhoff's current and voltage laws, cable theory, time and space constants, numerical integration methods for neural modeling, use of the NEURON simulation environment.

Network Dynamics and Learning: Synchronization of coupled model neurons, central pattern generators, models of synaptic plasticity and learning, quantification of spike train variability and firing rate codes. Sensory Processing and Neural Networks: Early vision and simple/complex cells in visual cortex, model reduction techniques, relationship between biological neural models and artificial neural networks, applications and case studies in computational and cognitive neuroscience.

Textbooks / References

1. P. Dayan and L. F. Abbott, *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*, MIT Press, 2001.
2. F. Gabbiani and S. J. Cox, *Mathematics for Neuroscientists*, 2nd ed., Academic Press, 2017.
3. G. B. Ermentrout and D. H. Terman, *Mathematical Foundations of Neuroscience*, Springer, 2010.
4. G. B. Ermentrout, *Simulating, Analyzing, and Animating Dynamical Systems: A Guide to XPPAUT for Researchers and Students*, SIAM, 2002.
5. N. T. Carnevale and M. L. Hines, *The NEURON Book*, Cambridge University Press, 2006.
6. W. Gerstner, W. M. Kistler, R. Naud, and L. Paninski, *Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition*, Cambridge University Press, 2014.
7. E. M. Izhikevich, *Dynamical Systems in Neuroscience*, MIT Press, 2007.
8. H. R. Wilson, *Spikes, Decisions, and Actions: Dynamical Foundations of Neuroscience*, Oxford University Press, 1999.
9. P. Miller, *An Introductory Course in Computational Neuroscience*, MIT Press, 2018.

10. C. Koch, *Biophysics of Computation: Information Processing in Single Neurons*, Oxford University Press, 1999.
11. C. Koch and I. Segev, Eds., *Methods in Neuronal Modeling: From Ions to Networks*, MIT Press, 1998.
12. S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, CRC Press, 2018.
13. Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, Springer, 2004.

CAE410 Scientific Machine Learning [3-0-0-3]

Course Prerequisites:

Python Programming, Fundamentals of Machine Learning, Probability, Statistics and Linear Algebra, Calculus and Numerical Methods

Course Objectives

- To introduce the algorithmic and computational foundations of Scientific Machine Learning.
- To understand the role of scientific computing and differential equations in machine learning-based modelling.
- To study automatic differentiation and optimization techniques used in scientific machine learning.
- To explore Physics-Informed Neural Networks and Physics Informed Gaussian Processes for solving scientific and engineering problems.

Course Outcomes

- CO1: Upon successful completion of the course, students will be able to:
- CO2: Understand the algorithmic and computational foundations of Scientific Machine Learning.
- CO3: Apply basic scientific computing techniques for modelling systems governed by differential equations.
- CO4: Understand automatic differentiation and its role in optimization and neural network training.
- CO5: Analyse Physics-Informed Neural Networks and Physics-Informed Gaussian Processes for solving scientific and engineering problems.

Syllabus

Introduction to Scientific Machine Learning. Scientific laws, mathematical models, and governing equations. Review of Machine Learning. Least-squares parametric regression using linear and neural network models. Non-parametric regression using Gaussian Processes. Applications of Scientific Machine Learning in science and engineering.

Basic notions of Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs). Initial and boundary value problems. Numerical solutions using classical discretization schemes. Introduction to scientific computing for differential equation-based modelling.

Introduction to automatic differentiation. Forward-mode and reverse-mode automatic differentiation. Computational graphs and gradient computation. Backpropagation and optimization in neural network training.

Physics-Informed Neural Networks (PINNs). PINN architecture and training methodology. Initialization, gradient-based optimization, adaptive learning rates, learning rate scheduling, and second-order optimization methods. Self-Adaptive PINNs. Forward and inverse problem modelling using PINNs.

Physics-Informed Gaussian Processes (PIGPs). Introduction to Reproducing Kernel Hilbert Spaces (RKHS). Solving differential equations using Gaussian Processes. Gaussian Processes as optimal recovery in RKHS. Comparison of PINNs and PIGPs for scientific modelling.

Textbooks / References

1. Baker et al., *Workshop report on Basic Research needs for scientific machine learning: Core Technologies for Artificial Intelligence*, 2019.
2. Raissi, M., Perdikaris, P., and Karniadakis, G. E., *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving non-linear partial differential equations*, Journal of Computational Physics, 2019.
3. Yifan et al., *Solving and learning nonlinear PDEs with Gaussian processes, Journal of Computational Physics*, 2021.
4. Iserles, A., *A first course in the numerical analysis of differential equations*, Cambridge University Press, 2009.
5. Boyd, J. P., *Chebyshev and Fourier Spectral Methods*, Second Revised Edition, Dover Publications, 2013.
6. Griewank, A. and Walther, A., *Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation*, SIAM, 2008.

OPEN ELECTIVE COURSES

CSO003 Quantum Computing for Engineers [2-1-0-3]

Course Prerequisites

Linear Algebra, Probability and Statistics, Basic Programming, and familiarity with Discrete Mathematics.

Course Objectives

- Introduce the requisite mathematical concepts that are relevant to computing.
- Explore the core concepts that distinguish quantum computation from classical computation.
- Examine various quantum computing models and algorithms.

- Develop knowledge of quantum gates, circuits, and error correction techniques.
- Familiarize students with quantum programming using modern tools for practical implementation.

Course Outcomes

After successful completion of the course, students should be able to:

- CO1: Understand the fundamental principles of mathematics relevant to computing.
- CO2: Explain the basic concepts of quantum computing
- CO3: Demonstrate quantum computing models and algorithms.
- CO4: Examine quantum gates, circuits, and error correction techniques.
- CO5: Formulate quantum inspired algorithms to solve computationally challenging problems.

Syllabus

Complex numbers, Tensor products, Unitary and Hermitian matrices, Introduction to Quantum Computing: Classical vs. Quantum Computing, Mathematical representations for Quantum Computing, Quantum Bits (Qubits) and Quantum Registers, Quantum Parallelism and Quantum Speedup, No-Cloning Theorem, Measurement, Quantum Decoherence and Noise, Platforms for Exploring Quantum Computing (Classiq, IBM Quantum, Google Quantum)

Quantum Gates and Circuits: Quantum Gates: Pauli, Hadamard, CNOT, Phase, and Toffoli Gates, Quantum Circuits and Their Representations, Reversible and Universal Gate Sets, Quantum Error Correction Techniques, Quantum Teleportation and Superdense Coding

Quantum Algorithms and Programming: Introduction to Quantum Algorithms, Deutsch Josza algorithm, Quantum Fourier Transform, Phase Estimation, Shor's Algorithm for Factorization, Grover's Algorithm, Amplitude Amplification, Quantum Random walk, Variational Quantum Algorithms

Introduction to Quantum Programming Languages Qiskit, Cirq, PennyLane, etc., Quantum Software Engineering and Applications: Quantum Software Development Life Cycle, Hardware Implementations and Current Limitations, Future Trends in Quantum Computing.

Textbooks / References

1. Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008.
2. Hiu Yung Wong, Introduction to quantum computing: from a layperson to a programmer in 30 steps. Springer Nature, 2023.
3. Thomas G. Wong, Introduction to classical and quantum computing. Omaha, NE, USA: Rooted Grove, 2022.
4. Robert S. Sutor, Dancing with Qubits: How quantum computing works and how it can change the world. Packt Publishing Ltd, 2019.
5. Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010.
6. Robert Loredo, Learn Quantum Computing with Python and IBM Quantum Experience: A hands-on introduction to quantum computing and writing your own quantum programs with Python. Packt Publishing Ltd, 2020.
7. Willi-Hans Steeb, and Yorick Hardy. Problems and solutions in quantum computing and quantum information. World Scientific Publishing Company, 2018.
8. Anshul Saxena, Javier Mancilla, Iraitz Montalban, and Christophe Pere. Financial Modeling Using Quantum Computing: Design and manage quantum machine learning solutions for financial analysis and decision making. Packt Publishing Ltd, 2023.

CSO006 Internet of Things: Concepts and Applications [3-0-0-3]

Course Objectives

- To understand the architecture, ecosystem, and enabling technologies of IoT systems.
- To study sensing, communication, and service integration mechanisms in IoT environments.
- To understand IoT protocols, middleware platforms, and data management frameworks.
- To analyze security, privacy, and interoperability challenges in IoT deployments.
- To examine real-world IoT applications and deployment strategies.
- To develop the ability to design and evaluate IoT-enabled solutions.

Course Outcomes

Upon successful completion of the course, students will be able to:

- CO1: Explain IoT architectures, enabling technologies, and ecosystem components.
- CO2: Analyze sensing, communication, and interoperability requirements in IoT systems.
- CO3: Evaluate IoT communication protocols and platform technologies.
- CO4: Analyze data management, security, and privacy challenges in IoT deployments.
- CO5: Design IoT-based solutions for application-specific environments.
- CO6: Evaluate emerging trends and future directions of IoT technologies.

Syllabus

Foundations of the Internet of Things: Evolution of Internet of Things, Characteristics of IoT systems, IoT ecosystem and stakeholders, Enabling technologies for IoT, Machine-to-Machine (M2M) communication, IoT World Forum (IoTWF) Reference Architecture, Simplified IoT

Architecture, Core IoT Functional Stack, Functional blocks of an IoT ecosystem, Smart objects and connected environments, Sensors and actuators, Introduction to Fog, Edge, and Cloud paradigms in IoT.

IoT Connectivity and Communication Protocols: Communication requirements in IoT systems, Constrained devices and constrained networks, IEEE 802.15.4 overview, IEEE 802.11ah overview, LoRaWAN architecture and applications, IPv6 for IoT, 6LoWPAN concepts and architecture, Application layer protocols: MQTT and CoAP, SCADA systems in industrial environments, Protocol selection considerations for IoT deployments.

IoT System Design and Development: IoT design methodology, IoT system building blocks, Device integration and deployment workflow, Fundamentals of embedded computing for IoT, Overview of microcontrollers and System-on-Chip (SoC) platforms, IoT hardware ecosystem, Raspberry Pi platform overview, Arduino platform overview, Design considerations for scalable IoT systems.

IoT Data Management, Security and Supporting Services: IoT data lifecycle, Structured and unstructured data, Data in motion and data at rest, Data acquisition and organization in IoT environments, IoT data analytics challenges, Data visualization concepts, Middleware and supporting services, Cloud-enabled IoT applications, Security and privacy challenges in IoT environments, Identity and trust management, Sustainable IoT deployment considerations, Responsible and ethical IoT systems.

IoT Applications and Emerging Trends: Smart homes and smart appliances, Smart buildings and infrastructure, Smart healthcare systems, Smart agriculture, Environmental monitoring, Industrial IoT (IIoT), Industry 4.0 and 5.0 concepts, Smart energy systems, Digital twins, Human-centered IoT applications, Sustainable IoT systems, Future trends and research challenges in IoT.

Textbooks / References

1. Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Introduction to IoT, Cambridge University Press, 2021.
2. Arshdeep Bahga and Vijay Madisetti, Internet of Things: A Hands-on Approach, Universities Press, 2014.
3. Olivier Hersent, David Boswarthick, and Omar Elloumi, The Internet of Things: Key Applications and Protocols, Wiley, 2011.
4. Adrian McEwen and Hakim Cassimally, Designing the Internet of Things, Wiley, 2013.
5. Pethuru Raj and Anupama Raman, The Internet of Things: Enabling Technologies, Platforms and Use Cases, CRC Press, 2017.
6. Honbo Zhou, The Internet of Things in the Cloud: A Middleware Perspective, CRC Press, 2013.

CAO002 Deep Learning [3-0-0-3]

Course Prerequisites

- Machine Learning
- Basic Linear Algebra, Probability and Statistics
- Programming using Python and familiarity with scientific computing libraries

Course Objectives

- To provide a comprehensive understanding of the theoretical foundations and practical aspects of deep learning.
- To study artificial neural networks, deep neural networks, and modern optimization techniques used in deep learning.
- To understand the architecture and applications of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and attention-based models.
- To explore deep generative models and unsupervised learning techniques.
- To develop the ability to implement deep learning solutions using modern frameworks such as TensorFlow and PyTorch for real-world applications.

Course Outcomes

- CO1: Explain the mathematical foundations and learning principles underlying neural networks and deep learning models.
- CO2: Design, train, and evaluate deep neural networks using appropriate architectures, optimization methods, and regularization techniques.
- CO3: Apply convolutional and recurrent neural network architectures to solve problems in computer vision, sequence modeling, and time-series analysis.
- CO4: Implement deep generative models, unsupervised learning methods, and representation learning techniques.
- CO5: Develop deep learning applications using TensorFlow, PyTorch, or equivalent frameworks for classification, prediction, and pattern recognition tasks.
- CO6: Analyze and solve real-world problems in computer vision, natural language processing, and intelligent systems using advanced deep learning techniques.

Syllabus

Introduction to Neural Networks: Biological neurons and inspiration for artificial neural networks, computational units, McCulloch–Pitts neuron model, threshold logic, linear perceptron, perceptron learning algorithm, linear separability, convergence theorem for perceptron learning, limitations of single-layer networks, introduction to multilayer learning systems.

Artificial Neural Networks and Feedforward Learning: Artificial neurons, activation functions, single-layer and multi-layer perceptrons, feedforward neural networks, multilayer perceptron (MLP), gradient descent optimization, backpropagation algorithm, empirical risk minimization, loss functions, overfitting and underfitting, regularization techniques, model evaluation and performance metrics.

Deep Neural Networks and Optimization: Challenges in training deep neural networks, vanishing and exploding gradient problems, greedy layer-wise training, weight initialization methods, optimization algorithms including Adagrad, Adadelta, RMSProp, Adam, Nesterov Accelerated Gradient (NAG), second-order optimization methods, saddle point problem, dropout, dropconnect, batch normalization, early stopping, and hyperparameter tuning.

Convolutional and Recurrent Neural Networks: Convolutional Neural Networks (CNNs), convolution and pooling operations, feature maps, LeNet, AlexNet, VGG, ResNet architectures, transfer learning, Recurrent Neural Networks (RNNs), backpropagation through time (BPTT), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), bidirectional RNNs and LSTMs, sequence modeling and temporal learning applications.

Generative Models and Unsupervised Deep Learning: Restricted Boltzmann Machines (RBMs), Markov Chain Monte Carlo (MCMC) methods, Gibbs sampling, Deep Boltzmann Machines (DBMs), autoencoders, sparse autoencoders, denoising autoencoders, contractive autoencoders, Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), deep representation learning, multi-task learning, and multi-view deep learning.

Applications of Deep Learning: Deep learning for computer vision including image classification, image segmentation, object detection, image captioning, image generation using GANs, video-to-text generation using LSTMs, attention mechanisms for vision tasks, natural language processing applications, vector space models of semantics, word embeddings, Continuous Bag-of-Words (CBOW), Skip-Gram models, GloVe embeddings, transformer architectures, self-attention mechanisms, large language models, and emerging trends in deep learning.

Suggested Laboratory / Programming Exercises

- Implementation of perceptron learning and multilayer neural networks.
- Training and evaluation of feedforward neural networks using TensorFlow/PyTorch.
- Hyperparameter tuning and regularization experiments.
- Implementation of CNNs for image classification tasks.
- Development of RNN, LSTM, and GRU models for sequence prediction.
- Construction of autoencoders and variational autoencoders.
- Training and evaluation of Generative Adversarial Networks (GANs).
- Word embedding generation using CBOW, Skip-Gram, and GloVe models.
- Implementation of transformer-based architectures for NLP tasks.
- Mini-project involving deep learning applications in computer vision, healthcare, finance, speech processing, or natural language processing.

Textbooks / References

1. I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.
2. C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
3. R. Rojas, *Neural Networks: A Systematic Introduction*, Springer, 1996.
4. F. Chollet, *Deep Learning with Python*, Manning Publications.
5. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media.
6. D. Foster, *Generative Deep Learning*, O'Reilly Media.
7. A. Zhang, Z. C. Lipton, M. Li, and A. J. Smola, *Dive into Deep Learning*, Cambridge University Press.
8. Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature*, Vol. 521, 2015.
9. D. Jurafsky and J. H. Martin, *Speech and Language Processing*, Stanford University Draft Edition.
10. A. Vaswani et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

CAO003 Design Thinking and Innovation [3-0-0-3]

Course Objectives

- To introduce students to the principles, processes, and applications of design thinking and innovation.
- To develop skills in user-centered problem identification, empathy building, and user research methods.
- To foster creativity and innovation through brainstorming, ideation, prototyping, and testing activities.
- To equip students with practical skills in interaction design, usability evaluation, and design communication.
- To encourage interdisciplinary collaboration, ethical thinking, and socially responsible innovation through project based learning.

Course Outcomes

- CO1:Explain the concepts, stages, and significance of the design thinking process in solving real-world problems.
- CO2:Apply user research methods, empathy techniques, and problem-definition strategies to understand user needs effectively.
- CO3: Generate innovative ideas using brainstorming and ideation techniques and develop prototypes for testing and evaluation.
- CO4: Design user-centered solutions using interaction design principles, usability evaluation methods, and iterative improvements.
- CO5: Collaborate effectively in interdisciplinary teams and communicate design solutions through presentations, reports, and project demonstrations.

Syllabus

Introduction to Design Thinking - Introduction to de-

sign - Characteristics of good and bad design - Importance of design thinking in engineering and problem solving - Human-centered design approach - The five stages of design thinking: Empathize, Define, Ideate, Prototype, and Test - Design thinking versus traditional problem-solving approaches - Individual and team-based problem solving.
User Empathy and Problem Identification - Understanding user needs and stakeholder analysis - Empathy development techniques - User research methods including interviews, observations, surveys, and focus groups - Problem definition and reframing techniques - User-centered design principles - Ergonomics and accessibility considerations.
Ideation and Concept Development - Brainstorming methods and innovation heuristics - Creativity enhancement techniques - Affinity diagrams, empathy maps, mind maps, and customer journey maps - Concept generation and evaluation - Scenario building and storytelling techniques - Collaborative ideation activities.
Prototyping and Usability Testing - Low-fidelity and high-fidelity prototyping methods - Prototype development tools and techniques - Wireframes - Usability testing methods - Gathering and analyzing user feedback - Iterative design improvement - Heuristic evaluation techniques - Basics of data analysis using SPSS for design evaluation.
Interaction Design and User Experience - Fundamentals of interaction design - History and evolution of interaction design - User Interface (UI) and User Experience (UX) principles - Information architecture - Personas and task flows - Laws and principles of interaction design - Qualitative and quantitative research methods - Usability evaluation techniques.
Innovation, Ethics, and Social Impact - Principles and processes of successful innovation - Modes of design innovation - Social entrepreneurship and innovation for social good - Ethics in design and responsible innovation - Inclusive and sustainable design practices - Industry case studies on innovation and design impact - Familiarization with modern design tools - Design-based mini project development and presentation.

Textbooks / References

1. Kumar, K., and Kurni, M. (Eds.), *Design Thinking: A Forefront Insight*,CRC Press, 2022.
2. Vijay Kumar, *101 Design Methods: A Structured Approach for Driving Innovation in Your Organization*, John Wiley and Sons, 2013.
3. Dan Saffer, *Designing for Interaction: Creating Innovative Applications and Devices*, New Riders Publishing, 2010.
4. Lewrick, M., Link, P., and Leifer, L, *The Design Thinking Toolbox: A Guide to Mastering the Most Popular and Valuable Innovation Methods*, Wiley, 2020.
5. Metts, M. J., and Welfle, A, *Writing is Designing: Words and the User Experience*, Rosenfeld Media, 2020.
6. Uebernickel, F., Jiang, L., Brenner, W., Pukall, B., Naef, T., and Schindlholzer, B, *Design Thinking: The Handbook*, World Scientific, 2020.
7. Gallagher, A., and Thordarson, K., *Design Thinking in Play: An Action Guide for Educators*, ASCD, 2020.
8. Regine Gilber, *Inclusive Design for a Digital World: Designing with Accessibility in Mind*, Apress, 2019.
9. Michael G. Luchs, *Design Thinking: New Product Development Essentials from the PDMA*, Wiley, 2015.
10. D'Source, *Digital Online Environment for Design Learning*, – https://www.dsource.in.

CAO004 Large Language Models [3-0-0-3]

Course Objectives

- To understand the transformer architecture in depth, including self-attention, positional encoding, and encoder-decoder design.
- To distinguish encoder-only, decoder-only, and encoder-decoder model families based on their training objectives.
- To gain understanding of tokenisation algorithms including BPE, WordPiece, and SentencePiece.
- To analyze data pipelines and quality filtering strategies used in major LLM pre-training efforts.

Course Outcomes

- CO1: Derive and implement the self-attention mechanism and explain its computational complexity.
- CO2: Trace a complete pre-training pipeline from raw data to a deployable language model.
- CO3: Compare positional encoding schemes such as sinusoidal embeddings, ALiBi, RoPE, and NoPE based on length-generalisation trade-offs.
- CO4: Configure a distributed training setup using data, tensor, and pipeline parallelism with ZeRO optimisation.

Syllabus

Transformer Architecture: Introduction to the transformer architecture, Self-attention mechanism, Scaled dot-product attention, Multi-head attention, Encoder layers and encoder stack, Decoder layers, Teacher forcing and masked (causal) attention, Position encoding using sinusoidal embeddings, Feed-forward sublayers, Residual connections, Batch normalisation, Layer normalisation.

LLM Families and Training Objectives: Language modelling objectives, Causal language modelling, Decoder-only Large Language Models (LLMs), GPT architecture and training, Pre-training and fine-tuning, Decoding strategies including greedy search, beam search, top-k sampling, and top-p sampling, Encoder-only LLMs, BERT architecture, Masked Language Model (MLM) and Next Sentence Prediction objectives, Encoder-decoder models, BART, T5 text-to-text framework, GPT-2 zero-shot learning.

Tokenisation and Data Pipelines: Word-level and character-level tokenisation challenges, Sub-word tokenisation, Byte

Pair Encoding (BPE), WordPiece, SentencePiece, Vocabulary size trade-offs, Data sources including C4, mC4, Pile, RedPajama, SlimPajama, The Stack, and Sangraha, Data pipelines used in Gopher, OPT, LLaMA, Falcon, and SETU, Quality filtering, Deduplication, Clean data and training effectiveness.

Efficient Attention and Positional Encodings: Time and space complexity of standard attention mechanisms, Sub-quadratic attention methods including local, dilated, random, block-sparse, low-rank, and kernel approaches, FlashAttention, KV-caching, Multi-Query Attention (MQA), Grouped-Query Attention (GQA), Paged attention, Limitations of sinusoidal positional embeddings, ALiBi, RoPE, NoPE, Length generalisation, Pre-normalisation and post-normalisation.

Training and Scaling Large Models: Scaling laws and Chinchilla compute-optimal training, Adam and LION optimisers, Learning rate schedules, Gradient clipping, Training failure modes, Mixed-precision training using FP16 and BF16, Activation checkpointing, CPU offloading, Data parallelism, Tensor parallelism, Pipeline parallelism, ZeRO (Zero Redundancy Optimizer) stages 1–3.

Textbooks / References

1. Vaswani, A. et al., "Attention Is All You Need," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
2. Devlin, J. et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *NAACL*, 2019.
3. Brown, T. et al., "Language Models are Few-Shot Learners," *NeurIPS*, 2020.
4. Raffel, C. et al., "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer," *Journal of Machine Learning Research (JMLR)*, 2020.
5. Touvron, H. et al., "LLaMA: Open and Efficient Foundation Language Models," *arXiv*, 2023.
6. Hoffmann, J. et al., "Training Compute-Optimal Large Language Models," *NeurIPS*, 2022.
7. Dao, T. et al., "FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness," *NeurIPS*, 2022.
8. Rajbhandari, S. et al., "ZeRO: Memory Optimizations Toward Training Trillion Parameter Models," *SC20*, 2020.

IHS314 Project Management and Quality Assurance [3-0-0-3]

Course Prerequisites

Basic knowledge of Management Principles, Economics, Probability and Statistics.

Course Objectives

- To understand the fundamentals of projects, project life cycle, risk analysis, and decision-making under uncertainty.
- To develop competence in project appraisal, feasibility analysis, and project selection techniques.
- To gain knowledge of government policies, tax benefits, industrial incentives, and the role of financial institutions in project financing.
- To learn project evaluation, scheduling, and resource allocation techniques including CPM and PERT.
- To understand project implementation, monitoring, control, quality assurance, and successful project completion strategies.

Course Outcomes

- CO1: Explain the concepts, characteristics, objectives, life cycle, and theoretical foundations of project management, and analyze project risks using utility theory and decision-making techniques under uncertainty.
- CO2: Conduct project appraisal and feasibility studies by evaluating technical, financial, economic, operational, legal, and market aspects of projects.
- CO3: Assess the impact of government policies, tax benefits, industrial incentives, and financial institutions on project financing and development.
- CO4: Apply project evaluation, selection, scheduling, and network analysis techniques including NPV, IRR, ARR, Benefit-Cost Ratio, CPM, and PERT.
- CO5: Plan, implement, monitor, and control projects using appropriate tendering, contracting, vendor management, MIS tools, and quality assurance mechanisms.
- CO6: Demonstrate effective project management practices for achieving project objectives within scope, cost, time, quality, and resource constraints.

Syllabus

Project Fundamentals, Risk Analysis, and Utility Theory: Meaning and definition of projects, characteristics and objectives of projects, temporary activities, project goals and constraints, project life cycle, project identification, formulation, appraisal, implementation, monitoring, closure, systems approach to project management, decision-making concepts, resource allocation, uncertainty in projects, financial risk, market risk, technical risk, operational risk, political and environmental risks, sensitivity analysis, scenario analysis, decision tree analysis, probability analysis, utility theory, utility functions, risk preferences, and investment decision-making under uncertainty.

Project Appraisal and Feasibility Analysis: Concepts of project appraisal and feasibility studies, technical feasibility, financial feasibility, economic feasibility, operational feasibility, legal feasibility, project identification methods, investment opportunities, preliminary screening of projects, market demand analysis, industrial policy and its objectives, employment generation, balanced regional development, export promotion, market potential assessment, resource availability, demand forecasting, competitor analysis, pricing analysis, consumer behavior studies, technical analysis, plant location, machinery selection, production process, capacity planning, technology selection, financial analysis methods, Net Present Value (NPV), Internal Rate of Return (IRR), Payback Period, profitability analysis, social cost-benefit analysis, environmental and social impacts of projects.

Tax Benefits, Incentives, and Financial Institutions: Income tax benefits, depreciation allowances, tax holidays, investment allowances, exemptions and deductions, industrial incentives, capital subsidies, export incentives, concessional finance, infrastructure support, regional development incentives, MSME support programs, entrepreneurship development schemes, project financing mechanisms, long-term and short-term financing, development financial institutions, technical consultancy services, risk assessment and financial guidance, role of IDBI, IFCI, SIDBI, NABARD, and other financial institutions in project development.

Project Evaluation, Selection, and Scheduling: Project evaluation methods, project selection criteria, profitability analysis, risk-return trade-offs, social benefit evaluation, resource allocation techniques, Net Present Value (NPV), Internal Rate of Return (IRR), Accounting Rate of Return (ARR), Benefit-Cost Ratio, Critical Path Method (CPM), network scheduling, critical path analysis, project completion time estimation, Program Evaluation and Review Technique (PERT), optimistic, pessimistic, and most likely time estimates, project management organizations, functional, matrix, and projectized structures, project coordination and communication, planning, organizing, staffing, directing, controlling, quality management, and resource optimization.

Project Implementation, Monitoring, Control, and Quality Assurance: Project implementation activities, resource mobilization, installation and commissioning, project delays and cost overruns, tendering procedures, open, limited, and global tenders, procurement systems, contracting processes, legal agreements, lump-sum contracts, cost-plus contracts, turnkey contracts, vendor selection and supplier evaluation, quality standards, delivery schedules, project planning and scheduling tools, Gantt charts, CPM and PERT applications, Management Information Systems (MIS) for project management, data collection and reporting, decision support systems, project monitoring and control systems, review mechanisms, corrective actions, performance evaluation, feedback systems, quality assurance principles, continuous improvement, project efficiency, and successful project completion.

Textbooks / References

1. H. Kerzner, *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*, Wiley.
2. P. Chandra, *Project Planning, Analysis, Selection, Financing, Implementation and Review*, McGraw-Hill Education.
3. V. Desai, *Project Management*, Himalaya Publishing House.
4. I. M. Pandey, *Financial Management*, Vikas Publishing House.
5. A. E. Boardman, D. H. Greenberg, A. R. Vining, and D. L. Weimer, *Cost-Benefit Analysis: Concepts and Practice*, Cambridge University Press.
6. D. Salvatore, *Managerial Economics*, Oxford University Press.
7. M. Y. Khan, *Indian Financial System*, McGraw-Hill Education.
8. L. M. Bhole and J. Mahakud, *Financial Institutions and Markets*, McGraw-Hill Education.
9. H. A. Taha, *Operations Research: An Introduction*, Pearson Education.
10. W. J. Stevenson, *Operations Management*, McGraw-Hill Education.
11. K. C. Laudon and J. P. Laudon, *Management Information Systems*, Pearson.
12. K. K. Chitkara, *Construction Project Management: Planning, Scheduling and Controlling*, Tata McGraw-Hill.
13. *A Guide to the Project Management Body of Knowledge (PMBOK Guide)*, Project Management Institute (PMI).
14. NPTEL Course Materials on Project Management.
15. Industry Reports and Case Studies from Government and Financial Institutions.

IHS315 Introduction to Indian Knowledge Systems [3-0-0-3]

Course Objectives

- To introduce and explain the foundational concepts, classifications, and historical development of Indian Knowledge Systems.
- To examine major Indian traditions in philosophy, mathematics, science, astronomy, medicine, and governance.
- To analyze key concepts related to Indian epistemology, ethics, sustainability, and holistic approaches to knowledge.
- To evaluate the relevance and application of Indian Knowledge Systems in contemporary interdisciplinary and sustainable practices.

Course Outcomes

- CO1: Explain the historical foundations, classification, and scope of Indian Knowledge Systems
- CO2: Describe and examine important contributions of Indian traditions in philosophy, mathematics, science, astronomy, and medicine
- CO3: Analyze key concepts related to Indian epistemology, ethics, governance, and holistic knowledge traditions
- CO4: Evaluate the contemporary relevance of Indian

Knowledge Systems in interdisciplinary studies, sustainability, and global knowledge practices.

Syllabus

Foundations of Indian Knowledge Systems: Introduction to Indian Knowledge Systems - Nature and scope of traditional knowledge - Vedas, Vedangas, Upanishads, Itihasas, and Puranas - Overview of Indian philosophical systems (Darshanas) - Oral traditions and knowledge transmission- Concepts of Dharma, Artha, Kama, and Moksha - Gurukula system.

Science and Knowledge Traditions in India: Indian mathematics and numeral systems - Contributions of Aryabhata and other scholars - Astronomy and Indian calendar systems - Traditional medicine and wellness systems - Indian approaches to logic and reasoning (Pramana) - Sanskrit and scientific knowledge traditions - Indian contributions to metallurgy, architecture, and engineering.

Society, Culture, and Contemporary Relevance: Indian ethics and governance traditions - Environmental consciousness and sustainability in IKS - Yoga and holistic wellbeing - Indian arts, aesthetics, and performance traditions - Indian Knowledge Systems and modern interdisciplinary studies - Influence of IKS on global thought and culture - Contemporary applications and relevance of IKS.

Textbooks / References

1. Chattopadhyay, S., "Reviving Timeless Insights: A Deep Dive into the Indian Knowledge System and its Modern Significance," *International Journal of Law Management and Humanities*, vol. 7, no. 6, pp. 2022–2035, 2024.
2. Das, T., "Relevance of Indian Knowledge System in the Present," in *Indian Knowledge Systems: Journey from Vedic Wisdom to Digital Repositories*, 2026, pp. 160–170.
3. Datta, R., and Mete, J., "Exploring the Foundations and Modern Relevance of Indian Knowledge Systems."
4. Kapoor, K., and Singh, A. K. (eds.), *Indian Knowledge Systems*, Vol. 1, New Delhi: Indian Institute of Advanced Study, 2005.
5. Khan, S., and Sharma, M., "An Overview on Indian Knowledge System," *Integrated Journal for Research in Arts and Humanities*, vol. 4, no. 4, pp. 42–46, 2024.
6. Kumari, P., and Niyogi, A., "The Indian Knowledge System and Indigenous Pedagogies: A Historical and Contemporary Review," *IUJ Journal of Management*, vol. 13, no. 1, pp. 9–27, 2025.
7. Mahadevan, B., Bhat, V. R., and Nagendra, P. R. N., *Introduction to Indian Knowledge System: Concepts and Applications*, 2022.
8. Medhi, U. N., "Epistemic Foundation of Indian Knowledge System: A Study in Primary Narratives and Secondary Exegesis," *International Journal of Social Science Research (IJSSR)*, vol. 2, no. 4, pp. 339–349, 2025.
9. Mehta, T., and Singh, N. V., "An Exploration of the Indian Knowledge System: Roots, Significance, and Contemporary Relevance," *Bhartiya Knowledge Systems*, pp. 33–39, 2024.
10. Mohanapriya, B., and Suriya, M., *Indian Knowledge Systems: Principles and Practices*, SSS Publications, 2025.
11. Nath, S., "Bridging the Past and Present: Philosophical Insights from the Indian Knowledge System," *Bharati International Journal of Multidisciplinary Research and Development*, vol. 2, no. 9, pp. 83–88, 2024.
12. Priya, B., and Thakur, G. K., *Indian Knowledge System: Mathematical Foundations and Engineering Applications*, GEH Press, 2026.
13. Rai, P., and Pandey, R. K., "Discourse of Indian Knowledge System," *World Research of Political Science Journal*, vol. 8, no. 1, pp. 67–74, 2025.
14. Rao, S. S., "Indigenous Knowledge Organization: An Indian Scenario," *International Journal of Information Management*, vol. 26, no. 3, pp. 224–233, 2006.
15. Singh, N., Kalhotra, S. K., and Duwarah, L., *Indian Knowledge System*, Addition Publishing House, 2025.
16. Singh, D., "Indian Knowledge Systems: Foundations, Transformations, and Contemporary Relevance," *International Journal of Emerging Voices in Education*, vol. 2, no. 1, pp. 21–37, 2026.
17. Swamy, M. K., *Indian Knowledge Systems: Tradition, Scientific Inquiry, and Philosophy*, Academic Guru Publishing House, 2025.
18. Yadav, A., "Indian Knowledge System as the Foundation of Viksit Bharat," *The Voice of Creative Research*, vol. 6, no. 4, pp. 81–86, 2024.

IHS316 Fundamentals of Sanskrit [3-0-0-3]

Course Objectives

- To introduce and explain the structure, pronunciation system, and linguistic features of Sanskrit language.
- To develop foundational skills in Sanskrit reading, writing, transliteration, and simple communication.
- To examine and apply basic Sanskrit grammatical concepts including Sandhi, Samasa, nouns, and verb forms in sentence construction.
- To interpret and appreciate selected Sanskrit literary and philosophical traditions and their contribution to Indian knowledge systems.

Course Outcomes

- CO1: To explain the basic linguistic structure and phonetic system of Sanskrit language.
- CO2: To demonstrate elementary skills in Sanskrit pronunciation, transliteration, reading, writing, and communication
- CO3: To apply fundamental Sanskrit grammatical concepts such as Sandhi, Samasa, and sentence formation in simple language exercises.

- CO4: To analyze and interpret selected Sanskrit literary and philosophical texts in the context of Indian intellectual traditions.

Syllabus

Introduction to Sanskrit Language and Tradition: Historical overview of Sanskrit - Sanskrit as a language of knowledge and culture-Introduction to sounds (Dhvani) and alphabets (Varna) -Transliteration using IAST system.

Basic vocabulary and pronunciation - Difference between Bhasa, Vak, and Vani - Oral traditions and preservation of knowledge - Sanskrit and Indian intellectual traditions.

Fundamentals of Sanskrit Grammar and Communication: Introduction to Sanskrit grammar traditions - Basic sentence structure -Introduction to nouns, gender, and number - Basic verb forms and sentence making - Introduction to Sandhi and Samasa - Conversational Sanskrit expressions -Reading and writing simple Sanskrit sentences-Introduction to Panini and grammatical systems.

Sanskrit Literature and Contemporary Relevance: Overview of Sanskrit literary traditions - Introduction to Vedas, Upanishads, and classical literature - Selected Subhashitas and simple verses -Sanskrit and Indian philosophy -Sanskrit in science and Natural Language Processing (NLP) -Introduction to epics and poetic traditions- Reading and interpretation of selected passages - Relevance of Sanskrit in contemporary interdisciplinary studies- Introduction to the software BARAHA.

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